

3.4 Interest Rates

Learning Outcomes

By the end of this section, you will be able to:

- Explain the relationship between the nominal interest rate and inflation.
- Calculate the real rate of interest.
- Explain the relationship between interest rates and risk.

Market for Loanable Funds

An interest rate is the rental price of money. The concepts of supply, demand, and equilibrium apply in this market just as they do in other markets. This market is referred to as the *market for loanable funds*.

In the market for loanable funds, the suppliers of funds are economic entities that currently have a surplus in their budget. In other words, they have more income than they currently want to spend; they would like to save some of their money and spend it in future time periods. Instead of just putting these savings in a box on a shelf for safekeeping until they want to spend it, they can let someone else borrow that money. In essence, they are renting that money to someone else, who pays a rental price called the *interest rate*.

The suppliers of loanable funds, also known as lenders, are represented by the upward-sloping curve in [Figure 3.17](#). A higher interest rate will encourage these lenders to supply a larger quantity of loanable funds.

The demanders of funds in the loanable funds market are economic entities that currently have a deficit in their budget. They want to spend more than they currently have in income. For example, a grocery store chain that wants to expand into new cities and build new grocery stores will need to spend money on land and buildings. The cost of buying the land and buildings exceeds the chain's current income. In the long run, its business expansion will be profitable, and it can pay back the money that it has borrowed.

The downward-sloping curve in [Figure 3.17](#) represents the demanders of loanable funds, also known as borrowers. Higher interest rates will be associated with lower quantities demanded of loanable funds. At lower interest rates, more borrowers will be interested in borrowing larger quantities of funds because the price of renting those funds will be cheaper.

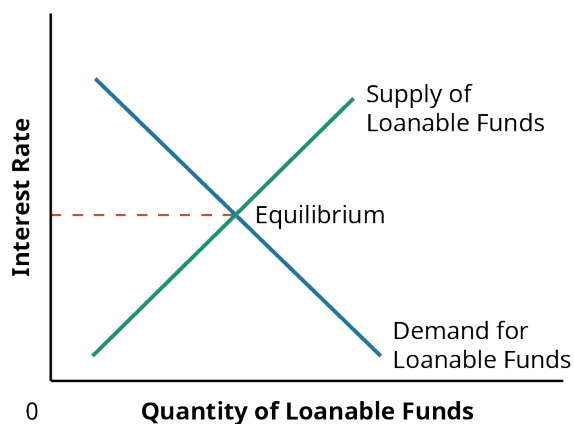


Figure 3.17 Equilibrium in the Loanable Funds Market

The equilibrium interest rate is determined by the intersection of the demand and supply curves. At that interest rate, the quantity supplied of loanable funds exactly equals the quantity demanded of loanable funds. There is no shortage of loanable funds, nor is there any surplus.

10 National Bureau of Economic Research. "US Business Cycle Expansions and Contractions." Last updated July 19, 2021. <https://www.nber.org/research/data/us-business-cycle-expansions-and-contractions>

Nominal Interest Rates

The **nominal interest rate** is the stated, or quoted, interest rate. If you want to borrow money to purchase a car and the bank quotes an interest rate of 5.5% on a four-year auto loan, the 5.5% is the nominal interest rate.

Or suppose you have \$1,000 you would like to place in a savings account. If the bank quotes an interest rate of 6% on its savings accounts, the 6% is a nominal interest rate. This means that if you place your \$1,000 in a savings account for one year, you will receive \$60 in interest for the year. At the end of the year, you will have a balance of \$1,060 in your savings account—your original \$1,000 plus the \$60 in interest that you earned.

Real Interest Rates

Suppose you are deciding between saving your \$1,000 for the year and using it to purchase a flat-screen TV. The advantage of spending the money on the TV today is that you can enjoy watching programs on it over the next year. The advantage of saving the money is that you will earn 6% nominal interest; in one year, you will have \$1,060 to spend.

If there is a 2% inflation rate, you would expect the TV that costs \$1,000 today to cost \$1,020 in one year. If you save the \$1,000, you will have \$1,060 in one year. You could purchase the TV for \$1,020 and have \$40 left over; then you could use the \$40 to order pizza to celebrate the first big game you are watching on the new TV.

Your choice comes down to enjoying a TV today or enjoying a TV and \$40 in one year. The \$40 is your reward for delaying consumption. It is your real return for saving money. The remaining \$20 of the interest you earned just covered the rate of inflation. This reward for delaying consumption is known as the **real interest rate**. The real interest rate is calculated as

$$\text{Real Interest Rate} = \text{Nominal Interest Rate} - \text{Inflation Rate}$$

The real interest rate, rather than the nominal interest rate, is the true determinant of the cost of borrowing and the reward for lending. For example, if a business had to pay 15% nominal interest rate in 1980, when the inflation rate was 12%, the real cost of borrowing for the firm was 3%. The company would have had to pay \$15 in interest each year for each \$100 it borrowed, but \$12 of that was simply compensating the lender for inflation. In real terms, the business was only paying \$3 to borrow \$100.

In recent years, a business may have paid 6% interest to borrow money. This nominal rate is half of what it was in 1980. However, inflation has been much lower. If inflation is 1% and the company pays 6% nominal interest, that results in a 5% real interest rate. For every \$100 the company borrows, it pays \$6 in interest; \$1 is compensating for inflation, and the remaining \$5 is the real cost of borrowing.

Risk Premiums

As we have discussed interest rates, we have talked about how the interest rate is determined by the demand and supply of loanable funds. This tells us the underlying interest rate in the economy. You will notice, however, if you look at the financial news, that there is more than one interest rate in the economy at any given time.

[Figure 3.18](#) shows the interest rates that three different types of borrowers have paid over the past 20 years. The bottom line shows the interest rate that the US government paid to borrow money for a three-month period. This rate is often referred to as the risk-free rate of interest. While theoretically it would be possible for the US government to default and not pay back those people who have loaned money to it, the chances of that are occurring are extremely low.

When someone borrows money, they enter into a contract to repay the money and the interest owed. However, sometimes, certain circumstances arise such that the lender has a difficult time collecting the money, even though the lender has the legal right to the money. For example, if a company goes bankrupt after borrowing the money and before paying back the loan, the lender may not be able to collect what is due. The chance that the lender may not be able to collect all of the money due at the time it is due is considered

credit risk. Lenders want to be rewarded for taking on this risk, so they charge a premium to borrowers who are higher risk.

Companies are more likely to go bankrupt and not be able to pay their bills on time than is the US government. So, corporations have to pay a higher interest rate than the US government. If a lender can earn 1% lending to the US government, the lender will only be willing to lend to the riskier corporation if they can earn more than 1%. In [Figure 3.18](#), the interest rates paid by very creditworthy companies is shown. The prime rate is the interest rate that banks charge their very best customers—large companies that are financially very strong and have a very low risk of default.

It is generally riskier for a lender to make a loan to an individual than to a corporation. Individuals are more likely to become ill, lose their job, or experience some other financial setback that makes it difficult for them to repay their loans. In [Figure 3.18](#), the interest rate on credit card loans is much higher than the interest rate charged to corporate borrowers. That is because credit card loans are a high risk to the lender. Unlike other loans made to an individual, such as a car loan, the credit card company has no collateral if a consumer cannot pay back the loan. With car loans, if the borrower fails to repay the borrowed money, the lender can repossess the automobile and sell it to recoup some of the money it is owed. If you use a credit card to buy groceries and do not repay the loan, the credit card company cannot repossess the groceries that you purchased. Therefore, credit card loans have notoriously high interest rates to compensate the lender for the high risk.

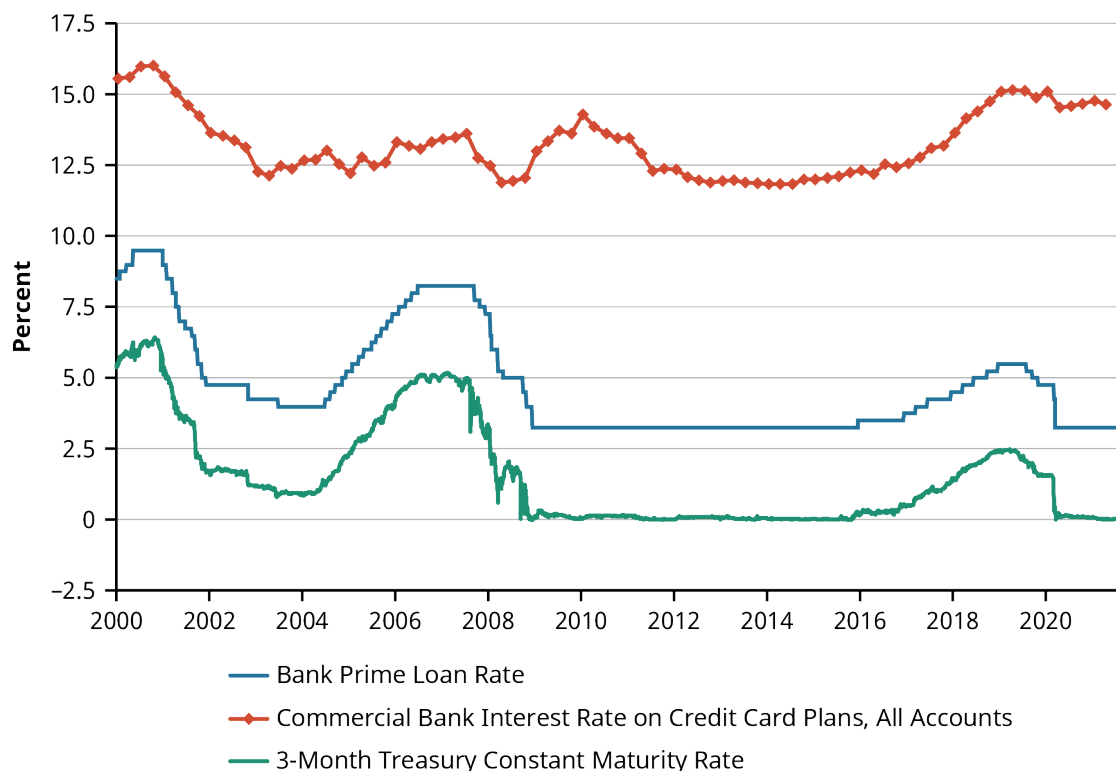


Figure 3.18 Interest Rate on US Treasury Bills and Credit Cards¹¹ This graph shows the rates posted by a majority of top 25 (by assets in domestic offices) insured US-chartered commercial banks. The prime rate is one of several base rates used by banks to price short-term business loans. For further information regarding Treasury constant maturity data, please refer to the Board of Governors and the Treasury.

¹¹ Data from Board of Governors of the Federal Reserve System (US). "Bank Prime Loan Rate (DPRIME)." FRED. Federal Reserve Bank of St. Louis, accessed July 8, 2021. <https://fred.stlouisfed.org/series/DPRIME>; Board of Governors of the Federal Reserve System (US). "3-Month Treasury Constant Maturity Rate (DGS3MO)." FRED. Federal Reserve Bank of St. Louis, accessed July 8, 2021. <https://fred.stlouisfed.org/series/DGS3MO>; Board of Governors of the Federal Reserve System (US). "Commercial Bank Interest Rate on Credit Card Plans, All Accounts (TERMCBCALLNS)." FRED. Federal Reserve Bank of St. Louis, accessed July 8, 2021. <https://fred.stlouisfed.org/series/TERMCBCALLNS>

3.5 Foreign Exchange Rates

Learning Outcomes

By the end of this section, you will be able to:

- Explain what it means for a currency to appreciate or depreciate.
- Define spot exchange rate.
- Explain the risks involved in translation exposure.

Spot Exchange Rate

An exchange rate is simply the price of a currency. If you live in the United States and are going on a trip to Mexico, you will need pesos to pay for your food, hotel, and other items. You will need to purchase pesos. Suppose that you go to your bank to purchase pesos. Suppose the bank tells you that it will cost \$0.0625 to purchase one Mexican peso. If you want to take 10,000 Mexican pesos with you on your trip, it will cost you \$625 to purchase the desired pesos.

In this example, the price of one peso is six and one-quarter cents. This price will often be written in the form of

$$\text{MXN } 1 = \text{USD } 0.0625$$

MXN is an abbreviation for the Mexican peso, and USD is an abbreviation for the US dollar. This price is known as a currency exchange rate, or the rate at which you can exchange one currency for another currency. Because this is the price you would pay to purchase Mexican pesos right now, it is known as the **spot exchange rate**.

If you know the price of Mexican pesos in dollars, you can easily find the price of US dollars in Mexican pesos. Simply divide both sides of the equation by 0.0625, or the price of a peso:

$$\begin{aligned} \text{MXN } 1 &= \text{USD } 0.0625 \\ \frac{\text{MXN } 1}{0.0625} &= \frac{\text{USD } 0.0625}{0.0625} \\ \text{MXN } 16 &= \text{USD } 1 \end{aligned}$$

If you have Mexican pesos and you want to exchange them for dollars, you will be using the pesos to buy dollars. Each US dollar will cost you MXN 16.

LINK TO LEARNING

Currency Conversion

Do you want to know how many British pounds your \$100 would buy? Or would you like to see how much that 10,000-yen-per-night hotel room will cost you in your home currency? You can enter the amount of one currency in the [MSN Money currency converter \(https://openstax.org/r/currencyconverter\)](https://openstax.org/r/currencyconverter) to see what the equivalent amount is in another currency.

Currency Appreciation

Just as the price of gasoline changes, resulting in it costing more to purchase gasoline on some visits to the gas station than on other visits, the price of a currency also changes. Currency appreciation occurs when it costs more to purchase a currency than it did before.

If the next time you go to the bank to purchase pesos, the bank quotes an exchange rate of $\text{MXN } 1 = \text{USD } 0.0800$, it means that it now costs \$0.0800 (up from \$0.0625) to purchase a Mexican peso. Hence, the price of a peso has risen, or the peso has appreciated.

Currency prices are determined in the marketplace through the same types of supply-and-demand forces we discussed earlier in this chapter. What would cause the peso to appreciate? Either an increase in the demand for pesos or a decrease in the supply of pesos.

Currency Depreciation

Just as a currency can appreciate, it can depreciate. If the quote at the bank was $\text{MXN } 1 = \text{USD } 0.0500$, it would only cost \$0.0500 to purchase a Mexican peso. When it costs fewer dollars to purchase a peso, the peso has depreciated. Either a decrease in demand for pesos or an increase in supply of pesos will cause the peso to depreciate.

Because an exchange rate is the price of one currency expressed in terms of another currency, if one of the currencies depreciates, the other currency must, by definition, appreciate. If it costs \$0.0500 to purchase a Mexican peso, the price of a US dollar, in terms of a Mexican peso would be calculated as

$$\begin{aligned}\text{MXN } 1 &= \text{USD } 0.0500 \\ \frac{\text{MXN } 1}{0.0500} &= \frac{\text{USD } 0.0500}{0.0500} \\ \text{MXN } 20 &= \text{USD } 1\end{aligned}$$

So, the price of one US dollar would be 20 Mexican pesos.

Exchange Rate Risk

Businesses that engage in international business face currency exchange rate risk. As exchange rates change, a business can be impacted in a number of ways. One of these risks, **transaction exposure**, is the risk that the value of a business's expected receipts or expenses will change as a result of a change in currency exchange rates. A pottery-making business that has sold merchandise to a company in the United States for \$20,000, for example, will need to exchange the \$20,000 for pesos to be able to pay its workers and other expenses in pesos. How many pesos it will receive for \$20,000 will change depending on the exchange rate. If the exchange rate is $\text{MXN } 16 = \text{USD } 1$, the company will receive 320,000 pesos for the \$20,000. If the exchange rate is $\text{MXN } 20 = \text{USD } 1$, the company will receive 400,000 pesos for the \$20,000. Thus, as the peso depreciates (and the US dollar appreciates), the same number of dollars will provide more pesos. Conversely, as the peso appreciates (and the US dollar depreciates), the same number of dollars will provide fewer pesos.

Firms that hold assets in a foreign country also face **translation exposure**. When a company creates its financial statements, items are reported using one currency. As foreign exchange rates change, the value of how items are reported on these financial statements can change. This type of risk is an accounting risk.

Economic exposure is the risk that a change in exchange rates will impact a business's number of customers and sales. For example, tourists have the option of spending a week-long vacation at a resort in the United States or in Mexico. As the dollar appreciates, US citizens can exchange their dollars for more pesos, resulting in their purchasing power going further at a Mexican resort. Because an appreciating dollar also means a depreciating peso, it would mean that Mexicans who earn pesos will receive fewer dollars when they exchange their pesos. A Mexican who wants to stay at a \$200-per-night hotel in Colorado will need more pesos to pay for the room when the peso depreciates. The depreciating peso will likely mean that more Mexicans will spend their vacation week in Mexico, and fewer will vacation in the United States.

Even businesses that do not view themselves as involved in international business can face economic exposure. The ski lodge in Colorado will find that its customers from Mexico decrease when the dollar appreciates. Likewise, when the dollar appreciates, some of the ski lodge's US-based customers may choose instead to visit a resort in Mexico, where their purchasing power is strong.

3.6 Sources and Characteristics of Economic Data

Learning Outcomes

By the end of this section, you will be able to:

- Interpret economic data.
- Compute the percent change for economic variables.

FRED: Federal Reserve Economic Data

One of the largest sources of economic data is the Federal Reserve Economic Data (FRED) database.¹² This database is maintained by the Federal Reserve Bank of St. Louis and contains more than 765,000 economic time series. These time series are compiled by the Federal Reserve and come from a number of sources, including the Bureau of Labor Statistics and the US Census.

LINK TO LEARNING

FRED

Data included in the FRED database is divided into these broad categories:

- Money, Banking, and Finance
- Population, Employment, and Labor Markets
- National Accounts
- Production and Business Activity
- Prices
- International Data
- US Regional Data
- Academic Data

Watch this [FRED introduction video \(https://openstax.org/r/what-is-fred\)](https://openstax.org/r/what-is-fred) to learn more information about how to use FRED.

You can find statistics on employment, inflation, exchange rates, gross domestic product, interest rates, and many other economic variables in the FRED database. Although much of the data is about the US markets, macroeconomic data from other countries is also available. In addition to being viewable in graphical and text form on the FRED site, the data is easily downloaded into an Excel spreadsheet for analysis.

LINK TO LEARNING

Creating a Stacked Area Graph

Not only does FRED provide an excellent source for economic data, but it allows users to create custom graphs that can be used in presentations and reports. To learn more about how to create these graphs, read [this article on using stacking \(https://openstax.org/r/fred-gdp-stacking\)](https://openstax.org/r/fred-gdp-stacking). This tutorial provides step-by-step instructions for creating a stacked area graph of real GDP and its separate components and a real GDP area graph showing the percent that each of these components contributes to the total value of real GDP.

¹² Federal Reserve Bank of St. Louis. Economic Resources & Data. Accessed October 25, 2021. <https://www.stlouisfed.org/>

Levels versus Percentage Changes

The same information can be presented in graphs several different ways. The particular format you choose will depend on how you are using the data.

[Figure 3.19](#) shows the real GDP of Japan for 2010–2020. This chart is created showing the level of real GDP. The steep drop in 2020 highlights the economic decline associated with the COVID-19 pandemic. Looking at the chart, it is easy to see that after 10 years of a general upward trend, Japan's GDP quickly fell to a level not seen in the previous decade as COVID-19 began spreading in early 2020.

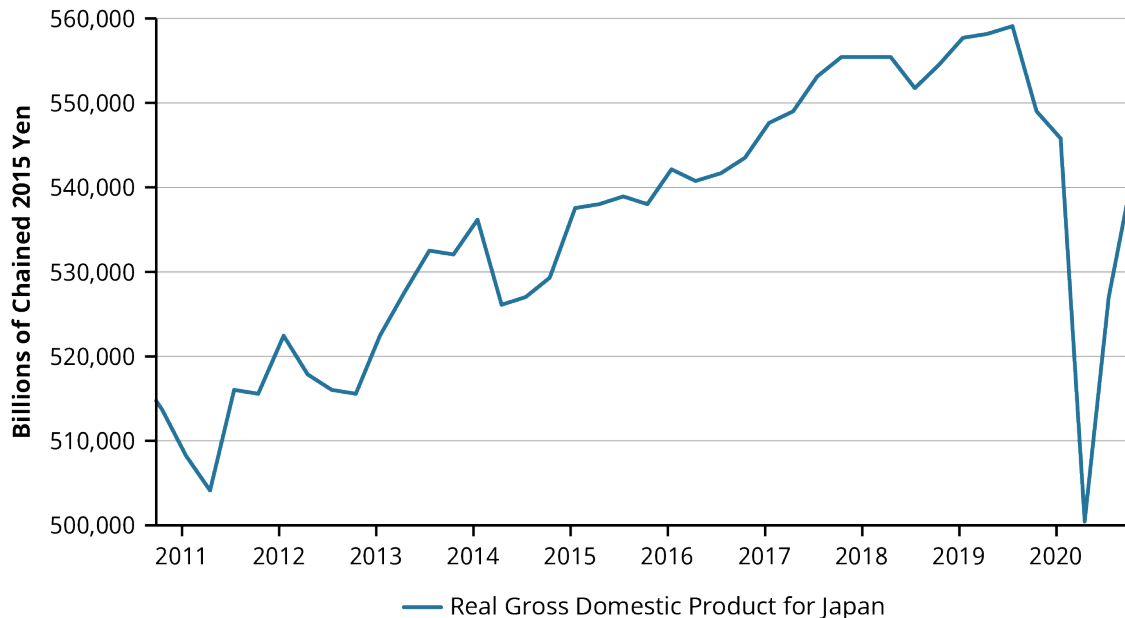


Figure 3.19 Real Gross Domestic Product for Japan, 2010–2020¹³

The vertical axis in [Figure 3.19](#) is measured in yen. Over the time period shown, the real GDP ranged from 500 trillion yen to 560 trillion yen. The general trend (until COVID-19) was upward, indicating growth in the Japanese economy. However, the growth was not consistent from year to year.

[Figure 3.20](#) also contains information about Japanese real GDP from 2010 to 2020. This chart measures the percent change for each quarter on the vertical axis. It is created using the same underlying data as [Figure 3.19](#). [Figure 3.20](#) demonstrates a way of highlighting the growth (or contraction) of an economy at a particular point in time.

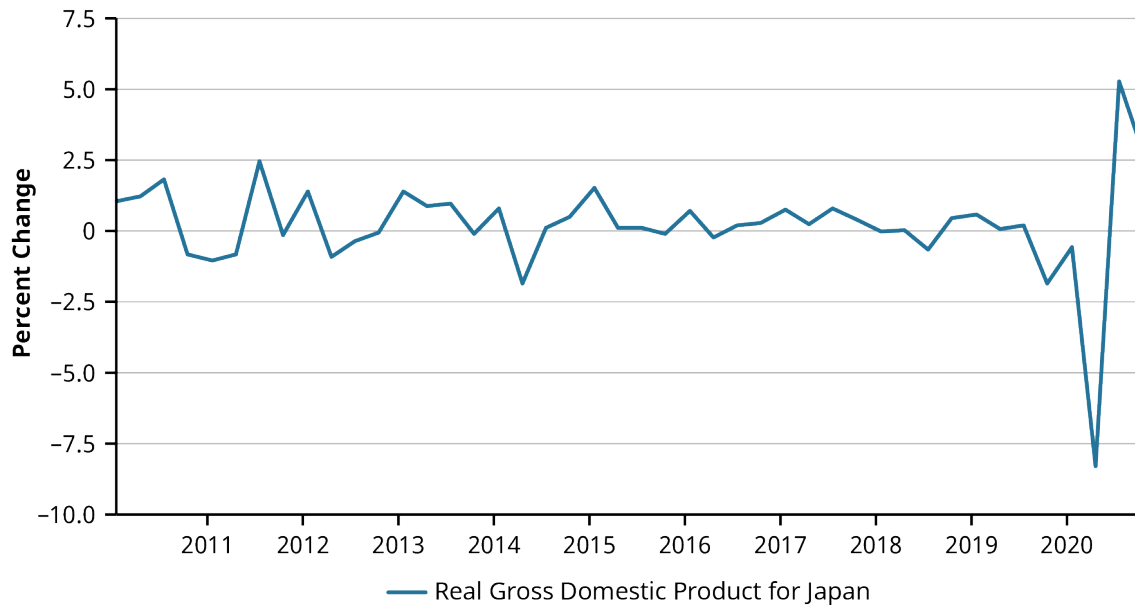


Figure 3.20 Percent Change for Gross Domestic Product for Japan, 2010–2020¹⁴

The formula to calculate the percentage change from one quarter to the next is

$$\text{Percentage Change} = \frac{\text{Quarter}_2 - \text{Quarter}_1}{\text{Quarter}_1}$$

In the first quarter of 2013, the real GDP for Japan was 522,594.2 billion yen. In the second quarter of 2013, the real GDP had risen to 527,277.0 billion yen. Thus, the percentage change in real GDP from quarter one to quarter two was

$$\text{Percentage Change} = \frac{\text{Quarter}_2 - \text{Quarter}_1}{\text{Quarter}_1} = \frac{527,277.0 - 522,594.2}{522,594.2} = 0.00896 = 0.896\%$$

As long as the percentage change for a quarter is positive, the real GDP in [Figure 3.20](#) will rise; this indicates that the economy is growing. If the percentage change shown in [Figure 3.20](#) is negative, then real GDP will fall; this indicates that the economy is contracting. Looking at the percentage change in [Figure 3.20](#) is helpful for detecting when the economy is growing but the growth is slowing. If the percentage change is positive but lower than it was for the previous quarter, then GDP is growing, but the growth rate is slowing.

Indexes

An index is created to track the performance of a particular aspect of the economy or the financial markets. An index helps compare the level of a variable at one point in time relative to another point in time. Indexes are often used when movement over time is more important than the absolute level of the variable at any one point in time.

Earlier in this chapter, we looked at the rate of change in the CPI to measure the rate of inflation. In its raw form, the CPI is an index. Remember that the CPI is a measure of the cost of a market basket of goods. When the index is created, the total cost of the market basket, whether it is \$300 or \$950, is irrelevant. What economists are interested in is the magnitude of the difference in cost of the same market basket at a later date.

In order to focus on the change over time, a base year is identified. The cost of the market basket in the base

¹³ Data from JP, Cabinet Office. "Real Gross Domestic Product for Japan (JPNRGDPEXP)." FRED. Federal Reserve Bank of St. Louis, accessed July 7, 2021. <https://fred.stlouisfed.org/series/JPNRGDPEXP>

¹⁴ Data from JP, Cabinet Office. "Real Gross Domestic Product for Japan (JPNRGDPEXP)." FRED. Federal Reserve Bank of St. Louis, accessed July 7, 2021. <https://fred.stlouisfed.org/series/JPNRGDPEXP>

year is given an index level of 100. Let's assume that the market basket costs \$300 in the base year. If the same basket of goods costs \$330 the following year, then the index level the following year would be 110. The index level increases by 10% when the cost of the market basket increases by 10%. This makes it easy to compare different measures of inflation.

For example, suppose a market basket costs 40,000 yen in the first year and 42,000 yen in the second year. In the base year, the CPI in Japan would be set at 100; the following year, the index would rise to 105 (because of the 5% rise in the market basket cost). Comparing the levels of the index in Japan with the index in the United States allows you to compare inflation trends in the two countries.

Table 3.6 contains the CPI for the United States, Japan, and Switzerland for each decade since 1970. A base year of 1970 is used for all three countries, so the index level is 100 for all three countries in 1970. You can see that Japan has experienced virtually no inflation for the last several decades. If the index level remains the same from one year to the next, there is a zero rate of inflation. Negative rates of inflation, or deflation, would be associated with a falling index level.

Year	United States	Switzerland	Japan
1970	100.0	100.0	100.0
1980	212.3	162.3	232.5
1990	336.5	226.4	291.2
2000	443.5	278.2	322.1
2010	561.6	304.3	313.6
2020	666.6	302.3	331.8

Table 3.6 CPI Levels for the United States, Switzerland, and Japan¹⁵

Using an index level helps us compare the impact that inflation has had on the cost of living in the three countries. Prices were rising rapidly in Japan in the 1970s, outpacing price increases in both the United States and Switzerland. By the mid-1980s, however, price increases in Japan tapered off. Although prices in Switzerland rose much more slowly in the 1970s, the price level continued to rise over the next couple of decades. Even though the price increases have followed different patterns in Switzerland and Japan, the overall price level today is about three times what it was in 1970 in both of those countries. However, the price level in the United States has continued to rise; today, the price level in the United States is about seven times higher than it was in the 1970s.

¹⁵ Data from US Bureau of Labor Statistics. "Consumer Price Index for All Urban Consumers: All Items in US City Average (CPIAUCNS)." FRED. Federal Reserve Bank of St. Louis, accessed July 31, 2021. <https://fred.stlouisfed.org/series/CPIAUCNS>; Organization for Economic Co-operation and Development. "Consumer Price Index: All Items for Switzerland (CHECPIALLMINMEI)." FRED. Federal Reserve Bank of St. Louis, accessed July 31, 2021. <https://fred.stlouisfed.org/series/CHECPIALLMINMEI>; Organization for Economic Co-operation and Development. "Consumer Price Index of All Items in Japan (JPNCPPIALLMINMEI)." FRED. Federal Reserve Bank of St. Louis, accessed July 31, 2021. <https://fred.stlouisfed.org/series/JPNCPPIALLMINMEI>

Summary

3.1 Microeconomics

Microeconomics is the study of individual economic decision makers. In the marketplace, buyers and sellers come together. The buyers are represented by a downward-sloping demand curve; lower prices are associated with a larger quantity demanded. The sellers are represented by an upward-sloping supply curve; higher prices are associated with a large quantity supplied. The point of intersection of the supply and demand curves determines the equilibrium price and quantity.

3.2 Macroeconomics

Macroeconomics looks at the economy as a whole. It focuses on broad issues such as inflation, unemployment, and growth of production. The consumer price index is a common measure of inflation. Unemployment equals the percent of the labor force that is without a job but looking for work. Gross domestic product is a measure of the growth of production and growth of the economy.

3.3 Business Cycles and Economic Activity

The business cycle is the repeated pattern of the economy experiencing expansion, a peak, contraction, and a trough. During the expansion, the economy is growing and real GDP is rising. During a contraction, also known as a recession, the economy is slowing down.

3.4 Interest Rates

The interest rate is the rental price of money. The nominal interest rate is the quoted, or stated, rate of interest. The real rate of interest equals the nominal interest rate minus the inflation rate. Lenders charge a risk premium to compensate them for lending to higher-risk borrowers.

3.5 Foreign Exchange Rates

A foreign exchange rate is the price of a currency in terms of another currency. If a currency will purchase more of a foreign currency than it would previously, the purchasing currency is said to have appreciated. If a currency will purchase less of a foreign currency than it would previously, the purchasing currency is said to have depreciated.

3.6 Sources and Characteristics of Economic Data

One of the most comprehensive sources of economic data is the FRED database, maintained by the Federal Reserve Bank of St. Louis. Over 765,000 time series of data are available. In addition to being viewable in graphical and text form on the FRED site, the data is easily downloaded into an Excel spreadsheet for analysis.

Key Terms

business cycle a period of economic expansion followed by economic contraction

ceteris paribus holding all other things constant

consumer price index (CPI) a measure of inflation based on the cost of a market basket of goods that a typical urban family of four might purchase

core inflation index a measure of inflation that removes food and energy prices from the CPI

demand the quantity of a good or service that consumers are willing and able to purchase during a given time period, ceteris paribus

economic exposure the risk that a change in exchange rates will impact a business's sales and number of customers

equilibrium the point at which the demand and supply curves for a good or service intersect

equilibrium price the price at which quantity demanded equals quantity supplied

expansion the part of the business cycle in which GDP is growing

GDP deflator a measure of inflation tracked by the Bureau of Economic Analysis, intended to measure what GDP would be if prices did not change from one year to the next

gross domestic product (GDP) the market value of all goods and services produced within an economy during a year

inflation a general increase in prices

law of demand the principle that the quantity of purchases varies inversely with price; the higher the price, the lower the quantity

macroeconomics the study of the economy as a whole, focusing on unemployment, inflation, and total output

microeconomics the study of the economy at the individual level, focusing on how individuals and businesses choose to allocate scarce resources

nominal interest rate the quoted or stated interest rate

producer price index (PPI) a measure of the prices that producers of goods and services pay for their supplies and raw materials

real interest rate the nominal interest rate minus the rate of inflation

recession the part of the business cycle characterized by contraction

spot exchange rate the price to immediately buy one currency in terms of another currency

supply the quantity of a good or service that firms are willing to sell in the market during a given time period, *ceteris paribus*

transaction exposure the risk that the value of a business's expected receipts or expenses will change as a result of a change in currency exchange rates

translation exposure the risk that a change in exchange rates will impact the financial statements of a company

unemployed members of the labor force who are not currently working but are actively seeking a job



CFA Institute

This chapter supports some of the Learning Outcome Statements (LOS) in this [CFA® Level I Study Session \(https://openstax.org/r/cfa-study-session-4-economics\)](https://openstax.org/r/cfa-study-session-4-economics). Reference with permission of CFA Institute.



Multiple Choice

1. Demand is the _____.
 - a. amount of a good or service that consumers need
 - b. amount of a good or service that consumers want to purchase at the equilibrium price
 - c. quantity of a good or service that consumers want to purchase minus the amount that producers are currently supplying
 - d. quantity of a good or service that consumers are willing to purchase at various prices over a given time period, *ceteris paribus*
2. An increase in supply will lead to _____.
 - a. a lower equilibrium price and a lower equilibrium quantity
 - b. a higher equilibrium price and a lower equilibrium quantity
 - c. a higher equilibrium price and a higher equilibrium quantity
 - d. a lower equilibrium price and a higher equilibrium quantity
3. An increase in the price of a basket of goods is known as _____.
 - a. inflation
 - b. falling GDP
 - c. rising GDP

- d. a change in quantity demanded
4. The unemployment rate equals _____.
a. the number of people unemployed divided by the labor force
b. the number of people seeking work divided by the adult population
c. the number of people unemployed divided by the number of people employed
d. the number of people unemployed plus the number of people working part-time divided by the adult population
5. To be considered unemployed, an individual must be _____.
a. out of work
b. actively seeking a job
c. able to work
d. All of the above
6. When measuring GDP, purchases are divided into the four broad categories of _____.
a. interest rates, inflation, unemployment, and investment
b. demand, inflation, interest rates, and government spending
c. imports, exports, loanable funds, and government spending
d. consumer spending, investment, government spending, and net exports
7. A business cycle is measured from _____.
a. one peak to the next trough
b. one trough to the next trough
c. one trough to the next peak
d. the time of the highest unemployment rate to the time of the lowest unemployment rate
8. Which of the following economic environments would most likely be associated with a recession?
a. Unemployment falling to 30-year low
b. GDP growing at an annual rate of 4.2%
c. Unemployment increasing from 5% to 9% during the year
d. New businesses opening in record numbers while new housing starts reach a 10-year high
9. The interest rate is the _____.
a. rental cost of money
b. increase in the price of a market basket of goods
c. measure of economic activity
d. speed at which the money supply is increasing
10. If the nominal interest rate is 9% and the rate of inflation is 2%, the real rate of interest is approximately _____.
a. $\frac{2}{9}\%$
b. 7%
c. 11%
d. 18%
11. A foreign exchange rate is _____.
a. the rate of inflation in a foreign country
b. the amount of imports a country has relative to its exports
c. the price of one currency in terms of another currency

- d. the rate of unemployment in one country compared to the rate in another country
12. In 2020, the CPI in a country is 120. In 2021, the CPI in the same country is 126. This would mean that inflation is _____.
- 5%
 - 6%
 - 26%
 - 46%



Review Questions

1. Explain the difference between an increase in demand and an increase in quantity demanded.
2. You see that price of laptop computers is falling at the same time that the quantity of laptop computers is rising. Would you attribute this to a change in supply or a change in demand for computers? Explain.
3. Outline the stages of the business cycle.
4. Use a graph of the supply of loanable funds and demand for loanable funds to explain what will happen if the Federal Reserve increases the money supply. What would you expect to happen to the interest rate when this occurs?
5. Your bank offers you a car loan with an interest rate of 6%. You expect inflation to be 2%. What is the real interest rate on this loan?
6. You see that First National Bank is willing to make you a four-year \$40,000 car loan with an interest rate of 4.2%. However, the same bank will charge you 16.5% interest on a credit card that it issues. Why is the interest rate on the credit card so much higher than the interest rate on the car loan?
7. Last year, the exchange rate between the Korean won and the US dollar was KWN 1020 = USD 1. This year, the exchange rate is KWN 1150 = USD 1. Has the Korean won appreciated or depreciated over the past year? Explain.



Problems

1. You are planning a budget for an upcoming trip to Japan. Your hotel will cost 12,000 yen per night, and you will stay in the hotel for six nights. If the current exchange rate is YEN 98 = USD 1, how much will the hotel stay cost you in US dollars?
2. Visit the Federal Reserve Bank of St. Louis's FRED site. Create a graph that plots the unemployment rate for men and the unemployment rate for women since 1948. Turn recession shading on so that you can tell when recession occurred. Are there any patterns that you detect when looking at your chart?



Video Activity

Should Investors Prepare for Inflation or Hyperinflation?

[Click to view content \(https://openstax.org/r/inflation-or-hyperinflation\)](https://openstax.org/r/inflation-or-hyperinflation)

Inflation is a general increase in the price level. If inflation is extremely high, hyperinflation is said to exist. The video *Should Investors Prepare for Inflation or Hyperinflation?* by TD Ameritrade explores economic conditions that existed in the spring of 2021 that led some economists to be concerned that investors should be preparing for high inflation.

1. How might the Federal Reserve's response to the economic conditions that existed in the wake of the COVID-19 pandemic lead to a higher rate of inflation than the United States has experienced over the

previous few decades?

2. What suggestions does the video provide to investors who want to protect the value of their savings from being negatively impacted by inflation? Do some research to find out how much inflation has occurred since the spring of 2021 and how an investor who followed the advice would have fared.

Unemployment Explained

[Click to view content \(https://openstax.org/r/unemployment-explained\)](https://openstax.org/r/unemployment-explained)

The video *Unemployment Explained*, produced by the International Monetary Fund (IMF), explores how economists think about unemployment. The basic economic tools of supply and demand are used to explain what causes unemployment. In this video, you will also learn about how economists categorize different types of unemployment and why a zero rate of unemployment is not likely or even desirable.

3. List the types of unemployment presented in the video, giving an example of each type.
4. Choose three countries and find out what unemployment has been in each of those countries over the past decade. Do you detect any patterns in the unemployment rates in those countries over time or across the three different countries? Tell which type of unemployment you think has been most prevalent in each country, and explain your reasoning.

Investment in The Kraft Heinz Company (Fair Value: March 31, 2017 - \$29,553; December 31, 2016 - \$28,000)	March 31, 2017	December 31, 2016
Other	12,491	12,491
Receivables	10,280	10,280
Inventories	10,280	10,280
Property, plant and equipment	10,280	10,280
Goodwill	10,280	10,280
Other intangible assets	10,280	10,280
Deferred charges reinsurance assumed	10,280	10,280
Other	10,280	10,280
Railroad, Utilities and Energy:		
Cash and cash equivalents	1,000	1,000
Property, plant and equipment	10,280	10,280
Goodwill	10,280	10,280
Regulatory assets	10,280	10,280
Other	10,280	10,280
Finance and Financial Products:		
Cash and cash equivalents and U.S. Treasury Bills:		
Cash and cash equivalents	1,000	1,000
U.S. Treasury Bills	1,000	1,000
Total cash, cash equivalents and U.S. Treasury	2,000	2,000
Investments in equity and fixed maturity securities	10,280	10,280
Other investments	10,280	10,280

4

Accrual Accounting Process

Figure 4.1 Accrual accounting reports revenue when goods are delivered or services are performed but are not necessarily paid for. (credit: modification of work "Reviewing Financial Statements" by Instructional Institutions)

Chapter Outline

- 4.1 Cash versus Accrual Accounting
- 4.2 Economic Basis for Accrual Accounting
- 4.3 How Does a Company Recognize a Sale and an Expense?
- 4.4 When Should a Company Capitalize or Expense an Item?
- 4.5 What Is "Profit" versus "Loss" for the Company?



Why It Matters

Emma, an accounting major, was on her way to her Wednesday morning accounting class with her friend Sam. Sam, a finance major, kept complaining to Emma that he didn't enjoy their accounting class at all. "Why do we have to spend so much time worried about debits, credits, accounts, and all this other accounting jargon anyway? I'm a finance major. I won't be spending my days recording entries and counting pennies. I'm dreading class today. I looked at the syllabus, we're studying the accrual method. So what? Just give me the financial statements so I have the data I need."

Emma smiled at her friend and thought for a moment before responding. Emma loves the details of debits, credits, and "all that other accounting jargon," as Sam put it. She knows that while not everyone enjoys the details or understands them, they do play a big role in the timing and accuracy of financial statements. "Yeah, yeah, Sam, accounting isn't always easy or fun, but it's important. The financial statements you will use in finance are a part of the foundation for many decisions. They are an important tool. Even though you don't have to create financial statements in a finance role, it's still necessary to understand the accounting principles they are based on. We haven't covered the details yet, but I do know that income or loss in a period can be significantly different if the income statement is prepared under cash versus accrual methods. I would think that would be important to know if you are the one using the income statement."

4.1 Cash versus Accrual Accounting

Learning Outcomes

By the end of this section, you will be able to:

- Outline the key concepts of the cash-basis accounting method.
- Explain the key characteristics of the accrual-basis accounting method.
- Identify businesses for which cash accounting or accrual accounting is more appropriate.

In this section, we will explore the basic elements of cash and accrual accounting and the businesses that are most likely to use each one. Some private companies may choose to use **cash-basis accounting** rather than **accrual-basis accounting** to report financial information.

Cash-Basis Accounting

In business, cash is certainly important. In fact, it's so important that it dictates one of two ways we can account for our business transactions. The *cash method* is just as the name implies—it records transactions only when cash flows. We track cash inflows and outflows as they occur. This method is most commonly used by small businesses that deal primarily in cash transactions. The other method, called the *accrual method*, records transactions when they occur, rather than waiting for cash to be accumulated. Using the accrual method, we match cash inflows and the outflows required to generate them. We call this the *matching principle*. This method is used by most publicly traded companies. In this chapter, you'll explore both methods, see how each impacts financial statements differently, note the role of timing in each method, and learn how and when to record capital and expense transactions.

Let's look at an example. Chris just finished the first month of her landscaping business operations at the end of August, and she used the cash method of accounting to figure out her **net income**. Most small start-up companies use the cash method of accounting because it is easy to understand, requires no special training, and helps them focus on one big key to their survival—cash. This means that she simply recorded the cash that came in and the cash that went out of her business. She brought in \$1,400 in revenue in her first month, which she felt was substantial given that it was her first month. But after deducting her expenses, she had only \$250 left, so she worried about the future of her business. Would she have to increase her sales exponentially in order to start bringing in a decent profit each month?

As you move through the chapter, you'll get to see the impact of the two methods of accounting and how these methods impact the insights and decisions Chris made for her new business.

Cash-basis accounting is a method of accounting in which transactions are not recorded in the financial statements until there is an exchange of cash. Cash-basis accounting sometimes impacts the timing of **revenue** and **expense** reporting until cash receipts or outlays occur. For example, as you saw above, Chris measured the performance of her landscaping business for the month of August using cash flows. Cash accounting is far simpler to track than accrual-basis accounting.

Accrual-Basis Accounting

Public companies reporting their financial positions use either US generally accepted accounting principles (GAAP) or International Financial Reporting Standards (IFRS), as allowed under the Securities and Exchange Commission (SEC) regulations. GAAP is a set of accounting standards created by the Financial Accounting Standards Board (FASB) and the Governmental Accounting Standards Board (GASB). It's key to note that though they are similar in many areas, there are still key areas that differ between GAAP and IFRS. Therefore, when using financial statements, it's important to be aware of the standards under which they were prepared. However, public or private companies using GAAP or IFRS must prepare their financial statements using the rules of accrual accounting. Accrual-basis accounting prescribes that revenues and expenses must be recorded in the accounting period in which they were earned or incurred, no matter when cash receipts or payments occur. It is because of accrual accounting that we have the revenue recognition principle and the

expense recognition principle (also known as the matching principle).

The accrual method is considered to better match revenues and expenses and standardizes reporting information for comparability purposes. Having comparable information is important to external users of information trying to make investment or lending decisions and to internal users trying to make decisions about company performance, budgeting, and growth strategies.

Who Uses Each Method?

Cash-basis accounting can be more efficient and well-suited for certain types of businesses, such as farming or professional services provided by lawyers and doctors. However, the accrual basis of accounting is theoretically preferable to the cash basis of accounting because it takes into account the timing of the transactions (when goods and services are provided and when the cash involved in the transactions is received). Cash can often be received a significant amount of time after the initial transaction. Considering this amount allows accountants to provide, in a timely manner, relevant and complete information to stakeholders.

There are several reasons accrual-basis accounting is preferred to cash-basis accounting. Accrual-basis accounting is required by GAAP because it typically provides a better sense of the financial well-being of a company. Accrual-based accounting information allows management to analyze a company's progress, and management can use that information to improve their business. Accrual accounting is also used to assist companies in securing financing because banks will typically require a company to provide accrual-basis financial income statements. The Internal Revenue Service requires businesses to report using accrual-basis information when preparing tax returns. In addition, companies with inventory must use accrual-based accounting for income tax purposes, though there are exceptions to the general rule.

So why might a company use cash-basis accounting? Companies that do not sell stock publicly can use cash-basis instead of accrual-basis accounting for internal management purposes or because they are exempt from such requirements in agreements such as a bank loan. Cash-basis accounting is a simpler accounting system to use than an accrual-basis accounting system when tracking real-time revenues and expenses.

THINK IT THROUGH

Cash- or Accrual-Basis Accounting?

You are a new accountant at a beauty salon. The salon had previously used cash-basis accounting to prepare its financial records but is now considering switching to an accrual-basis method. You have been tasked with determining if this transition is appropriate.

When you go through the records, you notice that this transition will greatly impact how the salon reports revenues and expenses. The salon will now report some revenues and expenses before it receives or pays cash.

How will this change positively impact its business reporting? How will it negatively impact its business reporting? If you were the accountant, would you recommend the salon transition from cash basis to accrual basis?

Solution:

Accrual accounting creates a more accurate picture of profit or loss, so the salon's owner can have a better understanding of its profitability from period to period. However, it can be more work to record under accrual accounting. If the salon is small and the profits and costs are easily understood, it might not be worth the extra effort to the owner to use accrual-basis accounting. If the salon is seeking ways to better understand profits and costs, accrual-basis accounting would be a great choice.

4.2 Economic Basis for Accrual Accounting

Learning Outcomes

By the end of this section, you will be able to:

- Assess the impact of business transactions on cash flow.
- Define double-entry accounting and explain how it supports the accounting equation.

How and when we record our transactions can have a significant impact on financial statements, especially the **income statement (net income)**. In this section you will explore the impact business transactions have on financial statements under each method. In doing so, you'll be introduced to **double-entry accounting** and see how it functions to support the accounting equation.

Timing of Business Activity versus Cash Flow

The first financial statement prepared is the income statement, a statement that shows the organization's financial performance *for a given period of time*. We already saw that Chris, who is a sole proprietor, started a summer landscaping business on August 1, 2020. She categorized her business as a service entity and used the cash-basis method of accounting to record the initial operations for her business. Although Chris was using her family's tractor to get her work done, she was responsible for paying for fuel and any maintenance costs. So, on August 31, Chris realized she had only \$250 in her checking account.

This balance was lower than expected because she had spent only slightly less (\$1,150 for brakes, fuel, and insurance) than she earned (\$1,400)—leaving a net income of \$250. While she would like the checking balance to grow each month, she realized that most of the August expenses were infrequent (brakes and insurance) and the insurance, in particular, was an unusually large expense. She knew that the checking account balance would likely grow more in September because she would earn money from some new customers; she also anticipated having fewer expenses.

This simple landscaping example can be used to discuss the elements of the income statement, which are revenues, expenses, **gains**, and **losses** for a particular period of time (see [Figure 4.2](#)). Together, these determine whether the organization has net income (where revenues and gains are greater than expenses and losses) or net loss (where expenses and losses are greater than revenues and gains). Revenues, expenses, gains, and losses are further defined in the Income Statement provided.

Chris's Landscaping Income Statement For the Month Ended August 31		
Revenue	\$1,400	
Total revenue		\$1,400
Expenses		
Tractor brake repair	100	
Tractor fuel	50	
Business insurance	<u>1,000</u>	
Total expenses		<u>1,150</u>
Net income		<u>\$ 250</u>

Figure 4.2 Income Statement for Chris's Landscaping

The income statement can also be visualized by the formula:

$$\text{Revenue} - \text{Expenses} = \text{Net Income or Net Loss}$$

Let's change this example slightly and assume the \$1,000 payment to the insurance company will be paid in

September rather than in August. In this case, the ending balance in Chris’s checking account would be \$1,250, a result of earning \$1,400 and only spending \$100 for the brakes on the tractor and \$50 for fuel. This stream of cash flows is an example of cash-basis accounting because it reflects when payments are received and made, not necessarily the time period that they affect. At the end of this section, you will address accrual accounting, which does reflect the time period that payments affect.

The Accounting Equation

It may be helpful to think of the **accounting equation** from a “sources and claims” perspective. Under this approach, the assets (items owned by the organization) were obtained by incurring liabilities or were provided by owners. Stated differently, every asset has a claim against it—by creditors and/or owners.

$$\text{Assets} = \text{Liabilities} + \text{Owner's Equity}$$

You may recall from mathematics courses that an equation must always be in balance. Therefore, we must ensure that the two sides of the accounting equation are always equal. We will explore the components of the accounting equation in more detail shortly. First, we need to examine several underlying concepts that form the foundation for the accounting equation: the double-entry accounting system, debits and credits, and the “normal” balance for each account that is part of a formal accounting system.

THINK IT THROUGH

The Accounting Equation

On a sheet of paper, use three columns to create your own accounting equation. In the first column, list all the things you own (assets). List only the asset itself; don’t worry about any associated liabilities (expenses) in that column. In the second column, list any amounts owed (the liabilities). When you are done, total up all the assets. Then total up all the liabilities.

Now, use the accounting equation to calculate the net amount of the asset (equity). To do so, subtract the total assets from the total liabilities. This figure makes the accounting equation balance and represents equity, or an estimate of your net worth.

Here is something else to consider: Is it possible to have negative equity? It sure is . . . ask any college student who has taken out loans. At first glance there is no asset directly associated with the amount of the loan. But is that, in fact, the case? You might ask yourself why you should make an investment in a college education—what is the benefit (asset) to going to college? The answer lies in the difference in lifetime earnings with a college degree versus without a college degree. This is influenced by many things, including the supply and demand of jobs and employees. It is also influenced by the earnings for the type of college degree pursued.

Solution:

Answers will vary but may include vehicles, clothing, electronics (include cell phones and computer/gaming systems), and sports equipment. They may also include money owed on these assets, most likely vehicles and perhaps cell phones. In the case of a student loan, there may be a liability with no corresponding asset (yet). Responses should be able to evaluate the benefit of investing in college and the wage differential between earnings with and without a college degree.

Let’s continue our exploration of the accounting equation, focusing on the equity component in particular. Recall that we defined equity as the net worth of an organization. It is helpful to also think of net worth as the accounting *value* of the organization. Recall, too, that revenues (inflows as a result of providing goods and services) *increase* the accounting value of the organization. So every dollar of revenue an organization generates increases the overall value of the organization.

Likewise, expenses (outflows as a result of generating revenue) *decrease* the value of the organization. So each dollar of expenses an organization incurs decreases the overall value of the organization. The same approach can be taken with the other elements of the financial statements:

- Gains *increase* the value (equity) of the organization.
- Losses *decrease* the value (equity) of the organization.
- Investments by owners *increase* the value (equity) of the organization.
- Distributions to owners *decrease* the value (equity) of the organization.
- Changes in assets and liabilities can *either* increase or decrease the value (equity) of the organization depending on the net result of the transaction.

Let's look at Chris's Landscaping business again and do the same quick exercise you did with your personal finances. If we were to total all of Chris's assets, we would find just one: \$250 in cash. She's using the family's tractor, but she doesn't own the tractor, so it is not her asset. Her liabilities are currently \$0, as she paid cash for all the expenses she incurred already. If we total her assets, we get \$250. Liabilities total \$0. Using the accounting equation, we find her equity to currently be \$250, or

$$\text{Assets } (\$250) - \text{Liabilities } (\$0) = \text{Equity } (\$250)$$

Double-Entry Accounting

Accounting is based on a double-entry accounting system, which requires the following:

- Each time we record a transaction, we must record a change in at least two different accounts. Having two or more accounts change will allow us to keep the accounting equation in balance.
- Not only will at least two accounts change, but there must also be at least one **debit** and one **credit** side impacted.
- The sum of the debits must equal the sum of the credits for each transaction.

In order for companies to record the myriad of transactions they have each year, there is a need for a simple but detailed system. Journals are useful tools to meet this need.

Debits and Credits

Each account can be represented visually by splitting the account into left and right sides as shown. The graphic representation of a general ledger account is known as a **T-account**. It is called this because it looks like a "T," as you can see with the T-account shown in [Figure 4.3](#).

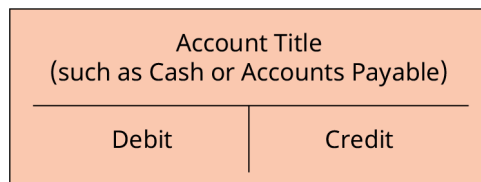


Figure 4.3 T-Account

A debit records financial information on the left side of each account. A credit records financial information on the right side of an account. One side of each account will increase, and the other side will decrease. The ending account balance is found by calculating the difference between debits and credits for each account. You will often see the terms *debit* and *credit* represented in shorthand, written as *DR* or *dr* and *CR* or *cr*, respectively. Depending on the account type, the sides that increase and decrease may vary. We can illustrate each account type and its corresponding debit and credit effects in the form of an *expanded* equation (see [Figure 4.4](#)).

Assets		=		Liabilities		+		Equity							
								Common Stock		- Dividends		+ Revenues		- Expenses	
Debit	Credit			Debit	Credit			Debit	Credit	Debit	Credit	Debit	Credit	Debit	Credit
Increase	Decrease			Decrease	Increase			Decrease	Increase	Increase	Decrease	Decrease	Increase	Increase	Decrease

Figure 4.4 Expanded Accounting Equation

As we can see from this expanded accounting equation, Assets accounts increase on the debit side and decrease on the credit side. This is also true of Dividends and Expenses accounts. Liabilities increase on the credit side and decrease on the debit side. This is also true of Common Stock and Revenues accounts. This becomes easier to understand as you become familiar with the *normal balance* of an account.

The balance sheet is a reflection of the accounting equation (see [Figure 4.5](#)). It has two sections, assets in one section and liabilities and equity in the other section. It's key to note that both assets and liabilities are broken down on the balance sheet into current and noncurrent classifications in order to provide more detail and transparency. Current assets are those that are consumed within a year. Assets that will be in use longer than a year are considered noncurrent. Current liabilities are those that will be due within a year. Noncurrent liabilities are those that are due more than a year into the future.

Assets				=	Liabilities				+	Owner's Equity			
Current		Noncurrent			Current		Noncurrent						
+	-	+	-		-	+	-	+		-		+	
										Distribution to Owners		Investments by Owners	
										Expenses		Revenues	
										Losses		Gains	
										Comprehensive Income		Comprehensive Income	

Figure 4.5 Graphical Representation of the Accounting Equation Both assets and liabilities are categorized as current and noncurrent. Also highlighted are the various activities that affect the equity (or net worth) of the business.

Notice each account subcategory (Current Assets and Noncurrent Assets, for example) has an “increase” side and a “decrease” side. As you may recall, these are called T-accounts, and they are used to analyze transactions.

The basic components of even the simplest accounting system are *accounts* and a *general ledger*. An account is a record showing increases and decreases to assets, liabilities, and equity—the basic components found in the accounting equation. Each of these categories, in turn, includes many individual accounts, all of which a company maintains in its general ledger. A general ledger is a comprehensive listing of all of a company's accounts with their individual balances.

4.3 How Does a Company Recognize a Sale and an Expense?

Learning Outcomes

By the end of this section, you will be able to:

- Explain the revenue recognition principle and how it relates to current and future sales transactions.
- Explain the expense recognition principle and how it relates to current and future purchase transactions.
- Assess the role of ethics in revenue and expense recognition.

You've learned the basics of each method as well as the accounting equation and double-entry accounting. Next, let's turn our attention to when we record transactions, as timing is key.

Revenue Recognition

Revenue is the value of goods and services the organization sold or provided to customers for a given period of time. In our current example, Chris's landscaping business, the “revenue,” or the value of services

performed, for the month of August would be \$1,400. It is the value Chris received in exchange for the services provided to her clients. Likewise, when a business provides goods or services to customers for cash at the time of the service or in the future, the business classifies the amount(s) as revenue. Just as the \$1,400 revenues from a business made Chris's checking account balance increase, revenues increase the value of a business. In accounting, **revenue recognition** involves recording sales or fees earned within the period earned. Just as earning wages from a business or summer job reflects the number of hours worked for a given rate of pay or payments from clients for services rendered, revenues (and the other terms) are used to indicate the dollar value of goods and services provided to customers for a given period of time.

THINK IT THROUGH

Coffee Shop Products

Think about a coffee shop in your area. Identify items the coffee shop sells that would be classified as revenues. Remember, revenues for the coffee shop are related to its primary purpose: selling coffee and related items. Or, better yet, make a trip to the local coffee shop and get a first-hand experience.

Solution:

Many coffee shops earn revenue through multiple revenue streams, including coffee and other specialty drinks, food items, gift cards, and merchandise.

Short-Term Revenue Recognition Examples

Two brief examples may help illustrate the difference between cash accounting and accrual accounting. Assume that a business sells \$200 worth of merchandise. In some businesses, there are two ways the customers pay: cash and credit (also referred to as "on account"). Cash sales include checks and credit cards and are paid at the time of the sale. Credit sales (not to be confused with credit card sales) allow the customer to take the merchandise but pay within a specified period of time, usually up to 45 days.

A cash sale would be recorded in the financial statements under both the cash basis and accrual basis of accounting. It makes sense because the customer received the merchandise and paid the business at the same time. It is considered two events that occur simultaneously (exchange of merchandise for cash).

A credit sale, however, would be treated differently under each of these types of accounting. Under the cash basis of accounting, a credit sale would not be recorded in the financial statements until the cash is received, under terms stipulated by the seller. For example, assume that in the next year of Chris's landscaping business, on April 1, she provides \$500 worth of services to one of her customers. The sale is made on account, with the payment due 45 days later. Under the cash basis of accounting, the revenue would not be recorded until May 16, when the cash was received. Under the accrual basis of accounting, this sale would be recorded in the financial statements at the time the services were provided, April 1. The reason the sale would be recorded is that, under accrual accounting, the business reports that it provided \$500 worth of services to its customer. The fact that the customers will pay later is viewed as a separate transaction under accrual accounting (see [Figure 4.6](#)).

Cash ←	Statement of Cash Flows For Month Ended <u>May 31</u>	Statement of Cash Flows For Month Ended <u>April 30</u>	→ Accrual
	Cash Flow from Operations Net Earnings \$500	Cash Flow from Operations Net Earnings \$500	

Figure 4.6 Credit versus Cash On the left is a credit sale recorded under the cash basis of accounting. On the right, the same credit sale is recorded under the accrual basis of accounting.

Let's now explore the difference between the cash basis and accrual basis of accounting using an expense.

Assume a business purchases \$160 worth of printing supplies from a supplier (vendor). Similar to a sale, a purchase of merchandise can be paid for at the time of sale using cash (also a check or credit card) or at a later date (on account). A purchase paid with cash at the time of the sale would be recorded in the financial statements under both cash basis and accrual basis of accounting. It makes sense because the business received the printing supplies from the supplier and paid the supplier at the same time. It is considered two events that occur simultaneously (exchange of merchandise for cash).

If the purchase was made on account (also called a credit purchase), however, the transaction would be recorded differently under each of these types of accounting. Under the cash basis of accounting, the \$160 purchase on account would not be recorded in the financial statements until the cash is paid, as stipulated by the seller's terms. For example, if the printing supplies were received on July 17 and the payment terms were 15 days, no transaction would be recorded until August 1, when the goods were paid for. Under the accrual basis of accounting, this purchase would be recorded in the financial statements at the time the business received the printing supplies from the supplier (July 17). The reason the purchase would be recorded is that the business reports that it bought \$160 worth of printing supplies from its vendors. The fact that the business will pay later is viewed as a separate issue under accrual accounting. [Table 4.1](#) summarizes these examples under the different bases of accounting.

Transaction	Under Cash-Basis Accounting	Under Accrual-Basis Accounting
\$200 sale for cash	Recorded in financial statements at time of sale	Recorded in financial statements at time of sale
\$200 sale on account	<i>Not</i> recorded in financial statements until cash is received	Recorded in financial statements at time of sale
\$160 purchase for cash	Recorded in financial statements at time of purchase	Recorded in financial statements at time of purchase
\$160 purchase on account	<i>Not</i> recorded in financial statements until cash is paid	Recorded in financial statements at time of purchase

Table 4.1 How Transactions Are Viewed under Cash and Accrual Accounting

Businesses often sell items for cash as well as on account, where payment terms are extended for a period of time (for example, 30 to 45 days). Likewise, businesses often purchase items from suppliers (also called vendors) for cash or, more likely, on account. Under the cash basis of accounting, these transactions would not be recorded until the cash is exchanged. In contrast, under accrual accounting, the transactions are recorded when the transaction occurs, regardless of when the cash is received or paid.

CONCEPTS IN PRACTICE

Ethics in Revenue Recognition

Because each industry typically has a different method for recognizing income, revenue recognition is one of the most difficult tasks for accountants, as it involves a number of ethical dilemmas related to income reporting. To provide an industry-wide approach, Accounting Standards Update No. 2014-09 and other related updates were implemented to clarify revenue recognition rules. The American Institute of Certified Public Accountants (AICPA) announced that these updates would replace US GAAP's current industry-specific revenue recognition practices with a principle-based approach, potentially affecting both day-to-day business accounting and the execution of business contracts with customers.¹ The AICPA and the International Federation of Accountants (IFAC) require professional accountants to act with due care and to remain abreast of new accounting rules and methods of accounting for different transactions, including revenue recognition.

The IFAC emphasizes the role of professional accountants working within a business in ensuring the quality of financial reporting: "Management is responsible for the financial information produced by the company. As such, professional accountants in businesses therefore have the task of defending the quality of financial reporting right at the source where the numbers and figures are produced!"² In accordance with proper revenue recognition, accountants do not recognize revenue before it is earned.

CONCEPTS IN PRACTICE

Gift Card Revenue Recognition

Gift cards have become an essential part of revenue generation and growth for many businesses. Although they are practical for consumers and are a low cost for businesses, navigating revenue recognition guidelines can be difficult. Gift cards with expiration dates require that revenue recognition be delayed until customer use or expiration. However, most gift cards now have no expiration date. So when do you recognize revenue?

Companies may need to provide an estimation of projected (or deferred) gift card revenue and usage during a period based on past experience or industry standards. There are a few rules governing reporting. If a company determines that a portion of all the issued gift cards will never be used, it may write this off to income. In some states, if a gift card remains unused, in part or in full, the unused portion of the card is transferred to the state government. It is considered unclaimed property for the customer, meaning that the company cannot keep these funds as revenue because, in this case, they have reverted to the state government.

Expense Recognition

An expense is a cost associated with providing goods or services to customers. In our opening example, the expenses that Chris incurred totaled \$1,150 (consisting of \$100 for brakes, \$50 for fuel, and \$1,000 for insurance). You might think of expenses as the opposite of revenue, in that expenses reduce Chris's checking account balance. Likewise, expenses decrease the value of the business and represent the dollar value of costs incurred to provide goods and services to customers for a given period of time.

¹ American Institute of Certified Public Accountants (AICPA). "Revenue Recognition." n.d. <https://www.aicpa.org/interestareas/frc/accountingfinancialreporting/revenuerecognition.html>

² Len Jui and Jessie Wong. "Roles and Importance of Professional Accountants in Business." *China Accounting Journal*. October 21, 2013. <https://www.ifac.org/news-events/2013-10/roles-and-importance-professional-accountants-business>

THINK IT THROUGH

Coffee Shop Expenses

While thinking about or visiting a coffee shop in your area, look around (or visualize) and identify items or activities that are the expenses of the coffee shop. Remember, expenses for the coffee shop are related to resources consumed while generating revenue from selling coffee and related items. Do not forget about any expenses that might not be so obvious—as a general rule, every activity in a business has an associated cost.

Solution:

Costs of the coffee shop that might be readily observed would include rent, wages for the employees, and the cost of the coffee, pastries, and other items/merchandise that may be sold. In addition, costs such as utilities, equipment, and cleaning or other supplies might also be readily observable. More obscure costs of the coffee shop would include insurance, regulatory costs such as health department licensing, point-of-sale/credit card costs, advertising, donations, and payroll costs such as workers' compensation, unemployment, and so on. There are also unseen costs, such as aging of the building (if owned by the coffee shop) and wear and tear or aging of the equipment.

4.4 When Should a Company Capitalize or Expense an Item?

Learning Outcomes

By the end of this section, you will be able to:

- Define the key characteristics of a fixed asset.
- Explain how the cost of a fixed asset is spread throughout its useful life via depreciation.
- Assess the impact to net income of expensing versus capitalizing an item.

Assets are items a business owns. For accounting purposes, assets are categorized as current versus long term and tangible versus intangible. Any asset that is expected to be used by the business for more than one year is considered a **long-term asset**. These assets are not intended for resale and are anticipated to help generate revenue for the business in the future. Some common long-term assets are computers and other office machines, buildings, vehicles, software, computer code, and copyrights. Although these are all considered long-term assets, some are tangible and some are intangible.

To better understand the nature of fixed assets, let's get to know Liam and their new business. Liam is excited to be graduating from their MBA program and looks forward to having more time to pursue their business venture. During one of their courses, Liam came up with the business idea of creating trendy workout attire. For their class project, they started silk-screening vintage album cover designs onto tanks, tees, and yoga pants. They tested the market by selling their wares on campus and were surprised how quickly and how often they sold out. In fact, sales were high enough that they decided to go into business for themselves. One of their first decisions involved whether they should continue to pay someone else to silk-screen their designs or do their own silk-screening. To do their own silk-screening, they would need to invest in a silk screen machine.

Liam will need to analyze the purchase of a silk screen machine to determine the impact on their business in the short term as well as the long term, including the accounting implications related to the expense of this machine. Liam knows that over time, the value of the machine will decrease, but they also know that an asset is supposed to be recorded on the books at its historical cost. They also wonder what costs are considered part of this asset. Additionally, Liam has learned about the matching principle (**expense recognition**) but needs to learn how that relates to a machine that is purchased in one year and used for many years to help generate revenue. Liam has a lot of information to consider before making this decision.

What Is a Fixed Asset?

An asset is considered a **tangible asset** when it is an economic resource that has physical substance—it can be seen and touched. Tangible assets can be either short term, such as inventory and supplies, or long term, such as land, buildings, and equipment. To be considered a long-term tangible asset, the item needs to be used in the normal operation of the business for more than one year, be of material value, and not be near the end of its useful life, and the company must have no plan to sell the item in the near future. The useful life is the time period over which an asset cost is allocated. Long-term tangible assets are known as fixed assets. It's also key to note that companies will **capitalize** a fixed asset if they have material value. A \$10 stapler to be used in the office, for example, may last for years, but the value of the item is not significant enough to warrant capitalizing it.

Businesses typically need many different types of these assets to meet their objectives. These assets differ from the company's products. For example, the computers that Apple, Inc. intends to sell are considered inventory (a short-term asset), whereas the computers Apple's employees use for day-to-day operations are long-term assets. In Liam's case, the new silk screen machine would be considered a long-term tangible asset as they plan to use it over many years to help generate revenue for their business. Long-term tangible assets are listed as noncurrent assets on a company's balance sheet. Typically, these assets are listed under the category of Property, Plant, and Equipment (PP&E), but they may be referred to as fixed assets or plant assets.

Apple, Inc. lists a total of \$36.766 million in total Property, Plant, and Equipment (net) on its September 2020 consolidated balance sheet (see [Figure 4.7](#)). As shown in the figure, this net total includes land and buildings, machinery, equipment and internal-use software, and leasehold improvements, resulting in a gross PP&E of \$103.526 million—less accumulated depreciation and amortization of \$66.760 million—to arrive at the net amount of \$36.766 million.

	2020	2019
Land and Buildings	\$ 17,952	\$ 17,085
Machinery, Equipment, and Internal-Use Software	75,291	69,797
Leasehold Improvements	10,283	9,075
Gross Property, Plant, and Equipment	103,526	95,957
Accumulated Depreciation and Amortization	(66,760)	(58,579)
Total Property, Plant, and Equipment, Net	\$ 36,766	\$ 37,378

Figure 4.7 Apple's Property, Plant, and Equipment, Net (September 2020, in \$ million)³ This report shows the company's consolidated financial statement details as of September 30, 2020, and September 30, 2019.

THINK IT THROUGH

Classifying Assets and Related Expenditures

You work at a business consulting firm. Your new colleague, Milan, is helping a client company organize its accounting records by types of assets and expenditures. Milan is a bit stumped on how to classify certain assets and related expenditures, such as capitalized costs versus expenses. They have given you the following list and asked for your help to sort through it. Help your colleague classify the expenditures as either capitalized or expensed, and note which assets are property, plant, and equipment.

Expenditures:

- Normal repair and maintenance on the manufacturing facility
- Cost of taxes on new equipment used in business operations
- Shipping costs on new equipment used in business operations
- Cost of a minor repair on existing equipment used in business operations

Assets:

- Land next to the production facility held for use next year as a place to build a warehouse
- Land held for future resale when the value increases
- Equipment used in the production process

Solution:**Expenditures:**

- Normal repair and maintenance on the manufacturing facility: expensed
- Cost of taxes on new equipment used in business operations: capitalized
- Shipping costs on new equipment used in business operations: capitalized
- Cost of a minor repair on existing equipment used in business operations: expensed

Assets:

- Land next to the production facility held for use next year as a place to build a warehouse: property, plant, and equipment
- Land held for future resale when the value increases: investment
- Equipment used in the production process: property, plant, and equipment

Why are the costs of putting a long-term asset into service capitalized and written off as expenses (depreciated) over the economic life of the asset? Let's return to Liam's start-up business as an example. Liam plans to buy a silk screen machine to help create clothing that they will sell. The machine is a long-term asset because it will be used in the business's daily operation for many years. If the machine costs Liam \$5,000 and it is expected to be used in their business for several years, GAAP require the allocation of the machine's costs over its useful life, which is the period over which it will produce revenues. Overall, in determining a company's financial performance, we would not expect that Liam should have an expense of \$5,000 this year and \$0 in expenses for this machine for future years in which it is being used. GAAP addressed this through the expense recognition (matching) principle, which states that expenses should be recorded in the same period with the revenues that the expense helped create. In Liam's case, the \$5,000 for this machine should be allocated over the years in which it helps to generate revenue for the business. Capitalizing the machine allows this to occur. As stated previously, to capitalize is to record a long-term asset on the balance sheet and expense its allocated costs on the income statement over the asset's economic life. Therefore, when Liam purchases the machine, they will record it as an asset on the financial statements (see journal entry in [Figure 4.8](#)).

Journal			
Date	Account	Debit	Credit
Jan. 1	Machine Cash	5,000	5,000

Figure 4.8 Journal Entry for Machine/Cash

When capitalizing an asset, the total cost of acquiring the asset is included in the cost of the asset. This includes additional costs beyond the purchase price, such as shipping costs, taxes, assembly, and legal fees. For example, if a real estate broker is paid \$8,000 as part of a transaction to purchase land for \$100,000, the land would be recorded at a cost of \$108,000.

Over time, as the asset is used to generate revenue, Liam will need to depreciate recognize the cost of the asset.

³ In the Chapter 4 financial statements, a number contained within parentheses is a negative number, such as the "Accumulated depreciation and amortization" line item.

What Is Depreciation?

When a business purchases a long-term asset (used for more than one year), it classifies the asset based on whether the asset is used in the business's operations. If a long-term asset is used in the business's operations, it will belong in property, plant, and equipment or intangible assets. In this situation, the asset is typically capitalized. Capitalization is the process by which a long-term asset is recorded on the balance sheet and its allocated costs are expensed on the income statement over the asset's economic life.

Long-term assets that are not used in daily operations are typically classified as an investment. For example, if a business owns land on which it operates a store, warehouse, factory, or offices, the cost of that land would be included in property, plant, and equipment. However, if a business owns a vacant piece of land on which the business conducts no operations (and assuming no current or intermediate-term plans for development), the land would be considered an investment.

Depreciation is the process of allocating the cost of a tangible asset over its useful life, or the period of time that the business believes it will use the asset to help generate revenue.

Fundamentals of Depreciation

As you have learned, when accounting for a long-term fixed asset, we cannot simply record an expense for the cost of the asset and record the entire outflow of cash in one accounting period. Like all other assets, when you purchase or acquire a long-term asset, it must be recorded at the historical (initial) cost, which includes all costs to acquire the asset and put it into use. The initial recording of an asset has two steps:

1. Record the initial purchase on the date of purchase, which places the asset on the balance sheet (as property, plant, and equipment) at cost, and record the amount as notes payable, accounts payable, or an outflow of cash.
2. At the end of the period, make an adjusting entry to recognize the depreciation expense. Depreciation expense is the amount of the asset's cost to be recognized, or expensed, in the current period. Companies may record depreciation expense incurred annually, quarterly, or monthly.

Following GAAP and the expense recognition principle, the depreciation expense is recognized over the asset's estimated useful life.

Recording the Initial Asset

Assets are recorded on the balance sheet at cost, meaning that all costs to purchase the asset and to prepare the asset for operation should be included. Costs outside of the purchase price may include shipping, taxes, installation, and modifications to the asset.

The journal entry to record the purchase of a fixed asset (assuming that a note payable, not a short-term account payable, is used for financing) is shown in [Figure 4.9](#).

Journal			
Date	Account	Debit	Credit
Jan. 1	Fixed Asset (truck, building, etc.) Cash/Notes Payable <i>To record purchase of fixed asset</i>	XXX	XXX

Figure 4.9 Journal Entry for Fixed Asset

Applying this to Liam's silk-screening business, we learn that they purchased their silk screen machine for \$54,000 by paying \$10,000 cash and the remainder in a note payable over five years. The journal entry to record the purchase is shown in [Figure 4.10](#).

Journal			
Date	Account	Debit	Credit
Jan. 1	Equipment Cash Notes Payable <i>To recognize purchase of silk-screening machine</i>	54,000	10,000 44,000

Figure 4.10 Journal Entry for Equipment/Cash

CONCEPTS IN PRACTICE

Estimating Useful Life and Salvage Value

Useful life and salvage value are estimates made at the time an asset is placed in service. It is common and expected that the estimates are inaccurate due to the uncertainty involved in estimating the future. Sometimes, however, a company may attempt to take advantage of estimating salvage value and useful life to improve earnings. A larger salvage value and longer useful life decrease annual depreciation expense and increase annual net income. An example of this behavior is Waste Management, which was disciplined by the SEC in March 2002 for fraudulently altering its estimates to reduce depreciation expense and overstate net income by \$1.7 billion.⁴

Components Used in Calculating Depreciation

The expense recognition principle that requires that the cost of the asset be allocated over the asset's useful life is the process of depreciation. For example, if we buy a delivery truck to use for the next five years, we would allocate the cost and record depreciation expense across the entire five-year period. The calculation of the depreciation expense for a period is not based on anticipated changes in the fair-market value of the asset; instead, the depreciation is based on the allocation of the cost of owning the asset over the period of its useful life.

The following items are important in determining and recording depreciation:

- Book value: the asset's original cost less accumulated depreciation.
- Useful life: the length of time the asset will be productively used within operations.
- Salvage (residual) value: the price the asset will sell for or be worth as a trade-in when its useful life expires. The determination of salvage value can be an inexact science since it requires anticipating what will occur in the future. Often, the salvage value is estimated based on past experiences with similar assets.
- Depreciable base (cost): the depreciation expense over the asset's useful life. For example, if we paid \$50,000 for an asset and anticipate a salvage value of \$10,000, the depreciable base is \$40,000. We expect \$40,000 in depreciation over the time period in which the asset was used, and then it would be sold for \$10,000.

Depreciation records an expense for the value of an asset consumed and removes that portion of the asset from the balance sheet. The journal entry to record depreciation is shown in [Figure 4.11](#).

⁴ United States Securities and Exchange Commission. "Waste Management, Inc. Founder and Five Other Former Top Officers Sued for Massive Earnings Management Fraud." March 26, 2002. <https://www.sec.gov/litigation/litreleases/lr17435.htm>

Journal			
Date	Account	Debit	Credit
Jan. 1	Depreciation Expense Accumulated Depreciation <i>To record depreciation on asset for period</i>	XXX	XXX

Figure 4.11 Journal Entry for Depreciation Expense

Depreciation expense is a common operating expense that appears on an income statement. It represents the amount of expense being recognized in the current period. Accumulated depreciation, on the other hand, represents the sum of all depreciation expense recognized to date, or the total of all prior depreciation expense for the asset. It is a contra account, meaning it is attached to another account and is used to offset the main account balance that records the total depreciation expense for a fixed asset over its life. In this case, the asset account stays recorded at the historical value but is offset on the balance sheet by accumulated depreciation. Accumulated depreciation is subtracted from the historical cost of the asset on the balance sheet to show the asset at book value. Book value is the amount of the asset that has not been allocated to expense through depreciation.

It is important to note, however, that not all long-term assets are depreciated. For example, land is not depreciated because depreciation is the allocating of the expense of an asset over its useful life. How can one determine a useful life for land? It is assumed that land has an unlimited useful life; therefore, it is not depreciated, and it remains on the books at historical cost.

Once it is determined that depreciation should be accounted for, there are three methods that are most commonly used to calculate the allocation of depreciation expense: the *straight-line method*, the *units-of-production method*, and the *double-declining-balance method*. A fourth method, the *sum-of-the-years-digits method*, is another *accelerated* option that has been losing popularity and can be learned in intermediate accounting courses. Note that these methods are for accounting and reporting purposes. The IRS allows firms to use the same or different methods to depreciate assets in calculating taxable income.

THINK IT THROUGH

Fixed Assets

You work for Georgia-Pacific as an accountant in charge of the fixed assets subsidiary ledger at a production and warehouse facility in Pennsylvania. The facility is in the process of updating and replacing several asset categories, including warehouse storage units, fork trucks, and equipment on the production line. It is your job to keep the information in the fixed assets subsidiary ledger up to date and accurate. You need information on original historical cost, estimated useful life, salvage value, depreciation methods, and additional capital expenditures. You are excited about the new purchases and upgrades to the facility and how they will help the company serve its customers better. However, you have been in your current position for only a few years and have never overseen extensive updates, and you realize that you will have to gather a lot of information at once to keep the accounting records accurate. You feel overwhelmed and take a minute to catch your breath and think through what you need. After a few minutes, you realize that you have many people and many resources to work with to tackle this project. Whom will you work with, and how will you go about gathering what you need?

Solution:

Though answers may vary, common resources would likely include purchasing managers (those actually buying the new equipment), maintenance managers (those who will repair and take care of the new equipment), and line managers (those in charge of the departments that will use the new equipment). To

gather the information needed, set up short meetings to visit with the individuals involved, walk around to see the equipment, and ask questions about functionality, life span, common problems or repairs, and more.

Assume that on January 1, Liam bought a silk screen machine for \$54,000. Liam pays shipping costs of \$1,500 and setup costs of \$2,500 and assumes a useful life of five years or 960,000 prints. Based on experience, Liam anticipates a salvage value of \$10,000. Recall that determination of the costs to be depreciated requires including all costs that prepare the asset for use by the company. Liam's example would include shipping and setup costs. Any costs for maintaining or repairing the equipment would be treated as regular expenses, so the total cost would be \$58,000, and after allowing for an anticipated salvage value of \$10,000 in five years, the business could take \$48,000 in depreciation over the machine's economic life (see [Figure 4.12](#)).

	Total Cost
Purchase Price	\$ 54,000
Shipping Costs	1,500
Set-up Costs	2,500
Total Cost	\$ 58,000
– Salvage Value	(10,000)
Depreciable Base	\$ 48,000

Figure 4.12 Purchase Price and Depreciable Base

Straight-line depreciation is a method of depreciation that evenly splits the depreciable amount across the useful life of the asset. Therefore, we must determine the yearly depreciation expense by dividing the depreciable base of \$48,000 by the economic life of five years, giving an annual depreciation expense of \$9,600. The journal entries to record the first two years of expenses are shown, along with the balance sheet information. Here are the journal entry and information for year one ([Figure 4.13](#)):

Journal			
Date	Account	Debit	Credit
Dec. 31	Depreciation Expense: Silk Screen Machine Accumulated Depreciation: Silk Screen Machine <i>To record depreciation on asset for period</i>	9,600	9,600

Silk Screen Machine	\$ 58,000
– Accumulated Depreciation: Silk Screen Machine	(9,600)
Net Book Value	\$ 48,400

Figure 4.13 Journal Entry for Silk Screen Machine Depreciation Expense, Year 1

After the journal entry in year one, the machine would have a book value of \$48,400. This is the original cost of \$58,000 less the accumulated depreciation of \$9,600. The journal entry and information for year two are shown in [Figure 4.14](#).

Journal			
Date	Account	Debit	Credit
Dec. 31	Depreciation Expense: Silk Screen Machine Accumulated Depreciation: Silk Screen Machine <i>To record depreciation on asset for period</i>	9,600	9,600

Silk Screen Machine	\$ 58,000
– Accumulated Depreciation: Silk Screen Machine	(19,200)
Net Book Value	\$ 38,800

Figure 4.14 Journal Entry for Silk Screen Machine Depreciation Expense, Year 2

Liam records an annual depreciation expense of \$9,600. Each year, the accumulated depreciation balance increases by \$9,600, and the machine's book value decreases by the same \$9,600. At the end of five years, the asset will have a book value of \$10,000, which is calculated by subtracting the accumulated depreciation of \$48,000 ($5 \times \$9,600$) from the cost of \$58,000.

Units-of-Production Depreciation

Straight-line depreciation is efficient accounting for assets used consistently over their lifetime, but what about assets that are used with less regularity? The units-of-production depreciation method bases depreciation on the actual usage of the asset, which is more appropriate when an asset's life is a function of usage instead of time. For example, this method could account for depreciation of a silk screen machine for which the depreciable base is \$48,000 (as in the straight-line method), but now the number of prints is important.

In our example, the machine will have total depreciation of \$48,000 over its useful life of 960,000 prints. Therefore, we would divide \$48,000 by 960,000 prints to get a cost per print of \$0.05. If Liam printed 180,000 items in the first year, the depreciation expense would be 180,000 prints $\times$ \$0.05 per print, or \$9,000. The journal entry to record this expense would be the same as with straight-line depreciation: only the dollar amount would have changed. The presentation of accumulated depreciation and the calculation of the book value would also be the same. Liam would continue to depreciate the asset until a total of \$48,000 in depreciation was taken after running 960,000 total prints.

THINK IT THROUGH

Deciding on a Depreciation Method

Liam is struggling to determine which depreciation method they should use for their new silk screen machine. They expect sales to increase over the next five years. They also expect (hope) that in two years they will need to buy a second silk screen machine to keep up with the demand for products of their growing company. Which depreciation method makes more sense for Liam: higher expenses in the first few years or keeping expenses consistent over time? Or would it be better for them to think not in terms of time, but rather in the usage of the machine?

Double-Declining-Balance Depreciation

The double-declining-balance depreciation method is the most complex of the three methods because it accounts for both time and usage and takes more expense in the first few years of the asset's life. Double declining considers time by determining the percentage of depreciation expense that would exist under straight-line depreciation. To calculate this, divide 100 percent by the estimated life in years. For example, a five-year asset would be $100/5$, or 20 percent a year. A four-year asset would be $100/4$, or 25 percent a year. Next, because assets are typically more efficient and are used more heavily early in their life span, the double-

declining method takes usage into account by doubling the straight-line percentage. For a four-year asset, multiply 25 percent ($100\%/4\text{-year life}$) $\times 2$, or 50 percent. For a five-year asset, multiply 20 percent ($100\%/5\text{-year life}$) $\times 2$, or 40 percent.

One unique feature of the double-declining-balance method is that in the first year, the estimated salvage value is not subtracted from the total asset cost before calculating the first year's depreciation expense. Instead, the total cost is multiplied by the calculated percentage. However, depreciation expense is not permitted to take the book value below the estimated salvage value, as demonstrated in [Figure 4.15](#).

Year	Depreciation Expense	Accumulated Depreciation	Book Value
			\$58,000
1: $\$58,000 \times 40\% =$	\$23,200	\$23,200	34,800
2: $\$34,800 \times 40\% =$	13,920	37,120	20,880
3: $\$20,880 \times 40\% =$	8,352	45,472	12,528
4: $\$12,528 - \$10,000 =$	2,528	48,000	10,000
5	0	48,000	10,000
Total	\$48,000	\$48,000	\$10,000

Figure 4.15 Depreciation Expense, Accumulated Depreciation, and Book Value, Years 1-5

Notice that in year four, the remaining book value of \$12,528 was not multiplied by 40 percent. This is because the expense would have been \$5,011.20, and since we cannot depreciate the asset below the estimated salvage value of \$10,000, the expense cannot exceed \$2,528, which is the amount left to depreciate (difference between the book value of \$12,528 and the salvage value of \$10,000). Since the asset has been depreciated to its salvage value at the end of year four, no depreciation can be taken in year five.

In our example, the first year's double-declining-balance depreciation expense would be $\$58,000 \times 40\%$, or \$23,200. For the remaining years, the double-declining percentage is multiplied by the remaining book value of the asset. Liam would continue to depreciate the asset until the book value and the estimated salvage value are the same (in this case, \$10,000). The net effect of the differences in straight-line depreciation versus double-declining-balance depreciation is that under the double-declining-balance method, the allowable depreciation expenses are greater in the earlier years than those allowed for straight-line depreciation. However, over the depreciable life of the asset, the total depreciation expense taken will be the same no matter which method the entity chooses. In the current example, both straight-line and double-declining-balance depreciation will provide a total depreciation expense of \$48,000 over its five-year depreciable life.

LINK TO LEARNING

McDonald's Corporation

See [Form 10-K \(https://openstax.org/r/2019-annual-report-pdf\)](https://openstax.org/r/2019-annual-report-pdf) that was filed with the SEC to determine which depreciation method McDonald's Corporation used for its long-term assets in 2019.

Hint: Use the search feature to search for key words in an annual report to find information more quickly. For example, search "depreciation." You should find the information you are looking for on page 40 of the annual report.

Based on the company's business model and the industry in which it operates, why do you think McDonald's chose this method of depreciating assets? Do you agree with this choice? Why or why not?

Summary of Depreciation

Table 4.2 and Figure 4.16 compare the three methods discussed. Note that although each time-based (straight-line and double-declining balance) annual depreciation expense is different, after five years the total amount depreciated (accumulated depreciation) is the same. This occurs because at the end of the asset's useful life, it was expected to be worth \$10,000: thus, both methods depreciated the asset's value by \$48,000 over that time period.

Depreciation Method	Calculation
Straight line	$(\text{Cost} - \text{Salvage Value}) / \text{Useful Life}$
Units of production	$(\text{Cost} - \text{Salvage Value}) \times (\text{Units Produced in Current Period} / \text{Estimated Total Units to Be Produced})$
Double-declining balance	$\text{Book Value} \times \text{Straight} - \text{Line Annual Depreciation Percentage} \times 2$

Table 4.2 Three Methods of Calculating Depreciation Expense

Period	Straight-Line Depreciation Method	Units of Production Method	Double-Declining-Balance Method
Year 1	\$ 9,600	(180,000 units) \$ 9,000	\$23,200
Year 2	9,600	(200,000 units) 10,000	13,920
Year 3	9,600	(210,000 units) 10,500	8,352
Year 4	9,600	(190,000 units) 9,500	2,528
Year 5	9,600	(180,000 units) 9,000	0
Total	\$48,000	\$48,000	\$48,000

Figure 4.16 Straight-Line Depreciation, Units of Production, Double-Declining Balance Method, Years 1–5

When analyzing depreciation, accountants are required to make a supportable estimate of an asset's useful life and its salvage value. However, "management teams typically fail to invest either time or attention into making or periodically revisiting and revising reasonably supportable estimates of asset lives or salvage values, or the selection of depreciation methods, as prescribed by GAAP."⁵ This failure is not an ethical approach to properly accounting for the use of assets.

Accountants need to analyze depreciation of an asset over the entire useful life of the asset. As an asset supports the cash flow of the organization, expensing its cost needs to be allocated, not just recorded as an arbitrary calculation. An asset's depreciation may change over its life according to its use. If asset depreciation is arbitrarily determined, the recorded "gains or losses on the disposition of depreciable property assets seen in financial statements"⁶ are not true best estimates. Due to operational changes, the depreciation expense needs to be periodically reevaluated and adjusted.

Any mischaracterization of asset usage is not proper GAAP and is not proper accrual accounting. Therefore, "financial statement preparers, as well as their accountants and auditors, should pay more attention to the quality of depreciation-related estimates and their possible mischaracterization and losses of credits and charges to operations as disposal gains."⁷ An accountant should always follow GAAP guidelines and allocate the expense of an asset according to its usage.

⁵ Howard B. Levy. "Depreciable Asset Lives: The Forgotten Estimate in GAAP." *The CPA Journal*. September 2016. cpajournal.com/2016/09/08/depreciable-asset-lives/

⁶ Ibid.

CONCEPTS IN PRACTICE

Ethical Considerations: How WorldCom’s Improper Capitalization of Costs Almost Shut Down the Internet

In 2002, telecommunications giant WorldCom filed for the largest Chapter 11 bankruptcy to date, a situation resulting from manipulation of its accounting records.⁸ At the time, WorldCom operated nearly a third of the bandwidth of the 20 largest US internet backbone routes, connecting over 3,400 global networks that serviced more than 70,000 businesses in 114 countries.⁹

WorldCom used a number of accounting gimmicks to defraud investors, mainly including capitalizing costs that should have been expensed. Under normal circumstances, this might have been considered just another accounting fiasco leading to the end of a company. However, WorldCom controlled a large percentage of backbone routes, a major component of the hardware supporting the internet, as even the Securities and Exchange Commission recognized. If such an event were to happen today, it could shut down international commerce and would be considered a national emergency.¹⁰ As demonstrated by WorldCom, the unethical behavior of a few accountants could have shut down the world’s online businesses and international commerce. An accountant’s job is fundamental and important: keep businesses operating in a transparent fashion.

(Sources: “WorldCom (UNNET).” *Cybertelecom*. n.d. <http://www.cybertelecom.org/industry/wcom.htm>; Dennis R. Beresford, Nicholas DeB. Katzenbach, and C. B. Rogers, Jr. “Report of the Special Investigative Committee of the Board of Directors of WorldCom, Inc.” US Securities and Exchange Commission. March 31, 2003. <https://www.sec.gov/Archives/edgar/data/723527/000093176303001862/dex991.htm>)

4.5 What Is “Profit” versus “Loss” for the Company?

Learning Outcomes

By the end of this section, you will be able to:

- Outline the components necessary to calculate profit or loss.
- Contrast revenue and gains versus expenses and losses.
- Differentiate revenue and expense versus receipt or payment of cash.

It’s a common misconception that if a business has cash they are making a profit, and if they are suffering a loss they must not have any cash. While this could be true, it’s not necessarily true. In this section you’ll explore key differences between cash flow, revenue, expense, profits, and losses.

Calculating Profit and Loss

Net income (net loss) is determined by comparing revenues and expenses. Net income is a result of revenues (inflows) being greater than expenses (outflows). A net loss occurs when expenses (outflows) are greater than revenues (inflows). In accounting it is common to present net income in the following format:

Recall that revenue is the value of goods and services a business provides to its customers and increases the value of the business. Expenses, on the other hand, are the costs of providing the goods and services and decrease the value of the business. When revenues exceed expenses, companies have net income. This means the business has been successful at earning revenues, containing expenses, or a combination of both. If, on the other hand, expenses exceed revenues, companies experience a net loss. This means the business was

⁷ Ibid.

⁸ Luisa Beltran. “WorldCom Files Largest Bankruptcy Ever.” *CNN Money*. July 22, 2002. https://money.cnn.com/2002/07/19/news/worldcom_bankruptcy/

⁹ Jeff Keefe. *Monopoly.com: Will the WorldCom-MCI Merger Tangle the Web?* Washington, DC: Economic Policy Institute, 1998. <https://www.epi.org/publication/monopoly-will-the-worldcom-mci-merger-tangle-the-web/>

¹⁰ Dan Schiller. *Bad Deal of the Century: The Worrisome Implications of the WorldCom-MCI Merger*. Washington, DC: Economic Policy Institute, 1998. https://www.epi.org/publication/studies_baddealfull/

unsuccessful in earning adequate revenues, sufficiently containing expenses, or a combination of both. While businesses work hard to avoid net loss situations, it is not uncommon for a company to sustain a net loss from time to time. It is difficult, however, for businesses to remain viable while experiencing net losses over the long term.

Shown as a formula, the net income (loss) function is:

$$\text{Revenues } (R) - \text{Expenses } (E) = \text{Net Income (when } R > E)$$

$$\text{Revenues } (R) - \text{Expenses } (E) = \text{Net Loss (when } E > R)$$

To be complete, we must also consider the impact of gains and losses. While gains and losses are infrequent in a business, it is not uncommon that a business would present a gain and/or loss in its financial statements. Recall that gains are similar to revenue and losses are similar to expenses. Therefore, the traditional accounting format would be as shown in [Figure 4.17](#).

Gains and Losses
Revenue (sometimes called Sales or Fees Earned)
+ Gains
- Expenses
- Losses
Net Income (or Net Loss)

Figure 4.17 Gains and Losses

Shown as a formula, the net income (loss) function, including gains and losses, is:

$$\text{Revenues } (R) + \text{Gains } (G) - \text{Expenses } (E) - \text{Losses } (L) = \text{Net Income [when } (R + G) > (E + L)]$$

$$\text{Revenues } (R) + \text{Gains } (G) - \text{Expenses } (E) - \text{Losses } (L) = \text{Net Loss [when } (E + L) > (R + G)]$$

When assessing a company's net income, it is important to understand the source of the net income. Businesses strive to attain "high quality" net income (earnings). High-quality earnings are based on sustainable earnings, also called permanent earnings, while relying less on infrequent earnings, also called temporary earnings. Recall that revenues represent the ongoing value of goods and services the business provides (sells) to its customers, while gains are infrequent and involve items ancillary to the primary purpose of the business. For example, assume a bakery sells the truck it uses to deliver wedding cakes and experiences a gain on the sale. The bakery is not in the business of buying and selling trucks. It is in the baked goods business. Thus, the gain on the sale of the truck would be ancillary to the primary purpose of the business and represent a gain rather than revenue. We should use caution if a business attains a significant portion of its net income as a result of gains rather than revenues. Likewise, net losses derived as a result of losses should be put into the proper perspective due to the infrequent nature of losses. While net losses are undesirable for any reason, net losses that result from expenses related to ongoing operations, rather than losses that are infrequent, are more concerning for the business.

Profit versus Cash Flow

Knowing the difference between the cash basis and accrual basis of accounting is necessary to understand the need for the statement of cash flows. Stakeholders need to know the financial performance (as measured by the income statement—that is, net income or net loss) and financial position (as measured by the balance sheet—that is, assets, liabilities, and owners' equity) of the business. This information is provided in the income statement, statement of owner's equity, and balance sheet. However, since these financial statements are prepared using accrual accounting, stakeholders do not have a clear picture of the business's cash activities. The statement of cash flows solves this inadequacy by specifically focusing on the cash inflows and cash outflows. It also helps better delineate the difference between revenues and cash flow in versus expenses and cash flow out. As mentioned in prior sections, revenue can occur without cash actually flowing. For

example, a customer may buy a good on account. Revenues would be recorded, but cash would not yet be received. The same is true on the expenses side. An expense can be incurred, such as an electric bill, but cash may not have been paid out yet. Thus, an expense is recorded and recognized on the income statement, but cash has not yet been given up. The statement of cash flows helps reconcile the difference between net income (a result of recorded revenues and expenses) and actual cash flow.



Summary

4.1 Cash versus Accrual Accounting

Cash-basis accounting records revenues and expenses only when cash is received or distributed. Accrual-basis accounting, on the other hand, records revenues and expenses when they are earned or incurred rather than waiting until cash changes hands. Most publicly traded companies are required by the SEC to use accrual-basis accounting. Generally, smaller businesses that deal primarily in cash are the best candidates for using the cash basis since the accrual method more accurately depicts net income or net loss each period.

4.2 Economic Basis for Accrual Accounting

Double-entry accounting means that each time a transaction is recorded, there are at minimum two accounts impacted by the entry. Each entry's debits must total its credits in order to support and maintain the balance in the accounting equation. The accounting equation is expressed as

$$\text{Assets} = \text{Liabilities} + \text{Owner's Equity}.$$

4.3 How Does a Company Recognize a Sale and an Expense?

In accrual accounting, the timing of revenues (when to record them) is governed by the revenue recognition principle. The principle indicates that revenue is only recognized on the income statement once it is earned. This means goods or services must have been delivered or rendered. Expenses, on the other hand, are guided by the expense recognition principle, which dictates that expenses must be recorded in the period in which they are incurred.

4.4 When Should a Company Capitalize or Expense an Item?

Determining when an item is actually an expense depends on whether it is capitalized (a fixed asset) or not. Fixed assets are those that are of significant value and last longer than one year. The cost of fixed assets is capitalized (placed on the balance sheet as an asset) and expensed over the useful life of the asset by recording depreciation. Depreciation can be calculated using straight-line, double-declining-balance, or units-of-production methods.

4.5 What Is “Profit” versus “Loss” for the Company?

Profit or loss for a company is calculated by subtracting expenses from revenue. The result is a profit if revenues are larger than expenses. Profit (or loss) is the money earned from the day-to-day general business operations. Gains and losses, on the other hand, occur when the business does something they don't normally do (like sell a piece of their equipment) and earns or loses money on the transaction. It's key to note that the timing of cash flows can vary from the timing of recording revenues or expenses. Thus a revenue does not necessarily equal cash in, and an expense does not necessarily equal a cash flow out.



Key Terms

accounting equation $\text{Assets} = \text{Liabilities} + \text{Owner's Equity}$

accrual-basis accounting an accounting system in which revenue is recorded or recognized when earned yet not necessarily received, and in which expenses are recorded when legally incurred and not necessarily when paid

capitalize the process in which a long-term asset is recorded on the balance sheet and its allocated costs are expensed on the income statement over the asset's economic life

cash-basis accounting a method of accounting in which transactions are not recorded in the financial statements until there is an exchange of cash

credit a record of financial information on the right side of an account

debit a record of financial information on the left side of each account

- depreciation** the process of allocating the costs of a tangible asset over the asset's economic life
- double-entry accounting** an accounting method that requires the sum of the debits to equal the sum of the credits for each transaction
- expense** a cost associated with providing goods or services
- expense recognition** (also, matching principle) matches expenses with associated revenues in the period in which the revenues were generated
- gains** increases in organizational value from activities that are "incidental or peripheral" to the primary purpose of the business
- income statement** a financial statement that measures the organization's financial performance for a given period of time
- long-term asset** asset used in the normal, ongoing course of business for more than one year that is not intended to be resold
- losses** decreases in organizational value from activities that are "incidental or peripheral" to the primary purpose of the business
- net income** income earned when revenues and gains are greater than expenses and losses
- revenue** inflows or other enhancements of assets of an entity or settlements of its liabilities (or a combination of both) from delivering or producing goods, rendering services, or other activities that constitute the entity's ongoing major or central operations
- revenue recognition** principle stating that a company must recognize revenue in the period in which it is earned; it is not considered earned until a product or service has been provided
- T-account** a graphic representation of a general ledger account in which each account is visually split into left and right sides
- tangible asset** an asset that has physical substance



Multiple Choice

- Which method of accounting must be used by publicly traded companies?
 - cash basis
 - accrual basis
 - a hybrid of cash and accrual basis
 - modified accelerated basis
- Which two accounting principles directly support the accrual method of accounting?
 - periodicity, transparency
 - historical cost, time period
 - conservatism, going concern
 - expense recognition, revenue recognition
- Which of the following account types has a normal debit balance?
 - cash
 - notes payable
 - wages payable
 - revenue
- Sara sells \$100 worth of inventory to her client on credit on January 15. She delivers the inventory to the client on January 20. The client pays for the inventory of February 26. On what date should Sara recognize the revenue from the sale?
 - January 15
 - January 20
 - January 31

- d. February 26
5. How is the book value of a fixed asset calculated?
 - a. original cost less accumulated depreciation
 - b. original cost divided by useful life
 - c. market value less depreciation expense
 - d. depreciable base less market value
 6. Which of the following balance sheet item is classified as a fixed asset?
 - a. inventory
 - b. goodwill
 - c. equipment
 - d. accounts receivable
 7. What is the formula to determine net income or net loss?
 - a. revenues less expenses
 - b. cash less expenses
 - c. revenues plus cash
 - d. revenues divided by expenses



Review Questions

1. Joe runs a small barbershop. Most of his customers pay with cash. He has only a few monthly expenses including wages for one employee and utilities. Which method of accounting should Joe use?
2. Which method of accounting generally provides the most accurate information on organizational performance (income or loss)?
3. Under the accrual basis of accounting, when is an expense recorded?
4. Describe the basic elements of a T-account.
5. What is double-entry accounting?
6. What is the expense recognition principle?
7. Why is ethics an important concept pertaining to revenue and expense recognition principles?
8. What is the revenue recognition principle?
9. What is the difference between a tangible and an intangible fixed asset? List common examples of each.
10. How is the cost of a fixed asset recorded and recognized over time?



Problems

1. Padma's Pools Inc. paid \$500 for office supplies. What was the impact of this transaction on Padma's cash flow?
2. Sally's BigBox Store issued 1,000 shares of common stock with par value of \$10 each and market value of \$16 each in exchange for a new building. What was the impact of this transaction on Sally's cash flow?
3. Jose sells \$500 worth of merchandise to a client on June 1. He delivers the product and invoices the customer on June 10. The customer pays Jose on July 9. What is Jose's revenue for June and July, respectively, under the cash and accrual methods of accounting?

4. Jamal's Car Repair purchases a new piece of equipment with a 10-year useful life for \$10,000. What is the impact to Jamal's net income in the year of purchase if he expenses the equipment? If he capitalizes it using straight-line depreciation?
5. Mariela's Shop had revenues of \$10,000 and expenses of \$6,000 and has cash on hand of \$5,000. What is Mariela's net income or net loss?

Video Activity

Depreciation Basics! With Journal Entries

[Click to view content \(https://openstax.org/r/depreciation-basics\)](https://openstax.org/r/depreciation-basics)

1. In what ways do depreciation and capitalization impact the income statement? If you were an investor considering investing a large sum of money into a company, what questions would you ask or what accounts would you look at to assess the impact to their income statement specifically? How would the characteristics of capitalization affect your feelings toward investing in the company?
2. If a company spends a large sum of cash to invest in equipment or another fixed asset, resulting in a loss that year on the income statement if they expense it, have they really experienced a loss? Or have they simply traded one asset for another? How does this delineation relate to capitalizing an asset and spreading out the cost throughout its useful life? Do you feel the depreciation process is ethical? Or do you feel it hides the true cost of business from being fully transparent on the financial statements? Explain your answer.

Difference between Cash Flow and Profit

[Click to view content \(https://openstax.org/r/difference-cashflow-profit\)](https://openstax.org/r/difference-cashflow-profit)

3. When a business incurs an expense (office supplies, utilities, or wages, for example), is their cash flow the same in both timing and amount as the expense they recognize?
4. What are the key components necessary to calculate profit or loss for a business? Are they the same elements necessary to calculate cash flow?

5.1 The Income Statement

Learning Outcomes

By the end of this section, you will be able to:

- Outline the purpose and importance of the income statement.
- Identify the structure and key elements of the income statement.
- Discuss the use of EBITDA as a measure of a company's profit.

Financial information flows from one financial statement to the next. Thus, the statements are prepared in a specific order. The first statement prepared is the income statement.

Functionality of the Income Statement

The **income statement** shows a firm's performance over a specific period of time. The statement helps financial statement users understand the sales generated during the period and the **expenses** incurred to generate those sales. If the expenses are smaller than the sales, the net result is profitability, or net income, rather than a net **loss**.

Breaking the income statement down into smaller pieces provides a more transparent view of the firm's performance, allowing users to see more clearly what areas of the business incurred expenses. This is helpful to management in driving improvements and to outside users in assessing performance.

Sales and Gross Profit

The first section of the income statement begins with sales. Though financial statements are required to follow a certain format, account names can differ slightly from one firm to another. You may see the first line, often referred to as *the top line*, called *sales*, *sales revenue*, **revenue**, *service revenues*, and other similar titles. All of these titles are meant to reflect the sales generated by selling product to customers in the day-to-day business. On Clear Lake's income statement in [Figure 5.2](#), we see its top line referred to as Sales.

Income from items that aren't common to the firm's day-to-day business are reported as **gains** and losses, and they are reported further down in the income statement rather than at the top line with its regular, core business activities. This is to ensure that anomalies like selling a machine or a loss on retiring a bond don't mislead financial statement users as to the general performance of the firm and impact their assumptions of future results.

Firms report their sales and any reductions to sales separately on the income statement. They begin with gross sales, which includes all sales to customers. Clear Lake reported gross sales of \$105,000 last year and \$126,000 this year. The next line is sales returns and allowances, which is deducted from gross sales in order to find net sales. Clear Lake's sales returns and allowances were \$5,000 and \$6,000 respectively, leaving the company with net sales of \$100,000 and \$120,000 respectively

$$(\$105,000 - \$5,000 \text{ and } \$126,000 - \$6,000).$$

Next, the cost of goods sold (COGS) is deducted from net sales in order to arrive at gross profit. (It is customary to refer to sales minus COGS as gross profit because $\text{gross margin} = \text{gross profit}/\text{sales}$.) Cost of goods sold includes the costs directly involved in making the product that was sold during the period. Common examples of costs included in cost of goods sold include direct labor, direct materials, and the overhead assigned to the product in production. For a service business, this would include its direct labor and any materials used to deliver its services. For a retail firm like Clear Lake Sporting Goods, this would include the costs of all the goods it purchased for resale. Clear Lake's COGS is seen at \$50,000 and \$60,000 for the prior and current years. Note that different types of companies will have different types of costs deducted in their Cost of Goods section. Clear Lake Sporting Goods is a retailer, or merchandiser that buys good to resell. Their cost of goods includes the cost of goods they purchased to resell. In the link to learning, you will explore Apple, a technology manufacturer. Their cost of goods would include the cost to manufacture the goods they

sell. Another type of firm is a service firm. A law office, for example, would include primarily the cost of labor in their cost of services.

Gross profit is a reflection of how profitable the firm's performance was in its core business function. It includes only the core business and direct costs of performing that business. If the company were a shoe company, gross profit would show how profitable the company was in simply making the shoes it sold. If it were a bakery, gross profit would show how profitable the company was in simply baking the goods it sold. Gross profit shows financial statement users how effective the business is at generating top-line profits on their core business function. It does not reflect the performance of other areas of the firm such as other operating costs to support the direct production process, indirect costs, and financing.

For Clear Lake Sporting Goods, we see its gross profit in [Figure 5.2](#). The company earned \$50,000 of gross profit (\$100,000 – \$50,000) the prior year and \$60,000 in the current year (\$120,000 – \$60,000).

Clear Lake Sporting Goods Comparative Year-End Income Statements		
	Prior Year	Current Year
Gross Sales	\$105,000	\$126,000
Sales Returns & Allowances	5,000	6,000
Net Sales	100,000	120,000
Cost of Goods Sold	50,000	60,000
Gross Profit	\$ 50,000	\$ 60,000

Figure 5.2 Income Statement through Gross Profit Line

LINK TO LEARNING

Gross Profit

Visit the [Apple, Inc. Annual Report \(https://openstax.org/r/2020-doc-financial-annual-report\)](https://openstax.org/r/2020-doc-financial-annual-report) for 2020 and locate the income statement (it begins on page 31). Review gross sales, sales returns and allowances, and net sales for the last few years. What can you learn about the company's recent performance in the area of sales? Is the gross sales line improving? How about the sales returns and allowances line? Has it improved or declined or simply changed incrementally with gross sales?

Review the firm's gross profit for the past few years. Has it improved or declined? Consider the company's improvement (or decline) in gross sales as compared to the improvement or decline in gross profit. Does it reflect any change in performance over time (earning more or less gross profit on its gross sales)?

Income from Operations

Gross profit is a very helpful measure, but it is only the first of several provided by the income statement. After gross profit is calculated, other operating expenses are deducted in order to calculate the firm's income from operations, also commonly called operating income. Common operating costs found in this section include building rent and utilities, property taxes, wages and salaries, and other overhead costs. In [Figure 5.3](#), we can see Clear Lake's operating expenses. To sell its hunting and fishing equipment in the current year, Clear Lake Sporting Goods paid rent for its building (\$5,500) and utilities for its retail and warehouse spaces (\$2,500); recorded **depreciation** on equipment, buildings, and store furnishings (shelves, racks, etc.) (\$3,600); and paid salaries to its indirect employees in accounting, purchasing, and human resources (\$5,400). The company's operating expenses are deducted from gross profit to arrive at operating income

$$(\$60,000 - \$5,500 - \$3,600 - \$5,400 - \$2,500 = \$43,000).$$

While gross profit reflects only how profitable the firm was in making its core product, operating income

reflects how profitable the firm's daily operations were as a whole. This still does not include other miscellaneous items outside the scope of a firm's normal business. Just as the name implies, it shows income from the core operations of the firm.

Clear Lake Sporting Goods Comparative Year-End Income Statements		
	Prior Year	Current Year
Gross Sales	\$105,000	\$126,000
Sales Returns & Allowances	5,000	6,000
Net Sales	100,000	120,000
Cost of Goods Sold	50,000	60,000
Gross Profit	50,000	60,000
Rent Expense	5,000	5,500
Depreciation Expense	2,500	3,600
Salaries Expense	3,000	5,400
Utility Expense	1,500	2,500
Operating Income	\$ 38,000	\$ 43,000

Figure 5.3 Income Statement through Income from Operations

We can see that the company was able to generate \$20,000 (\$120,000 – \$100,000) more in net sales in the current year than the prior year. However, it only generated \$10,000 (\$60,000 – \$50,000) in gross profit and \$5,000 (\$43,000 – \$38,000) of additional operating income. Further investigation shows that while net sales increased, so did the direct costs of its goods (COGS) and its operating expenses.

We will further explore how to assess each of these expenses later in the chapter using common-size analysis. Common-size analysis reflects each element of a financial statement as a percentage of the base. In the case of the income statement, the base is net sales. Here we would see that COGS was 50 percent of net sales in both the current (\$60,000/\$120,000) and prior year (\$50,000/\$100,000), indicating there wasn't any significant change in performance we could detect from the information provided in the income statement.

LINK TO LEARNING

Operating Expenses

Visit the [Apple, Inc. Annual Report \(https://openstax.org/r/2020-doc-financial-annual-report\)](https://openstax.org/r/2020-doc-financial-annual-report) for 2020 and locate the company's income statement (the income statement begins on page 31). Review Apple's operating income for the last few years. Has it improved or declined? Does this fall in line with your expectations based on your previous review of the company's net sales and gross profit?

Based on the improvement or decline in operating income, review Apple's operating expenses to investigate. Have any of the operating expenses changed significantly? How did that impact the company's operating income? What might it tell you about the company's performance?

Net Income

Finally, we move on to **net income**, or what is commonly referred to as *the bottom line*. Net income (or loss) reflects the net impact of all financial transactions for the firm, including those that are caused by events outside the normal course of business. The most common items deducted from operating income to arrive at net income include interest expense, gains/losses, and income tax expense. Remember, gains and losses are those that result from unusual transactions outside the normal course of business. Examples include selling a piece of old equipment or a loss on retiring debt.

We can see in [Figure 5.4](#) that Clear Lake Sporting Goods has outstanding debt, so it incurred interest expense

of \$2,000 in the current year and \$3,000 the prior year. Since it recorded net income (not a loss), it must also record income tax expense of \$6,000 in the current and \$5,000 in the prior year.

Clear Lake Sporting Goods Comparative Year-End Income Statements		
	Prior Year	Current Year
Gross Sales	\$105,000	\$126,000
Sales Returns & Allowances	5,000	6,000
Net Sales	100,000	120,000
Cost of Goods Sold	50,000	60,000
Gross Profit	50,000	60,000
Rent Expense	5,000	5,500
Depreciation Expense	2,500	3,600
Salaries Expense	3,000	5,400
Utility Expense	1,500	2,500
Operating Income	38,000	43,000
Interest Expense	3,000	2,000
Income Tax Expense	5,000	6,000
Net Income	\$ 30,000	\$ 35,000

Figure 5.4 Comparative Year-End Income Statement

EBITDA (Earnings before Interest, Taxes, Depreciation, and Amortization)

Now that we have a full income statement, we can look at another commonly used measure of financial performance called EBITDA. EBITDA stands for earnings before interest, taxes, depreciation, and amortization. Amortization is similar to depreciation. It is the spreading of the cost of an intangible asset over the course of its useful life. Intangible assets are long-term assets that lack physical substance, such as patents and copyrights.

Since EBITDA removes noncash items from the net income equation, it is considered a useful measure in assessing the cash flows provided by operating activities. We will assess cash flows using the **statement of cash flows** and various other cash flow measures later in this chapter as well.

As shown in [Figure 5.5](#), Clear Lake Sporting Goods' EBITDA in the prior year was

$$\$19,500 (\$30,000 - \$3,000 - \$5,000 - \$2,500)$$

and in the current year was

$$\$23,400 (\$35,000 - \$2,000 - \$6,000 - \$3,600).$$

Clear Lake Sporting Goods Comparative Year-End Income Statements		
	Prior Year	Current Year
Gross Sales	\$105,000	\$126,000
Sales Returns & Allowances	5,000	6,000
Net Sales	100,000	120,000
Cost of Goods Sold	50,000	60,000
Gross Profit	50,000	60,000
Rent Expense	5,000	5,500
Depreciation Expense	2,500	3,600
Salaries Expense	3,000	5,400
Utility Expense	1,500	2,500
Operating Income	38,000	43,000
Interest Expense	3,000	2,000
Income Tax Expense	5,000	6,000
Net Income	\$ 30,000	\$ 35,000

Figure 5.5 EBITDA (Earnings before Interest, Taxes, Depreciation, and Amortization)

THINK IT THROUGH

Net Income

Visit the [Apple, Inc. Annual Report \(https://openstax.org/r/2020-doc-financial-annual-report\)](https://openstax.org/r/2020-doc-financial-annual-report) for 2020 and locate the company's income statement (the income statement begins on page 31). Review net income for the last few years. Has it improved or declined? Does this fall in line with your expectations based on your previous review of the company's operating income?

Solution:

Net income improved in the current year over last year but declined from 2018 to 2019 (from \$59,531 to \$55,256). Apple's EBITDA in 2020 was 108.89. (Amounts are in millions.)

5.2 The Balance Sheet

Learning Outcomes

By the end of this section, you will be able to:

- Outline the purpose and importance of the balance sheet.
- Identify the structure and key elements of the balance sheet.

The Accounting Equation and the Classified Balance Sheet

Recall that the income statement shows the performance of a firm over the course of time. The classified balance sheet shows the financial state of a company as of a specific point in time. It is a key distinction between the two statements. The classified balance sheet is prepared in sections that align with the accounting equation.

Remember, the **accounting equation** reflects the **assets** (items owned by the organization) and how they were obtained (by incurring **liabilities** or provided by owners). Liabilities are debts owed to other parties. Stated differently, every asset has a claim against it—by creditors and/or owners.

$$\text{Assets} = \text{Liabilities} + \text{Owner's Equity}$$

The classified balance sheet is thus broken down into three sections; assets, liabilities, and owner's equity. If

prepared correctly, the total assets on the balance sheet equals the total liabilities and owner's equity sections of the balance sheet.

The classified balance sheet is considered or termed *classified* when the assets and liabilities within the balance sheet are grouped into even smaller sections: current and noncurrent (see [Figure 5.6](#)). Both assets and liabilities are broken down on the balance sheet into current and noncurrent classifications in order to provide more detail and transparency as well as abide by the convention of reporting in descending order of liquidity. **Current assets** are those that can be used or converted to cash within a year. Common examples of current assets include cash, inventory, accounts receivable, and short-term investments. Assets that will be in use longer than a year are considered noncurrent. Common examples of **noncurrent assets** include notes receivable, machinery and equipment, buildings, and land.

Just as we noted a few key differences in the income statements based on the type of firm, you may also notice a few slight differences in the balance sheet depending on the firm type. Clear Lake Sporting Goods is a retailer. Thus, you will see that their inventory for resale on their balance sheet is simply called "Inventory." This is the goods they have purchased for resale but have not yet sold. A manufacturer, like Apple, Inc. in the Link to Learning sections, will have a variety of inventory types including raw materials, work in progress, and finished goods inventory. These represent the various states of the inventory (ready to use, partially complete, and fully completed product). A service firm, on the other hand, may not have inventory at all. If it does, it may be simple goods it uses to help deliver its service. For example, a cleaning company may keep an inventory of cleaning supplies.



Figure 5.6 Graphical Representation of the Accounting Equation Both assets and liabilities are categorized as current and noncurrent.

Clear Lake Sporting Goods has cash, accounts receivable, inventory, short-term investments, and equipment. It rents its facilities, so it has no buildings on its balance sheets. Of all its assets, only the equipment is long term. The assets section for Clear Lake's classified balance sheet is shown in [Figure 5.7](#).

Clear Lake Sporting Goods Comparative Year-End Balance Sheets		
	Prior Year	Current Year
Assets		
Current Assets:		
Cash	\$ 90,000	\$110,000
Accounts Receivable	20,000	30,000
Inventory	35,000	40,000
Short-Term Investments	15,000	20,000
Total Current Assets	160,000	200,000
Noncurrent Assets:		
Equipment	40,000	50,000
Total Assets	<u>\$200,000</u>	<u>\$250,000</u>

Figure 5.7 Assets Section of Classified Balance Sheet

THINK IT THROUGH

Current and Noncurrent Assets

Visit the [Apple, Inc. Annual Report \(https://openstax.org/r/2020-doc-financial-annual-report\)](https://openstax.org/r/2020-doc-financial-annual-report) for 2020 and locate the company's balance sheet (the balance sheet begins on page 33). What types of current assets

does the company have? What types of noncurrent assets does it have? How has the total of each type of asset changed over time?

Solution:

Apple reports cash and cash equivalents, marketable securities, accounts receivable, inventories, vendor non-trade receivables, and “other” current assets on its balance sheet. The company’s current assets decreased from \$162,819 in 2019 to \$143,713 in 2020. Apple reports marketable securities, property, plant and equipment, and other noncurrent assets in the noncurrent asset section of its balance sheet. The company’s noncurrent assets increased from \$175,697 in 2019 to \$180,175 in 2020. (Amounts are in millions.)

We’ve covered the assets side of the accounting equation; now let’s turn our attention to the other side of the equation and the other two sections of the balance sheet: liabilities and equity. Just like the assets section, the liabilities section is broken down between current and noncurrent. **Current liabilities** are those that will be due within a year. Common examples of current liabilities include accounts payable, wages payable, and unearned revenue. **Noncurrent liabilities** are those that are due more than a year into the future. Notes payable is a common noncurrent liability.

Clear Lake Sporting Goods has accounts payable and has collected payments from a few customers that it hasn’t yet shipped its product to (unearned revenue). It also owes money on a note payable. Its accounts payable and unearned revenue are both current liabilities. The note payable is not due for several years, thus making it a noncurrent liability (see [Figure 5.8](#)).

	Prior Year	Current Year
Current Liabilities:		
Accounts Payable	\$ 60,000	\$ 75,000
Unearned Revenue	10,000	25,000
Total Current Liabilities	70,000	100,000
Noncurrent Liabilities:		
Notes Payable	40,000	50,000
Total Liabilities	<u>\$110,000</u>	<u>\$150,000</u>

Figure 5.8 Liability Section of Classified Balance Sheet

THINK IT THROUGH

Current and Noncurrent Liabilities

Visit the [Apple, Inc. Annual Report \(https://openstax.org/r/2020-doc-financial-annual-report\)](https://openstax.org/r/2020-doc-financial-annual-report) for 2020 and locate its balance sheet (the balance sheet begins on page 33). What types of current and noncurrent liabilities does the company have? How has the total of each type of liability changed over time?

Solution:

Apple has accounts payable, deferred revenue, commercial paper, and term debt listed as current liabilities. Its current liabilities declined by only a small amount from 2019 to 2020 (\$105,718 to \$105,392).

The company has term debt and “other” listed as noncurrent liabilities, which increased from 2019 to 2020 (\$142,310 to \$153,157). (Amounts are in millions.)

The stockholders’ equity section of the balance sheet for corporations contains two primary categories of accounts. The first is contributed capital, which is funds paid in by owners. The second category is earned

capital, which is funds earned by the corporation as part of business operations. On the balance sheet, **retained earnings** is a key component of the earned capital section, while the stock accounts such as common stock, preferred stock, and additional paid-in capital are the primary components of the contributed capital section.

Clear Lake Sporting Goods has just one contributed capital account—common stock—and one earned capital account—retained earnings. The equity section of its balance sheet is shown in [Figure 5.9](#).

	Prior Year	Current Year
Stockholder Equity		
Common Stock	\$ 75,000	\$ 80,000
Ending Retained Earnings	15,000	20,000
Total Stockholders' Equity	90,000	100,000
Total Liabilities and Stockholder Equity	<u>\$200,000</u>	<u>\$250,000</u>

Figure 5.9 Stockholders' Equity Section of Classified Balance Sheet

THINK IT THROUGH

Assets, Liability, and Equity

Visit the [Apple, Inc. Annual Report \(https://openstax.org/r/2020-doc-financial-annual-report\)](https://openstax.org/r/2020-doc-financial-annual-report) for 2020 and locate the company's balance sheet (the balance sheet begins on page 33). What is the amount of the company's total assets for the most recent year? What is the amount of its total liabilities and equity?

Solution:

Apple's total assets for 2020 were \$323,888, and its total liabilities and equity were also \$323,888. (Amounts are in millions.)

Importance of the Balance Sheet

Now that we have gone to all the work to carefully assemble a classified balance sheet, what do we use it for? The answer lies within the accounting equation itself. Think of the accounting equation from a “sources and claims” perspective. Under this approach, the assets (items owned by the organization) were obtained by incurring liabilities or were provided by owners. Stated differently, every asset has a claim against it—by creditors and/or owners. The balance sheet shows us what the firm has (its assets), who owns them (equity), and who the firm owes (its liabilities).

Limitations of the Balance Sheet

The balance sheet is indeed a very helpful financial statement, but it also poses challenges. First, assets on the balance sheet, under generally accepted accounting principles (GAAP), are recorded at historical cost. Historical cost is simply the cost paid for the item at the time it was purchased. Changes in market value of big-ticket items like land or buildings are not reflected in the balance sheet. Land remains at historical cost, and depreciable items like buildings are reflected at their current book value (historical cost less accumulated depreciation). If the asset has appreciated over time, the higher market value of the assets would not be seen on the balance sheet.

Estimates are another limitation of the balance sheet. Items on the balance sheet such as allowance for doubtful accounts and allowance for bad debt are based on estimates. The useful lives for calculating depreciation is another common estimate. If these estimates are incorrect, the net value of the asset can be under- or overstated.

Another key limitation is the fact that a balance sheet reflects balances at only one given point in time. This

means that the account value could have been quite different on the day before or the day after the date of the balance sheet. For example, if a firm were concerned with certain ratios or investor/lender expectations of its cash balance, it could choose to not pay several vendor payments in the last week of December. Thus, on December 31, the firm reflects a high cash balance on its balance sheet. However, by the end of the first week of January, it has caught up on late vendor payments and again shows a low cash balance.

Finally, there are many possible things of value that are not recorded on the balance sheet. Internally generated assets and the firm's human capital are two common examples. Internally generated assets can be anything from a website, a process, to an idea.

5.3 The Relationship between the Balance Sheet and the Income Statement

Learning Outcomes

By the end of this section, you will be able to:

- Identify connected elements between the balance sheet and the income statement.
- Differentiate between expenses and payables.

Net Income and Retained Earnings

As mentioned earlier, the financial statements are linked by certain elements and thus must be prepared in a certain order. The income statement was first since net income (or loss) is a required figure in preparing the balance sheet. During the period close process, all temporary accounts are closed to the income summary account, which is then closed to retained earnings. All revenue and expense accounts are closed since they are temporary. The net result is either net profit or net loss as the balance in the income summary account.

Remember that the retained earnings account reflects all income the firm has earned since its inception less any **dividends** paid out to shareholders. Thus the result (net income) of the income statement feeds the retained earnings account on the balance sheet. Retained earnings is also an element of the statement of stockholders' equity, which we will cover later in this chapter.

In [Figure 5.10](#), we see net income in the current year of \$35,000, which was added to the company's prior year retained earnings balance of \$15,000. Notice, however, that the prior year balance was \$15,000, and the current year balance is only \$20,000. It does not total \$50,000 as we might have expected.

Remember, retained earnings represents all earnings since inception less any dividends paid out. Clear Lake Sporting Goods must have paid out \$30,000 in dividends in the current year. We will see this information laid out in the statement of retained earnings. In the prior year they began with a \$10,000 balance in retained earnings. Income of \$30,000 increased retained earnings and dividends paid back out to investors reduced retained earnings, leaving an ending balance in the prior year of \$15,000. This rolls over and is the beginning balance for the current year. In the current year Clear Lake had net income of \$35,000 and paid \$30,000 of their earnings out to shareholders, essentially resulting in a \$5,000 increase to the retained earnings account.

Clear Lake Sporting Goods Statement of Retained Earnings		
	Prior Year	Current Year
Beginning Retained Earnings Balance Jan 1	\$10,000	\$15,000
Plus Net Income	30,000	35,000
Less Dividends Paid to Shareholders	25,000	30,000
Ending Retained Earnings Dec 31	<u>\$15,000</u>	<u>\$20,000</u>

Figure 5.10 Statement of Retained Earnings

Now we can see the full flow of information from the income statement to the statement of retained earnings ([Figure 5.10](#)) and finally to the balance sheet. Clear Lake's net income flows from the income statement into

retained earnings, which is reflected on the statement of retained earnings. The balance in retained earnings is then reflected on the balance sheet. This flow is depicted in Figure 5.11.

Clear Lake Sporting Goods Comparative Year-End Income Statements			Clear Lake Sporting Goods Comparative Year-End Balance Sheets		
	Prior Year	Current Year		Prior Year	Current Year
Gross Sales	\$105,000	\$126,000	Assets		
Sales Returns & Allowances	5,000	6,000	Current Assets:		
Net Sales	100,000	120,000	Cash	\$ 90,000	\$110,000
Cost of Goods Sold	50,000	60,000	Accounts Receivable	20,000	30,000
Gross Margin	50,000	60,000	Inventory	35,000	40,000
Rent Expense	5,000	5,500	Short-Term Investments	15,000	20,000
Depreciation Expense	2,500	3,600	Total Current Assets	160,000	200,000
Salaries Expense	3,000	5,400	Noncurrent Assets:		
Utility Expense	1,500	2,500	Equipment	40,000	50,000
Operating Income	38,000	43,000	Total Assets	200,000	250,000
Interest Expense	3,000	2,000	Current Liabilities:		
Income Tax Expense	5,000	6,000	Accounts Payable	60,000	75,000
Net Income	\$ 30,000	\$ 35,000	Unearned Revenue	10,000	25,000
			Total Current Liabilities	70,000	100,000
			Noncurrent Liabilities:		
			Notes Payable	40,000	50,000
			Total Liabilities	110,000	150,000
			Stockholder Equity		
			Common Stock	75,000	80,000
			Ending Retained Earnings	15,000	20,000
			Total Stockholders' Equity	90,000	100,000
			Total Liabilities and Stockholder Equity	\$200,000	\$250,000

Clear Lake Sporting Goods Statement of Retained Earnings		
	Prior Year	Current Year
Beginning Retained Earnings Balance Jan 1	\$10,000	\$15,000
Plus Net Income	30,000	35,000
Less Dividends Paid to Shareholders	25,000	30,000
Ending Retained Earnings Dec 31	<u>\$15,000</u>	<u>\$20,000</u>

Figure 5.11 Connections between Clear Lake Sporting Goods' Balance Sheet and Income Statement

LINK TO LEARNING

Apple's Income Statement and Balance Sheet

Visit the [Apple, Inc. Annual Report \(https://openstax.org/r/2020-doc-financial-annual-report\)](https://openstax.org/r/2020-doc-financial-annual-report) for 2020 and locate the company's income statement and balance sheet (they begin on page 31). What is the company's net income for the most recent year? What is the amount of retained earnings on the balance sheet for the current and prior years? Does it appear the company paid out dividends this year?

Expenses versus Payables

There is another key relationship between the income statement and the balance sheet can often be confusing to non-accountants: an expense versus a payable. The two are often assumed to be the same thing. However, it is important to note that the two are distinctly different.

Let's look at an example to outline the key differences. Clear Lake Sporting Goods incurred utility expenses during the current period (electric and gas). Its utilities totaled \$1,500. In the month that followed, the utilities vendor sent an invoice for \$1,500. Clear Lake has incurred an expense. It will reflect an expense of \$1,500 on the income statement for the utilities expense. This is the income statement impact of the transaction. So is it safe to assume that because Clear Lake has an expense, it also used cash? Not necessarily. Or is it safe to assume that if the company has an expense, it is the same as a payable? Again, the answer is no.

Remember, the accounting equation rests on the foundation of the double-entry accounting system. This means that every transaction has two sides: a debit and a credit. They must be equal. When Clear Lake records an expense of \$1,500, it must also record the other half of that transaction. In this case, the company incurred utilities expenses throughout the period "on account," which means it records an increase in their accounts payable. When the invoice comes due, another transaction must then be recorded to reduce accounts payable and reduce cash. Accounts payable is a liability found on the balance sheet, normally a current liability. The expense incurred caused the payable, but it is distinctly separate from the payable (see [Figure 5.12](#)).

Clear Lake Sporting Goods Comparative Year-End Income Statements			Clear Lake Sporting Goods Comparative Year-End Balance Sheets		
	Prior Year	Current Year		Prior Year	Current Year
Gross Sales	\$105,000	\$126,000	Assets		
Sales Returns & Allowances	5,000	6,000	Current Assets:		
Net Sales	100,000	120,000	Cash	\$ 90,000	\$110,000
Cost of Goods Sold	50,000	60,000	Accounts Receivable	20,000	30,000
Gross Profit	50,000	60,000	Inventory	35,000	40,000
Rent Expense	5,000	5,500	Short-Term Investments	15,000	20,000
Depreciation Expense	2,500	3,600	Total Current Assets	160,000	200,000
Salaries Expense	3,000	5,400	Equipment	40,000	50,000
Utility Expense	1,500	2,500	Total Assets	200,000	250,000
Operating Income	38,000	43,000	Current Liabilities:		
Interest Expense	3,000	2,000	Accounts Payable	60,000	75,000
Income Tax Expense	5,000	6,000	Unearned Revenue	10,000	25,000
Net Income	\$ 30,000	\$ 35,000	Total Current Liabilities	70,000	100,000
			Noncurrent Liabilities:		
			Notes Payable	40,000	50,000
			Total Liabilities	110,000	150,000
			Stockholder Equity		
			Common Stock	75,000	80,000
			Ending Retained Earnings	15,000	20,000
			Total Stockholders' Equity	90,000	100,000
			Total Liabilities and Stockholder Equity	\$200,000	\$250,000

Figure 5.12 Connections between Expenses and Accounts Payable

5.4 The Statement of Owner's Equity

Learning Outcomes

By the end of this section, you will be able to:

- Explain the concept of owner's equity.
- Outline the purpose and importance of the statement of owner's equity.
- Identify the structure and key elements of the statement of owner's equity.

What Is Equity?

Recall that the accounting equation can help us see what is owned (assets), who is owed (liabilities), and finally who the owners are (equity). The statement of owner's equity addresses the last segment of the accounting equation in detail by laying out the equity elements of the firm and highlighting changes in these elements throughout the period.

Equity represents the ownership of the firm. The stockholders' equity section of the balance sheet for corporations contains two primary categories of accounts. The first is paid-in capital or contributed capital—consisting of amounts paid in by owners. The second category is earned capital, consisting of amounts earned by the corporation as part of business operations. On the balance sheet, retained earnings is a key component of the earned capital section, while the stock accounts such as common stock, preferred stock, and additional paid-in capital are the primary components of the contributed capital section.

Common stock represents ownership in the firm. Common stockholders normally have voting rights. Preferred stock has unique rights that are “preferred,” or more advantageous, to shareholders than common stock. Unlike common stockholders, preferred shareholders typically do not have voting rights and do not share in the common stock dividend distributions. Instead, the “preferred” classification entitles shareholders to a dividend that is fixed (assuming sufficient dividends are declared). Treasury stock is shares that were outstanding and have been repurchased by the firm but not retired. Thus they are still issued, but not outstanding. Additional paid-in capital is the difference between the issue price and par value of the common stock. For example, if a firm issued and sold stock at a market price of \$20 that had a \$5 par value, \$5 for each share would be recorded into common stock and the excess \$15 per share would be recorded into the additional paid-in capital account.

If we review the balance sheet for Clear Lake Sporting Goods, we see just two elements of equity: common stock and retained earnings. Common stock in the prior year was \$75,000 and increased to \$80,000 in the current year, indicating Clear Lake issued additional common stock (see [Figure 5.13](#)).

Clear Lake Sporting Goods Comparative Year-End Balance Sheets		
	Prior Year	Current Year
Stockholder Equity		
Common Stock	\$ 75,000	\$ 80,000
Ending Retained Earnings	15,000	20,000
Total Stockholders' Equity	90,000	100,000
Total Liabilities and Stockholder Equity	<u>\$200,000</u>	<u>\$250,000</u>

Figure 5.13 Stockholder Equity Section of Balance Sheet

THINK IT THROUGH

Equity Accounts

Visit the [Apple, Inc. Annual Report \(https://openstax.org/r/2020-doc-financial-annual-report\)](https://openstax.org/r/2020-doc-financial-annual-report) for 2020 and

locate the company's balance sheet (it begins on page 33). What types of equity accounts does it report?

Solution:

Apple reports common stock, retained earnings, and accumulated other comprehensive income.

Distributions to Owners

When firms earn a profit, they have two options as to what to do with their earnings. They can keep (retain) them and reinvest them back into the business, or they can pay them out to their shareholders in the form of dividends. Dividends are commonly in the form of cash, but dividends can be paid out in the form of stock or other assets as well.

To pay a cash dividend, the firm must have enough cash on hand and sufficient retained earnings. They cannot pay out a dividend beyond the retained earnings available. Some companies issue shares of stock as a dividend rather than cash or property. This often occurs when the company has insufficient cash but wants to keep its investors happy. When a company issues a stock dividend, it distributes additional shares of stock to existing shareholders.

A property dividend occurs when the firm pays out dividends in the form of something other than stock or cash, often one of their assets or something they hold in inventory. For example, Walt Disney Company may choose to distribute tickets to visit its theme parks. A property dividend may be declared when a company wants to reward its investors but doesn't have the cash to distribute, or if it needs to hold on to its existing cash for other investments.

Remember, the retained earnings account reflects the cumulative earnings of a firm since they began business, less dividends paid out to shareholders. This includes all forms of dividends (cash, stock, and other assets). Note that dividends are distributed or paid only to shares of stock that are outstanding. Treasury shares are not outstanding, so no dividends are declared or distributed for these shares. Regardless of the type of dividend, the declaration always causes a decrease in the retained earnings account.

THINK IT THROUGH

Dividends

Visit the [Apple, Inc. Annual Report \(https://openstax.org/r/2020-doc-financial-annual-report\)](https://openstax.org/r/2020-doc-financial-annual-report) for 2020. Review the notes to the financial statements found on pages 19 and 20. What kind of dividends did the company pay (cash, property, stock)? Review the notes regarding dividends on page 26. What does Apple intend to do regarding dividends in the future, pending board approval?

Solution:

Apple issued cash dividends. On page 26, it notes that the company intends to increase the dividend annually, pending approval by the board.

Elements of the Statement of Owner's Equity

Now that we have covered the basic elements of equity and know what dividends are, we have the basic pieces we need to create the statement of **owner's equity**. The statement is broken out into columns, one for each element of equity: common stock, preferred stock, additional paid-in capital, retained earnings, and treasury stock.

The first line of the statement provides the balance of each segment as of the first day of the period. Each

following line provides information on any events during the period that changed the value of any of the accounts. Common examples of events found on the statement include net income or loss for the period, issuing common or preferred stock, purchasing or selling treasury stock, and declaring a dividend.

Clear Lake Sporting Goods has just common stock and retained earnings to report in their statement of owner's equity. They had just two events to report in their statement that impacted their equity accounts; they reported net income and they issued dividends (see [Figure 5.14](#)).

	Common Stock	Additional Paid-in Capital	Retained Earnings	Treasury Stock	Total
Beginning Balance	\$75,000	\$0	\$15,000	\$0	\$ 90,000
Net Income			35,000		35,000
Dividends Paid to Shareholders			(30,000)		(30,000)
Common Stock Issued	5,000				5,000
Ending Balance	\$80,000	\$0	\$20,000	\$0	\$100,000

Figure 5.14 Statement of Stockholder Equity for Clear Lake Sporting Goods

LINK TO LEARNING

Cash Flows

[Khan Academy \(https://openstax.org/l/50CashFlowsVid\)](https://openstax.org/l/50CashFlowsVid) explains cash flows in a unique way.

5.5 The Statement of Cash Flows

Learning Outcomes

By the end of this section, you will be able to:

- Outline the purpose and importance of the statement of cash flows.
- Identify the structure and key elements of the statement of cash flows.

The final financial statement is the statement of cash flows. It is a crucial statement, as it shows the sources of and uses of cash for the firm during the accounting period. Remember, under accrual accounting, transactions are recorded when they occur, not necessarily when cash moves. Thus, the income statement does not provide all the insights necessary to understand a firm's cash flows. To fully understand the firm's flow of cash, the statement of cash flows is needed.

Importance of the Statement of Cash Flows

Remember, most firms use accrual accounting. Revenues and expenses are recorded when they occur, not necessarily when cash moves. This can create timing differences between profits and cash flows. A firm can be profitable and still not have an adequate flow of cash. The opposite is also true: it can experience a net loss and still have cash on hand. Earning a profit is wonderful, but it is not the only goal an organization has. It must also have adequate cash flow to support daily operations. To support cash planning and to provide external financial statement users such as lenders and investors information about the firm's cash flow, the statement of cash flows is prepared. Cash flow is also a crucial metric for determining the value of a company.

External financial statement users also rely on the statement of cash flows to help them evaluate the quality of the firm's earnings. Users compare earnings to cash flow to assess the validity of the earnings data. For example, a firm reporting a strong profit but very little cash flow might raise some questions as to what was recorded to drive profits that isn't also driving cash flows.

The statement of cash flows also helps external users determine the driving forces behind the firm's cash flows. They can see if cash is generated primarily by daily operations or if cash is being generated or

consumed by events outside the firm's normal course of business.

There are two key methods of preparing the statement: direct and indirect. FASB (Financial Accounting Standards Board) favors the **direct method**. Despite that, the most common method used by far in general practice is the **indirect method**. The direct method lists cash flows directly from revenues and expenses, whereas the indirect method reconciles income to cash flows. The indirect method begins with net income and reconciles each account in order to arrive at net cash flow. It essentially reconciles **accrual basis** accounting to **cash basis**, or cash flow.

Operating Activities

To provide clear information about what areas of the business generated and used cash, the statement of cash flows is broken down into three key categories: operating, financing, and investing. The operating section reflects cash flows generated by and used by the day-to-day operations of the business. Investing activities include investments in other firms as well as investments in the firm itself (items like machinery, land, or other fixed assets). Finally, financing activities are those used to provide funds to run the business (loans, interest).

As mentioned, operating activities are those that are used or generated by the day-to-day operations of the firm. The operating activities section of the statement of cash flows begins with net income. All lines thereafter, in that section, are then adjustments to reconcile net income to actual cash flows by adding back **noncash expenses** like depreciation and adjusting for changes in asset and liability accounts. For example, depreciation is a noncash expense. Thus, depreciation is added back to net income.

To prepare the statement of cash flows for Clear Lake Sporting Goods, we need the beginning cash balance from the balance sheet, net income and depreciation expense from the income statement, and a set of comparative balance sheets to see the change in asset and liability accounts (see [Figure 5.15](#)).

Clear Lake Sporting Goods Comparative Year-End Income Statements			Clear Lake Sporting Goods Comparative Year-End Balance Sheets		
	Prior Year	Current Year		Prior Year	Current Year
Gross Sales	\$105,000	\$126,000	Assets		
Sales Returns & Allowances	5,000	6,000	Current Assets:		
Net Sales	100,000	120,000	Cash	\$ 90,000	\$110,000
Cost of Goods Sold	50,000	60,000	Accounts Receivable	20,000	30,000
Gross Margin	50,000	60,000	Inventory	35,000	40,000
Rent Expense	5,000	5,500	Short-Term Investments	15,000	20,000
Depreciation Expense	2,500	3,600	Total Current Assets	160,000	200,000
Salaries Expense	3,000	5,400	Noncurrent Assets:		
Utility Expense	1,500	2,500	Equipment	40,000	50,000
Operating Income	38,000	43,000	Total Assets	200,000	250,000
Interest Expense	3,000	2,000	Current Liabilities:		
Income Tax Expense	5,000	6,000	Accounts Payable	60,000	75,000
Net Income	\$ 30,000	\$ 35,000	Unearned Revenue	10,000	25,000
			Total Current Liabilities	70,000	100,000
			Noncurrent Liabilities:		
			Notes Payable	40,000	50,000
			Total Liabilities	110,000	150,000
			Stockholder Equity		
			Common Stock	75,000	80,000
			Ending Retained Earnings	15,000	20,000
			Total Stockholders' Equity	90,000	100,000
			Total Liabilities and Stockholder Equity	\$200,000	\$250,000

Figure 5.15 Comparative Income Statements and Balance Sheets

Clear Lake's statement of cash flows begins with the current year net income of \$35,000 from the income statement. Next, noncash expenses are deducted. Clear Lake's only noncash expense on their current year income statement is depreciation of \$3,600. Since depreciation is an expense that reduces income but is not actually paid out in cash in the current period, it must be added back to net income to reconcile net income to cash flow.

Next, changes in operational assets and liabilities are used to continue reconciling net income to actual cash flow. For example, Clear Lake's accounts receivable increased from the prior period to the current period. This means that there were more sales recorded but not yet received in cash in this period than there were in the prior period, making an increase in accounts receivable a reduction on the statement. Inventory increased, which means additional cash was spent to acquire it, making it a use of cash or reduction to net income to move closer to cash. Accounts payable and unearned revenue, both liability accounts, increased. Since these are liabilities, an increase would indicate that the liability was incurred but not as quickly paid out; thus it is an increase to the statement.

Tallying all these adjustments to net income shows Clear Lake's net cash flows provided by operating activities of \$53,600 (see [Figure 5.16](#)).

Clear Lake Sporting Goods Statement of Cash Flows Indirect Method		
Cash Flow from Operating Activities		
Net Income		\$35,000
Adjustment to Reconcile Net Income to Net Cash Flow from Operating Activities:		
Depreciation	\$ 3,600	
Accounts Receivable Increase	(10,000)	
Inventory Increase	(5,000)	
Accounts Payable Increase	15,000	
Unearned Revenue Increase	15,000	18,600
Net Cash Flow from Operating Activities		<u>\$53,600</u>

Figure 5.16 Operating Activities Section of the Statement of Cash Flows

Investing Activities

As mentioned, investing activities include investments in other firms as well as investments in the firm itself (items like machinery, land, or other fixed assets). These are items that are capitalized (placed on the balance sheet and depreciated over time) and thus did not reduce net income. They did, however, cause an impact to cash flow (see [Figure 5.17](#)).

During the current year, Clear Lake purchased an additional \$5,000 in short-term investments (see the comparative balances sheets; the balance in that account increased by \$5,000 since the prior year). They also purchased additional plant assets in the amount of \$13,600. This amount can be figured by comparing the difference in the current and prior plant assets accounts and including the impact of current year depreciation (\$50,000 current year balance less \$40,000 prior year balance and \$3,600 of depreciation = \$13,600 of new assets purchased).

Cash Flow from Investing Activities		
Purchase of Short-Term Investments	\$(5,000)	
Cost of New Plant Assets	(13,600)	
Net Cash Flow: Investing Activities		<u>\$(18,600)</u>

Figure 5.17 Investing Activities Section of the Statement of Cash Flows

Financing Activities

Recall that financing activities are those used to provide funds to run the business. Common items in this section of the statement include the payment of dividends, issuance of common or preferred stock, and issuance or payment of notes payable (see [Figure 5.18](#)).

In the current year, Clear Lake took out additional notes payable (a cash inflow). We can see this by the increase in their notes payable account from the prior year to current year (\$40,000 to \$50,000). This is an inflow of cash and thus an increase on the statement. Dividends of \$30,000 were paid to shareholders (found on the statement of retained earnings and the statement of owner's equity). Finally, we see that Clear Lake must have issued additional common stock, as their common stock balance increased from \$75,000 to \$80,000.

Cash Flow from Financing Activities		
Additional Notes Payable	\$10,000	
Issuance of Common Stock	5,000	
Payment of Dividends	(30,000)	
Net Cash Flow from Financing Activities		<u>\$(15,000)</u>

Figure 5.18 Financing Activities Section of the Statement of Cash Flows

LINK TO LEARNING

Statement of Cash Flows

Visit the [Apple, Inc. Annual Report \(https://openstax.org/r/2020-doc-financial-annual-report\)](https://openstax.org/r/2020-doc-financial-annual-report) for 2020 and locate the company's statement of cash flows (it begins on page 35). What is the cash provided by/used by operating activities? Review the key adjustments in the operating section. What are the biggest adjustments? Review the investing section of the statement. Did the company have a net cash inflow or net cash used by investing? What key items appear in this section? What was the net cash provided by or used by financing activities? What key items appear in this section for Apple?

The final task to wrap up the statement of cash flows is to tally net cash generated or used by summing all three sections. This amount is then used to adjust the beginning cash balance from the balance sheet. Assuming the statement was prepared correctly, the sum should equal the ending cash balance on the balance sheet.

In the full statement, we can see that Clear Lake has net cash flow of \$20,000. The beginning cash balance was \$90,000, making the ending cash balance \$110,000 (see [Figure 5.19](#)).

Clear Lake Sporting Goods Statement of Cash Flows Indirect Method			
Cash Flow from Operating Activities			
Net Income			\$ 35,000
Adjustment to Reconcile Net Income to Net Cash Flow from Operating Activities:			
Depreciation	\$ 3,600		
Accounts Receivable Increase	(10,000)		
Inventory Increase	(5,000)		
Accounts Payable Increase	15,000		
Unearned Revenue Increase	15,000	18,600	
Net Cash Flow from Operating Activities		53,600	
Cash Flow from Investing Activities			
Purchase of Short-Term Investment	(5,000)		
Cost of New Plant Assets	(13,600)		
Net Cash Flow: Investing Activities		(18,600)	
Cash Flow from Financing Activities			
Additional Notes Payable	10,000		
Issuance of Common Stock	5,000		
Payment of Dividends	\$(30,000)		
Net Cash Flow from Financing Activities		(15,000)	
Total Cash Flow Increase/Decrease		20,000	
Beginning Cash Balance		90,000	
Ending Cash Balance		\$110,000	

Figure 5.19 Full Statement of Cash Flows

Analyzing Performance

The statement of cash flows can be used in a number of ways to assess firm performance by both internal and external financial statement users. Internal users can assess sources of and uses of cash in order to aid in adapting, as necessary, to ensure adequate future cash flows. External users also use the statement. Recall that comparing net income to operational cash flows can help assess the quality of earnings. In the next section you'll explore operating cash flow and **free cash flow** to the firm, two key points of analysis in

assessing cash flows.

5.6 Operating Cash Flow and Free Cash Flow to the Firm (FCFF)

Learning Outcomes

By the end of this section, you will be able to:

- Calculate operating cash flow and free cash flow.
- Assess organizational cash management performance.

Operating Cash Flow

Now that we have a statement of cash flows prepared, we can move on to a few key elements of the statement used to assess organizational cash management performance. Operating cash flow, or net cash flow from operating activities, is calculated in the first section of the statement of cash flows. It depicts the cash generated (or used by) the primary business activities. Remember, operating cash flow is calculated under the indirect method by adjusting net income for noncash expenses like depreciation and adjustments for changes in current asset and liability accounts (changes in working capital).

$$\text{Operating Cash Flow} = \text{Operating Income} + \text{Depreciation} - \text{Taxes} + \text{Change in Working Capital}$$

Operating cash flow is helpful in assessing organizational cash management performance as it relates to the core business function—operations. Key management practices in this area can have a profound impact on the firm's cash flow. Practices and policies include customer payment terms, collection policies and practices, and vendor payment terms. Though changing a customer or vendor payment terms will not change the profit or loss for the firm, it will have an impact on the timing to cash flows. This is a key element of managing operational cash flow.

Free Cash Flow

Free cash flow (FCF) is calculated by taking operating cash flows less capital expenditures. Free cash flow is an important measure, as it depicts the cash available to support the business's operations and maintain its fixed assets. It is commonly used by investors as part of their overall evaluation of an investment, as it is a key measure of cash flow management practices and a firm's ability to generate enough cash to cover operations and capital assets and it shows if there is any left over for other considerations such as dividend payments, debt repayment, and contributions to increase working capital for future growth.

$$\text{Free Cash Flow (FCF)} = \text{Operating Cash Flow} - \text{Capital Expenditures}$$

Using the data for Clear Lake Sporting Goods, we can calculate its free cash flow as follows:

$$\text{Free Cash Flow (FCF)} = \$53,600 - \$13,600 = \$40,000$$

This means that Clear Lake Sporting Goods has \$40,000 of cash available to repay debt or pay cash dividends after having covered the cash needs of its operations and capital asset investments.

Managing Cash Flow

Managing cash flow is not an easy task. A firm has a myriad of places that its cash flows from or to. However, there are a few key areas to place attention in order to manage or improve cash flows. First, consider where cash is coming from—customers. Managing terms and collection efforts with customers can have a significant impact on a firm's cash flow. For example, a customer with terms of net 10 will likely yield payment quite quickly—10 days, give or take. A customer with terms of net 60, on the other hand, will require roughly 60 days to collect assuming they pay on time. The 50-day difference between these two examples means that the firm will go 50 additional days having expended the resources to provide the customer their good or service, but with no cash flow yet to cover it.

The same theory holds true on the opposite side with accounts payable and the vendors a firm uses. Accepting

net 10 terms requires the firm to give up its cash quickly, while pushing for more favorable terms like net 30 or net 60 allows the firm to wait much longer to give up its cash.

It is important to assess both sides of cash flow and the impact it has on the firm as well as the customer and vendor relationships it maintains. Some customers may not have difficulty negotiating a more favorable payment term in order for the firm to improve its cash flow. Others, however, may not be willing to accept shorter payment terms. In order to be competitive in the industry, the firm must assess the customer relationships, industry standards, and its own ability to support cash flow when considering its customer payment terms.

When there is a gap in cash flow, it is crucial that it is recognized early, with proper cash flow forecasting, so financing can be obtained to bridge the gap. A common tool used to manage the ebb and flow of cash flow for a firm is an open line of credit with a bank. This allows the firm to borrow and repay from month to month as cash flow fluctuates.

Now that you've become more familiar with the four basic financial statements, let's move on to a tool helpful in evaluating them: common-size statements. Common-size financial statements, also termed vertical analysis, restate the financial statement items as a percentage of a base item. Doing so helps reveal relationships between items, aids in assessing performance over time, and makes it easier to compare one company to another, regardless of size (thus the name common-size).

5.7 Common-Size Statements

Learning Outcomes

By the end of this section, you will be able to:

- Prepare common-size statements.
- Assess organizational performance using common-size statements.
- Use industry comparisons to assess organizational performance.

Common-Size Income Statements

A common-size income statement is created by restating each line as a percentage of net sales. Expressing each item on the income statement as a percentage rather than in absolute dollars makes it much easier to make comparisons, particularly to other divisions or competitors of varying sizes. The formula to calculate each item on the income statement is:

$$\text{Common-Size Item} = \frac{\text{Income Statement Line Item}}{\text{Net Sales}}$$

Using Clear Lake Sporting Goods' current year income statement, we can see how each line item in it is divided by net sales in order to assemble a common-size income statement (see [Figure 5.20](#)).

Clear Lake Sporting Goods Comparative Year-End Income Statements			
	Current Year	Current Year %	Formula
Gross Sales	\$126,000		
Sales Returns & Allowances	6,000		
Net Sales	120,000	100%	=120,000/120,000
Cost of Goods Sold	60,000	50%	=60,000/120,000
Gross Margin	60,000	50%	=60,000/120,000
Rent Expense	5,500	5%	=5,500/120,000
Depreciation Expense	3,600	3%	=3,600/120,000
Salaries Expense	5,400	5%	=5,400/120,000
Utility Expense	2,500	2%	=2,500/120,000
Operating Income	43,000	36%	=43,000/120,000
Interest Expense	2,000	2%	=2,000/120,000
Income Tax Expense	6,000	5%	=6,000/120,000
Net Income	\$ 35,000	29%	=35,000/120,000

Figure 5.20 Common-Size Income Statement

It may seem cumbersome to create a common-size statement. However, a simple tool like Microsoft Excel can be quite handy in making the process easier and faster. The same formula can be copied and replicated in each income statement line, making the calculations much faster. In [Figure 5.21](#), you can see the formulas used to create Clear Lake Sporting Goods' common-size income statement in Excel. Notice that the \$ can be inserted to anchor a cell reference, making it easier to copy and paste the same formula onto many lines or columns.

	A	B	C	D	E	F	G
		Prior Year	Prior Year %	Formula	Current Year	Current Year %	Formula
1							
2	Gross Sales	\$ 105,000			\$ 126,000		
3	Sales Returns	5,000			6,000		
4	Net Sales	100,000	100%	=B4/\$B\$4	120,000	100%	=E4/\$E\$4
5	Cost of Goods Sold	50,000	50%	=B5/\$B\$4	60,000	50%	=E5/\$E\$4
6	Gross Margin	50,000	50%	=B6/\$B\$4	60,000	50%	=E6/\$E\$4
7	Rent Expense	5,000	5%	=B7/\$B\$4	5,500	5%	=E7/\$E\$4
8	Depreciation Expense	2,500	3%	=B8/\$B\$4	3,600	3%	=E8/\$E\$4
9	Salaries Expense	3,000	3%	=B9/\$B\$4	5,400	5%	=E9/\$E\$4
10	Utility Expense	1,500	2%	=B10/\$B\$4	2,500	2%	=E10/\$E\$4
11	Operating Income	38,000	38%	=B11/\$B\$4	43,000	36%	=E11/\$E\$4
12	Interest Expense	3,000	3%	=B12/\$B\$4	2,000	2%	=E12/\$E\$4
13	Income Tax Expense	5,000	5%	=B13/\$B\$4	6,000	5%	=E13/\$E\$4
14	Net Income	\$ 30,000	30%	=B14/\$B\$4	\$ 35,000	29%	=E14/\$E\$4

Figure 5.21 Clear Lake Sporting Goods Common-Size Income Statements with Excel Formulas

Common-Size Balance Sheet

The common-size balance sheet functions much like the common-size income statement. Each line item on the balance sheet is restated as a percentage of total assets.

$$\text{Common-Size Item} = \frac{\text{Balance Sheet Line Item}}{\text{Total Assets}}$$

Using Clear Lake Sporting Goods' current balance sheet, we can see how each line item in its statement is divided by total assets in order to assemble a common-size balance sheet (see [Figure 5.22](#)).

Clear Lake Sporting Goods Comparative Year-End Balance Sheets			
	Current Year	Current Year %	Formula
Assets			
Current Assets:			
Cash	\$110,000	44%	=110,000/250,000
Accounts Receivable	30,000	12%	=30,000/250,000
Inventory	40,000	16%	=40,000/250,000
Short-Term Investments	20,000	8%	=20,000/250,000
Total Current Assets	<u>200,000</u>	80%	=200,000/250,000
Noncurrent Assets:			
Equipment	<u>50,000</u>	20%	=50,000/250,000
Total Assets	<u>250,000</u>	100%	=250,000/250,000
Current Liabilities:			
Accounts Payable	75,000	30%	=75,000/250,000
Unearned Revenue	<u>25,000</u>	10%	=25,000/250,000
Total Current Liabilities	<u>100,000</u>	40%	=100,000/250,000
Noncurrent Liabilities:			
Notes Payable	<u>50,000</u>	20%	=50,000/250,000
Total Liabilities	<u>150,000</u>	60%	=150,000/250,000
Stockholder Equity			
Common Stock	80,000	32%	=80,000/250,000
Ending Retained Earnings	<u>20,000</u>	8%	=20,000/250,000
Total Stockholders' Equity	<u>100,000</u>	40%	=100,000/250,000
Total Liabilities and Stockholder Equity	<u>\$250,000</u>	100%	=250,000/250,000

Figure 5.22 Common-Size Balance Sheet

Excel can also be used to create a common-size balance sheet. Once the formula is created, it can be copied into each line, making the process to create a common-size statement much easier (see [Figure 5.23](#)):

	A	B	C	D
1		Current Year	Current Year %	Formula
2	Assets			
3	Current Assets:			
4	Cash	\$ 110,000	44%	=B4/B\$11
5	Accounts Receivable	30,000	12%	=B5/B\$11
6	Inventory	40,000	16%	=B6/B\$11
7	Short-Term Investments	20,000	8%	=B7/B\$11
8	Total Current Assets	200,000	80%	=B8/B\$11
9	Noncurrent Assets:			
10	Equipment	50,000	20%	=B10/B\$11
11	Total Assets	<u>\$ 250,000</u>	100%	=B11/B\$11
12	Current Liabilities:			
13	Accounts Payable	75,000	30%	=B13/B\$11
14	Unearned Revenue	25,000	10%	=B14/B\$11
15	Total Current Liabilities	100,000	40%	=B15/B\$11
16	Noncurrent Liabilities:			
17	Notes Payable	50,000	20%	=B17/B\$11
18	Total Liabilities	<u>150,000</u>	60%	=B18/B\$11
19	Stockholder Equity			
20	Common Stock	80,000	32%	=B20/B\$11
21	Ending Retained Earnings	20,000	8%	=B21/B\$11
22	Total Stockholders' Equity	<u>100,000</u>	40%	=B22/B\$11
23	Total Liabilities and Stockholder Equity	<u>\$ 250,000</u>	100%	=B23/B\$11

Figure 5.23 Common-Size Balance Sheet with Excel Formulas

It is important to note that while we have provided two years of data here to explore the process, when performing analysis for a firm or investment, several years of data are commonly used to provide a better view of historical performance.

Analyzing Organizational Performance

Common-size financial statements facilitate the analysis of financial performance by converting each element of the statements to a percentage. This makes it easier to compare figures from one period to the next, compare departments within an organization, and compare the firm to other companies of any size as well as industry averages. On the income statement, analysts can see how much of sales revenue is spent on each type of expense. They can see this breakdown for each firm and compare how different firms function in terms of expenses, proportionally. They can also look at the percentage for each expense over time to see if they are spending more or less on certain areas of the business, such as research and development. On the balance sheet, analysts commonly look to see the percentage of debt and equity to determine capital structure. They can also quickly see the percentage of current versus noncurrent assets and liabilities.

In Clear Lake Sporting Goods' common-size income statement for the current and prior years, we can see that cost of goods as a percentage of sales remained the same (see [Figure 5.24](#)). This means that while sales

increased, so did cost of goods sold, but it increased at the same proportion as sales. No improvement or decline occurred in the company's cost of goods sold. The same is true for rent, depreciation, and utilities expenses. One key item did change slightly as a percentage: salaries expense. The 2 percent decrease in operating income from the prior year's 38 percent to the current year's 36 percent was caused by the increase in salaries expense as a percentage of sales.

Net income, however, only declined by 1 percent from 30 percent in the prior year to 29 percent in the current year because interest expense dropped by 1 percent, offsetting the 2 percent increase in salaries expense.

Clear Lake Sporting Goods Comparative Year-End Income Statements				
	Prior Year	Prior Year %	Current Year	Current Year %
Gross Sales	\$105,000		\$126,000	
Sales Returns & Allowances	5,000		6,000	
Net Sales	100,000	100%	120,000	100%
Cost of Goods Sold	50,000	50%	60,000	50%
Gross Profit	50,000	50%	60,000	50%
Rent Expense	5,000	5%	5,500	5%
Depreciation Expense	2,500	3%	3,600	3%
Salaries Expense	3,000	3%	5,400	5%
Utility Expense	1,500	2%	2,500	2%
Operating Income	38,000	38%	43,000	36%
Interest Expense	3,000	3%	2,000	2%
Income Tax Expense	5,000	5%	6,000	5%
Net Income	\$ 30,000	30%	\$ 35,000	29%

Figure 5.24 Common-Size Income Statement

LINK TO LEARNING

Common-Size Income Statement

Visit the [Apple, Inc. Common-Size Income Statement \(https://openstax.org/r/common-size-income-statement\)](https://openstax.org/r/common-size-income-statement) provided by the Stock Analysis on Net website. Review the company's cost of sales, gross profit, operating income, other income, and net income percentages for the current and prior year. What can you learn about the company's recent performance in these areas? Has it improved or declined, or has it simply changed incrementally with gross sales?

On the Clear Lake Sporting Goods' common-size balance sheet, we see that current assets remained at 80 percent of total assets from the prior to current year (see [Figure 5.25](#)). The mix of current assets that comprise that 80 percent changed only slightly with a 1 percent decrease in cash, 2 percent increase in accounts receivable, 2 percent decrease in inventory, and no change in short-term investments. Noncurrent assets includes only equipment. While the balance in the equipment account did change as a percentage of total assets, equipment remained the same at 20 percent.

On the debt and equity side of the balance sheet, however, there were a few percentage changes worth noting. In the prior year, the balance sheet reflected 55 percent debt and 45 percent equity. In the current year, that balance shifted to 60 percent debt and 40 percent equity. The firm did issue additional stock and showed an increase in retained earnings, both totaling a \$10,000 increase in equity. However, the equity increase was much smaller than the total increase in liabilities of \$40,000. Long-term debt increased by only \$10,000 by issuing additional notes payable. The remainder of that increase is seen in the 5 percent increase in current liabilities. In that increase, most of it was in unearned revenue.

Clear Lake Sporting Goods Comparative Year-End Balance Sheets				
	Prior Year	Prior Year %	Current Year	Current Year %
Assets				
Current Assets:				
Cash	\$ 90,000	45%	\$110,000	44%
Accounts Receivable	20,000	10%	30,000	12%
Inventory	35,000	18%	40,000	16%
Short-Term Investments	15,000	8%	20,000	8%
Total Current Assets	<u>160,000</u>	80%	<u>200,000</u>	80%
Noncurrent Assets:				
Equipment	<u>40,000</u>	20%	<u>50,000</u>	20%
Total Assets	<u>200,000</u>	100%	<u>250,000</u>	100%
Current Liabilities:				
Accounts Payable	60,000	30%	75,000	30%
Unearned Revenue	<u>10,000</u>	5%	<u>25,000</u>	10%
Total Current Liabilities	<u>70,000</u>	35%	<u>100,000</u>	40%
Noncurrent Liabilities:				
Notes Payable	<u>40,000</u>	20%	<u>50,000</u>	20%
Total Liabilities	<u>110,000</u>	55%	<u>150,000</u>	60%
Stockholder Equity				
Common Stock	75,000	38%	80,000	32%
Ending Retained Earnings	<u>15,000</u>	8%	<u>20,000</u>	8%
Total Stockholders' Equity	<u>90,000</u>	45%	<u>100,000</u>	40%
Total Liabilities and Stockholder Equity	<u>\$200,000</u>	100%	<u>\$250,000</u>	100%

Figure 5.25 Common-Size Balance Sheet

LINK TO LEARNING

Common-Size Assets and Common-Size Liabilities and Equity

Visit the [Apple, Inc. Common-Size Assets \(https://openstax.org/r/common-size-assets\)](https://openstax.org/r/common-size-assets) Balance Sheet and [Common-Size Liabilities and Equity \(https://openstax.org/r/common-size-liabilities\)](https://openstax.org/r/common-size-liabilities) Balance Sheet provided by the Stock Analysis on Net website. Review the composition of the company's assets, liabilities, and equity. How have assets changed? Has capital structure changed? If so, what elements impacted the change?

Industry Comparison

Recall that a key benefit of common-size analysis is comparing the firm's performance to the industry. Expressing the figures on the income statement and balance sheet as percentages rather than raw dollar figures allows for comparison to other companies regardless of size differences.

Clear Lake Sporting Goods, for example, might compare their financial performance on their income statement to a key competitor, Charlie's Camping World. Charlie is a much bigger retailer for outdoor gear, as Charlie has nearly seven times greater sales than Clear Lake. Common-size statements allow Clear Lake to compare their statements in a meaningful way (see [Figure 5.26](#)). Notice that Clear Lake spends 50 percent of its sales on cost of goods sold while Charlie spends 59 percent. This is a significant difference that would be an indicator that Clear Lake and Charlie have key differences in their operations, purchasing policies, or general performance in their core products.

We know that Charlie is a bigger retailer, and we see this clearly in the rent expense as a percentage of sales. Charlie spends 11 percent of its sales on building rent, while Clear Lake spends only 5 percent. A clear difference in performance is hinted at here, alluding to Charlie spending more per square foot on rent or using its retail space differently, causing it to rent more space for its product per sales dollar than Clear Lake does. Depreciation expense, though not a large figure, is smaller for Charlie, giving us a hint that Charlie has less capital equipment than Clear Lake, perhaps tied to the higher rent expense. It is possible that Charlie rents some of its equipment, which would help explain the higher rent percentage. Finally, Charlie's salaries percentage is significantly higher at 12 percent than Clear Lake's 5 percent. Though the simple percentage does not tell us why, it does provide us a hint and allow for further questions or investigation. Charlie may pay its employees a much higher wage than Clear Lake. They may also have far more sales associates on the floor in their larger spaces than Clear Lake does in their smaller retail spaces.

Note that although we have compared just two years of data for Charlie and Clear Lake, it is more common to use several years of data to get a more robust view of long-term trends.

Charlie's Camping World Common-Size Income Statement			Clear Lake Sporting Goods Common-Size Income Statement	
	Prior Year	Prior Year %	Current Year	Current Year %
Gross Sales	\$748,000		\$126,000	
Sales Returns & Allowances	41,000		6,000	
Net Sales	707,000	100%	120,000	100%
Cost of Goods Sold	418,000	59%	60,000	50%
Gross Profit	289,000	41%	60,000	50%
Rent Expense	75,000	11%	5,500	5%
Depreciation Expense	5,000	1%	3,600	3%
Salaries Expense	85,000	12%	5,400	5%
Utility Expense	25,000	4%	2,500	2%
Operating Income	99,000	14%	43,000	36%
Interest Expense	1,000	0%	2,000	2%
Income Tax Expense	14,000	2%	6,000	5%
Net Income	\$ 84,000	12%	\$ 35,000	29%

Figure 5.26 Comparison of Common-Size Income Statements

LINK TO LEARNING

Microsoft versus Apple

Locate the sales, cost of goods sold, and gross profit data for [Microsoft \(https://openstax.org/r/microsoft-financial\)](https://openstax.org/r/microsoft-financial)'s most current period using Yahoo! Finance. Calculate the common-size percentage for cost of goods sold and gross profit (cost of goods sold divided by net sales, and gross profit divided by net sales). Compare these percentages to those of [Apple, Inc \(https://openstax.org/r/common-size-income-statement\)](https://openstax.org/r/common-size-income-statement)., a key competitor, on its common-size income statement. What can you learn about the performance of each firm based on these two percentages?

Now that you have covered the basic financial statements and a little bit about how they are used, where do we find them? How often are they prepared? Who gets them? In this next section we will explore the requirements for what needs to be reported, when, and to whom.

5.8 Reporting Financial Activity

Learning Outcomes

By the end of this section, you will be able to:

- Identify the most common types of accounting periods.
- Outline key considerations and accounting principles that dictate the timing of financial reporting.
- Describe the SEC reporting requirements relevant to financial statements.
- Identify the key elements of a company annual report.

Defining Accounting Periods

An accounting period can be any period of time, but the most common accounting periods are months, quarters, and a year. Accounting periods are important, particularly in accrual accounting, so there are clear cutoffs for recording transactions.

It is important to note that Generally Accepted Accounting Principles (GAAP) require companies to provide quarterly financial statements. However, most firms, even those not covered by GAAP, prepare financial statements monthly in order to provide timely data to their financial statement users both internally and externally.

Time Period Principle

Providing information to financial statement users in a timely fashion brings us to our next key topic, which is the time period principle. In order for information to be useful, it must be timely. This means that financial statement users need to get the statements quickly enough to be able to make relevant decisions with them. Providing statements on a timely basis is the foundation of the time period accounting principle.

Fiscal versus Calendar Year

Though a firm may present financial statements monthly, quarterly, and annually, not all time periods are created equal. Firms have the option to choose between a fiscal and a calendar year. The calendar year, which begins January 1 and ends December 31, is the traditional year we are accustomed to. The calendar year, however, doesn't always work well with a firm's business cycle. Due to seasonality or other factors, a firm may choose to adopt a fiscal year with their own beginning and end date. For example, a firm may choose to start its fiscal year on June 1 and end it on May 31. A firm can opt to change from a fiscal to a calendar year or vice versa but must only do so for a justifiable reason.

THINK IT THROUGH

Fiscal versus Calendar Year

Visit the [Apple, Inc. Annual Report \(https://openstax.org/r/2020-doc-financial-annual-report\)](https://openstax.org/r/2020-doc-financial-annual-report) for 2020 and review the title of the table of contents on page 1. The title reads "Apple Inc. Form 10-K." The next line provides information on the company's chosen time period. Does Apple use a fiscal or calendar year?

Solution:

Apple uses a fiscal year. Their fiscal year ended on September 26, 2020.

The Annual Report

An annual report is part of the 10-K filing that publicly traded firms must provide to the SEC and investors each year. The report details the firm's operations and performance for the year and its current financial conditions. The report contains general company information, the firm's financial statements, notes to the financial

statements, various required disclosures pertaining to accounting policies, the management discussion and analysis (MD&A) statement, a letter from the CEO to the shareholders, and the firm's audit report.

THINK IT THROUGH

Annual Report Key Elements

Visit the [Apple, Inc. Annual Report \(https://openstax.org/r/2020-doc-financial-annual-report\)](https://openstax.org/r/2020-doc-financial-annual-report) for 2020 and review the title of the table of contents on page 1. What key elements would you be drawn to review if you were a potential investor? In Part II of the table of contents, find the MD&A report (page 20) and use the table of contents link to navigate to it. What key highlights and challenges did the company report for the year?

Solution:

Answers may vary. A few key items of note might include risk factors, financial statements, the MD&A letter, and disclosures about market risk. Apple's MD&A reported significant updates on the impact of the COVID-19 pandemic, a few financial updates, product updates, segment performance data, a conversation about its debt, contractual obligations, and accounting policies.

SEC Reporting Requirements

The Securities and Exchange Commission (SEC) requires publicly traded firms to regularly provide several reports. The two most common are the 10-K and quarterly report (10-Q). Certain unique events also require an additional filing called an 8-K. There are many events that might require reporting. A few key examples include changes to rights of shareholders, changes in control of the company, and amendments to the company charter or bylaws.

The quarterly report is much like the annual report already discussed. It contains the firm's financial statements, required accounting disclosures, statements on internal control, and a management discussion and analysis (MD&A) letter. It does not contain an auditor's report, as firms are not audited on a quarterly basis. Quarterly reports are helpful to investors because they provide information on a timely enough basis to be relevant.

LINK TO LEARNING

Investor Relations

Visit the [Apple, Inc. Investor Relations page, SEC Filings section \(https://openstax.org/r/sec-filings\)](https://openstax.org/r/sec-filings). Notice how many different reports Apple files regularly with the SEC. Locate the most recent quarterly (10-Q) and annual reports (10-K) and scan the table of contents. How does the quarterly report differ from the annual report? Do you notice key items in the annual report not provided in the quarterly report?



Summary

5.1 The Income Statement

The income statement reflects a firm's performance over a period of time. Most financial statements are prepared monthly, quarterly, and annually. The income statement reflects sales less cost of goods sold to arrive at gross profit. Operating costs are deducted to arrive at operating income. Finally, other nonoperational costs like interest and taxes are deducted to arrive at net income.

5.2 The Balance Sheet

The balance sheet reflects the financial position of a firm as of a particular point in time. It is laid out to clearly depict and support the accounting equation: $\text{assets} = \text{liabilities} + \text{owner's equity}$. A classified balance sheet breaks down the assets and liabilities sections into current and noncurrent for greater transparency.

5.3 The Relationship between the Balance Sheet and the Income Statement

The financial statements are all tied together. The income statement is generated first, as net income is needed in order to determine the ending balance of retained earnings, a key account in the equity section of the balance sheet.

5.4 The Statement of Owner's Equity

The statement of owner's equity is divided by each type of equity the firm has: common stock, preferred stock, additional paid-in capital, and retained earnings, for example. Beginning balances are provided, and all key transactions impacting equity are provided in order to show how ending balance were derived. Key transactions commonly include recording net income or loss, issuing additional stock, and paying out dividends.

5.5 The Statement of Cash Flows

Under accrual accounting, transactions are recorded when they occur, not necessarily when cash moves. This creates a timing difference. Net profit, therefore, does not necessarily mean a firm has cash, and a net loss doesn't mean they don't have any cash. To reconcile net income to actual cash flow and see how a firm generates and uses its funds, a statement of cash flows is prepared. The statement reflects cash flow from operating activities, financing activities, and investing activities.

5.6 Operating Cash Flow and Free Cash Flow to the Firm (FCFF)

Operating cash flow reflects the cash generated by (or used by) the core business function. Free cash flow to the firm (FCFF) or simply free cash flow (FCF) is calculated by deducting capital expenditures from operating cash flow. FCF reflects the cash available to repay debts, pay dividends to shareholders, and contribute to cash needs for growth.

5.7 Common-Size Statements

Common-size statements are a restatement of the financial statements with all dollar figures restated as a percentage. On the income statement, each line is restated as a percentage of net sales. On the balance sheet, each line is restated as a percentage of total assets. Common-size statements are useful for analysis and are particularly helpful in comparing firms of different sizes.

5.8 Reporting Financial Activity

Publicly traded firms must file company and financial data with the Securities and Exchange Commission (SEC) on a regular basis. Key reports include the quarterly report, called a 10-Q, and the annual report, called a 10-K.

Key Terms

accounting equation $\text{assets} = \text{liabilities} + \text{owner's equity}$

accrual basis accounting system in which revenue is recorded or recognized when earned yet not necessarily received, and in which expenses are recorded when legally incurred and not necessarily when paid

assets tangible or intangible resources owned or controlled by a company, individual, or other entity with the intent that they will provide economic value

cash basis method of accounting in which transactions are not recorded in the financial statements until there is an exchange of cash

current assets asset typically used up, sold, or converted to cash in one year or less

current liabilities debt or obligation due within one year or, in rare cases, a company's standard operating cycle, whichever is greater

depreciation process of allocating the costs of a tangible asset over the asset's economic life

direct method approach used to determine net cash flows from operating activities, whereby accrual basis revenue and expenses are converted to cash basis collections and payments

dividends portion of the net worth (equity) that is returned to owners of a corporation as a reward for their investment

expenses costs associated with providing goods or services

free cash flow operating cash, reduced by expected capital expenditures

gains increases in organizational value from activities that are "incidental or peripheral" to the primary purpose of the business

income statement financial statement that measures the organization's financial performance for a given period of time

indirect method approach used to determine net cash flows from operating activities, starting with net income and adjusting for items that impact net income but do not require outlay of cash

liabilities probable future sacrifice of economic benefits arising from present obligations of a particular entity to transfer assets or provide services to other entities in the future as a result of past transactions or events

loss decrease in organizational value from activities that are "incidental or peripheral" to the primary purpose of the business

net income revenues and gains that are greater than expenses and losses

noncash expenses expenses that reduce net income but are not associated with a cash flow; most common example is depreciation expense

noncurrent assets assets used in the normal course of business for more than one year that are not intended to be resold

noncurrent liabilities liabilities that are expected to be settled in more than one year

owner's equity residual interest in the assets of an entity that remains after deducting its liabilities

retained earnings cumulative, undistributed net income or net loss for the business since its inception

revenue inflows or other enhancements of assets of an entity or settlements of liabilities from delivering or producing goods, rendering services, or other activities that constitute the entity's ongoing major or central operations

statement of cash flows financial statement listing the cash inflows and cash outflows for the business for a period of time

CFA Institute

This chapter supports some of the Learning Outcome Statements (LOS) in this [CFA® Level I Study Session \(https://openstax.org/r/media-document-study-session\)](https://openstax.org/r/media-document-study-session). Reference with permission of CFA Institute.



Multiple Choice

1. Which of the following is a measure of the performance of a firm's daily operations?
 - a. gross profit
 - b. cost of goods sold
 - c. operating income
 - d. net income
2. In which section of the classified balance sheet would you find a note payable due in six months?
 - a. current assets
 - b. current liabilities
 - c. noncurrent liabilities
 - d. common stock
3. Which financial statement must be prepared first?
 - a. statement of retained earnings
 - b. balance sheet
 - c. statement of cash flows
 - d. income statement
4. Which of the following represents earned capital on the statement of owner's equity?
 - a. retained earnings
 - b. common stock
 - c. preferred stock
 - d. additional paid-in capital
5. Which section of the statement of cash flows reflects the cash generated from or used by a company's day-to-day operations?
 - a. investing activities
 - b. financing activities
 - c. operating activities
 - d. noncash activities
6. How do you calculate free cash flow (FCF)?
 - a. net income less dividends
 - b. operating income less capital expenditures
 - c. gross profit less depreciation
 - d. net income plus interest
7. How do you calculate common-size analysis on the income statement?
 - a. income statement line item/gross profit
 - b. income statement line item/net income
 - c. net sales/income statement line item
 - d. income statement line item/net sales
8. Which of the following does not represent a filing commonly required by the SEC?
 - a. annual report, 10-K
 - b. quarterly report, 10-Q
 - c. Form 8-K
 - d. 1040



Review Questions

1. What is the difference between gross profit and net income?
2. If a classified balance sheet has total assets of \$900,000 and total owner's equity of \$350,000, what must the company's total liabilities be?
3. What key element of the income statement flows through to the balance sheet?
4. What key columns are commonly found on the statement of owner's equity?
5. Ted's firm reported net income for the current period of \$65,750. Is it safe to assume that because Ted's firm reported such a large net income, it has plenty of cash to fund its operations? Why or why not?
6. What useful insights does free cash flow (FCF) provide in financial analysis?
7. Describe how common-size statements are useful.
8. What is the difference between a calendar year and a fiscal year?



Problems

1. Rickey's Retail has the following financial information for the most recent accounting period. Prepare an income statement.

Rickey's Retail Income Statement	
Gross Sales	\$864,740
Sales Returns & Allowances	47,399
Cost of Goods Sold	483,237
Rent Expense	86,705
Interest Expense	1,156
Income Tax Expense	16,185
Depreciation Expense	5,780
Salaries Expense	98,266
Utility Expense	\$ 28,902

2. Big Box has the following accounts. In which section of its classified balance sheet does each belong?
Cash
Wages payable
Taxes payable
Accounts receivable
Retained earnings
Common stock
Land
Note payable due in 10 years
Prepaid insurance
3. Big Box Outlet has \$10,350 of supplies expense on its income statement. Does this mean that there must also be a supplies payable account balance of \$10,350 on its balance sheet? Why or why not?
4. What are the three key types of dividends a firm might distribute to their shareholders? Describe each.
5. Big Box Outlet had an increase in its accounts payable account this period and a decrease in its accounts receivable, took out a long-term note payable, paid dividends to its shareholders, had depreciation on its

equipment, bought new equipment, increased its inventory account, and repaid a bond. In which section of the statement of cash flows would each of these items appear?

- Kokoya's Firm calculates its free cash flow at only \$2,000, which the company feels is quite low based on its historical performance and compared to others in its industry. What actions might Kokoya's Firm take to improve its overall cash flow?
- Complete the common-size income statement for Big Box Outlet using the information below:

Big Box Outlet Common-Size Income Statement		
	Current Year	Current Year %
Gross Sales	\$1,089,836	_____
Sales Returns & Allowances	59,737	_____
Net Sales	1,030,099	_____
Cost of Goods Sold	609,026	_____
Gross Profit	421,073	_____
Rent Expense	109,275	_____
Depreciation Expense	7,285	_____
Salaries Expense	123,845	_____
Utility Expense	36,425	_____
Operating Income	144,243	_____
Interest Expense	1,457	_____
Income Tax Expense	20,398	_____
Net Income	<u>\$ 122,388</u>	_____

- List at least eight items commonly found in a firm's annual report filed with the SEC.

Video Activity

The Income Statement Explained

[Click to view content \(https://openstax.org/r/income-statement-explain\)](https://openstax.org/r/income-statement-explain)

- In the video, you see two years of data for Imaginary Ltd. The company had the same sales and same operating profit in both years. However, its performance really wasn't the same from one year to the next. How do the details of the income statement help you see this? What key performance measures were you able to glean from the income statement?
- Though the key elements of the income statement can be summarized as simply all the firm's revenues and expenses, how would you describe the elements of the income statement in more detail? Assume you were consulting for a friend who owns a small dairy farm. What items would you expect to see on their income statement? Make a list and be as detailed as you can. Curious if your list is accurate? Consider doing an online search to find an income statement for a dairy farm or similar business. Compare its income statement to your list to see if you had the right idea.

How the Balance Sheet Works

[Click to view content \(https://openstax.org/r/balance-sheet-works\)](https://openstax.org/r/balance-sheet-works)

- In the video, before the example started, a fair amount of discussion time was devoted to the accounting equation, double-entry accounting, and the past versus the present data. How do all of these concepts tie together in the balance sheet and the type of data we can hope to glean from a balance sheet?
- Three key account types are represented on the balance sheet: assets, liabilities, and equity. Much like the income statement, however, there is a great deal more detail to a balance sheet than simply these three

figures. What additional accounts and detail can you find on the balance sheet? If you were to look up the balance sheet for the company that sells your favorite thing (e.g., coffee, your laptop, your phone, candy), what types of accounts do you think you will find on its balance sheet? Consider doing an online search for the company's balance sheet to see if your guess is correct.



6

Measures of Financial Health

Figure 6.1 Organizations must continually measure their financial health in order to remain successful. (credit: "Money" by Pictures of Money/flickr, CC BY 2.0)

Chapter Outline

- 6.1 Ratios: Condensing Information into Smaller Pieces
- 6.2 Operating Efficiency Ratios
- 6.3 Liquidity Ratios
- 6.4 Solvency Ratios
- 6.5 Market Value Ratios
- 6.6 Profitability Ratios and the DuPont Method



Why It Matters

Financial analysis is a crucial element of business, but it can be used in personal finance as well. It differs depending on the role and perspective of those performing the analysis. For example, your personal accountant will have different goals and needs in making recommendations to you about your personal finances. All accounting professionals use financial analysis to check for validity, accurate data, compliance in reporting, and more.

Some tactics for managing your personal finances can be the same as for managing business finances. For example, reducing expenses and maximizing returns on long-term investments are always good practices. Debt can also be a beneficial tool in both personal and professional finances when used appropriately. Debt is neither inherently good nor bad; it simply needs to be properly managed in order to achieve a reasonable return in exchange for the cost and risk it poses.

Though the process and tools may be similar, financial analysis from a business perspective has different goals and needs. Investors are looking to identify firm performance, financial health, and profitability. Financial analysts closely review information found on financial statements so they can make informed business decisions. The income statement, statement of retained earnings, balance sheet, and statement of cash flows, among other financial information, are analyzed for internal and external stakeholders and provide a company with valuable information about its overall performance and specific areas for improvement. The analysis can

help with budgeting and making decisions about where the company could cut costs, how it might increase revenues, and what capital investment opportunities it should pursue.

LINK TO LEARNING

Financial Analyst

Lots of individuals and companies perform financial analysis. One of these roles is that of a financial analyst. The skills and qualifications of a financial analyst vary widely from one industry to another, but there are a number of similarities in individuals who hold these roles. As you watch the [video about financial analysts \(https://openstax.org/r/video-about-financial-analysts\)](https://openstax.org/r/video-about-financial-analysts), consider your own career path and how your skills, abilities, and interests may fit this role.

6.1 Ratios: Condensing Information into Smaller Pieces

By the end of this section, you will be able to:

- Explain the importance of financial statement analysis in making informed decisions about business opportunities.
- Outline the limitations of financial statement analysis in making investment decisions.

When considering the outcomes from analysis, it is important for a company to understand that the data generated needs to be compared to similar data within the industry at large as well as that of close competitors. The company should also consider its past experience and how it corresponds to current and future performance expectations.

Importance of Ratios and Analysis

Financial ratios help internal and external stakeholders make informed decisions about actions like investing, becoming a supplier, making a loan, or altering internal operations, among other things. The information resulting from ratio analysis can be used to examine trends in performance, establish benchmarks for success, set budget expectations, and compare industry competitors. There are four main types of ratios: liquidity, solvency, efficiency, and profitability. While outcomes for some ratios may seem more ideal, the industry in which the business operates can change the influence of these outcomes on stakeholder decisions.

There are several benefits to analyzing financial statements. The information can show trends over time, which can help in making future business decisions. Converting information to percentages or ratios eliminates some of the disparities between competitors' sizes and operating abilities, making it easier for stakeholders to make informed decisions. It can assist with understanding the makeup of current operations within the business and which shifts need to occur internally to increase productivity.

Limitation of Financial Statement Analysis

Though useful, it's important to note that there are limitations to financial statement analysis as well. Stakeholders need to remember that past performance does not always predict future performance. Economic influences, such as inflation or a recession, could skew the data being analyzed. Additionally, the way a company reports information may change over time. For example, there could be changes in where and when certain transactions are recorded, and this may not be immediately evident to financial statement users. It is also key to note that though all publicly traded companies in the United States are required to follow Generally Accepted Accounting Principles (GAAP), there are many estimates and flexibility in how some standards are applied. This means that firms can still follow accounting standards appropriately but present some information differently from other firms.

It makes good sense for a company to use financial statement analysis to guide future operations so it can

budget properly, control costs, increase revenues, and make long-term expenditure decisions. As long as stakeholders understand the limitations of financial statement analysis, it is a useful way to predict growth and financial strength.

Despite limitations, ratios are still a valuable tool if used appropriately. The next section discusses several operating efficiency ratios including accounts receivable turnover, total asset turnover, inventory turnover, and days' sales in inventory. Operating efficiency ratios help users see how well management is using the financial assets of the firm.

6.2 Operating Efficiency Ratios

By the end of this section, you will be able to:

- Calculate accounts receivable turnover to assess a firm's performance in managing customer receivables.
- Evaluate management's use of assets using total asset turnover and inventory turnover.
- Assess organizational performance using days' sales in inventory calculations.

Efficiency ratios show how well a company uses and manages its assets, one key element of financial health. Important areas of efficiency are the management of sales, accounts receivable, and inventory. A company that is efficient will usually be able to generate revenues quickly using the assets it has acquired. Let's examine four **efficiency ratios**: accounts receivable turnover, total asset turnover, inventory turnover, and **days' sales in inventory**.

Accounts Receivable Turnover

For our discussion of financial statement analysis, we will look at Clear Lake Sporting Goods. Clear Lake Sporting Goods is a small merchandising company (a company that buys finished goods and sells them to consumers) that sells hunting and fishing gear. [Figure 6.2](#) shows the comparative income statements and balance sheets for the past two years.

Clear Lake Sporting Goods Comparative Year-End Income Statements			Clear Lake Sporting Goods Comparative Year-End Balance Sheets		
	Prior Year	Current Year		Prior Year	Current Year
Net Sales	\$100,000	\$120,000	Assets:		
Cost of Goods Sold	50,000	60,000	Cash	\$ 90,000	\$110,000
Gross Profit	50,000	60,000	Accounts Receivable	20,000	30,000
Rent Expense	5,000	5,500	Inventory	35,000	40,000
Depreciation Expense	2,500	3,600	Short-Term Investments	15,000	20,000
Salaries Expense	3,000	5,400	Total Current Assets	160,000	200,000
Utility Expense	1,500	2,500	Equipment	40,000	50,000
Operating Income	38,000	43,000	Total Assets	\$200,000	\$250,000
Interest Expense	3,000	2,000	Liabilities:		
Income Tax Expense	5,000	6,000	Accounts Payable	\$ 60,000	\$ 75,000
Net Income	<u>\$ 30,000</u>	<u>\$ 35,000</u>	Unearned Revenue	10,000	25,000
			Total Current Liabilities	70,000	100,000
			Notes Payable	40,000	50,000
			Total Liabilities	<u>\$110,000</u>	<u>\$150,000</u>
			Stockholder Equity		
			Common Stock	75,000	80,000
			Ending Retained Earnings	15,000	20,000
			Total Stockholders' Equity	90,000	100,000
			Total Liabilities and Stockholder Equity	\$200,000	\$250,000

Figure 6.2 Comparative Income Statements and Year-End Balance Sheets Note that the comparative income statements and

balance sheets have been simplified here and do not fully reflect all possible company accounts.

To begin an analysis of receivables, it's important to first understand the cycles and periods used in the calculations.

Operating Cycle

A period is one **operating cycle** of a business. The operating cycle includes the cash conversion cycle plus the accounts receivable cycle (discussed below). Essentially it is the time it takes a business to purchase or make inventory and then sell it. For example, assume Clear Lake Sporting Goods orders and receives a shipment of fishing lures on June 1. It stocks the shelves with lures and tracks its inventory and sales. By July 15, all the lures from that shipment are gone. In this example, Clear Lake's operating cycle is 45 days.

Cash Conversion Cycle

Cash, however, doesn't necessarily flow linearly with accounting periods or operating cycles. The cash conversion cycle is the time it takes to spend cash to purchase inventory, produce the product, sell it, and then collect cash from the customer. Accounts receivable is one section of that cycle. Referring to Clear Lake's June 1 shipment of lures that sold by July 15, assume that some of the customers were fishing guides that keep an open account with Clear Lake. This company did not pay for its lures until August 15 when it settled its account. In this example, Clear Lake's cash cycle is 75 days.

Let's take a look at the accounts receivable turnover ratio, which helps assess that element of the cash conversion cycle.

Accounts Receivable Turnover Ratio

Receivables ratios show company performance in relation to current receivables (what is due from customers), as well as credit policy effect on sales growth. One receivables ratio is called the **accounts receivable turnover ratio**. This ratio determines how many times (i.e., how often) accounts receivable are collected during a year and converted to cash. A higher number of times indicates that receivables are collected quickly. This quick cash collection may be viewed as a positive occurrence because liquidity improves, and the company may reinvest in its business sooner when the value of the dollar has more buying power (time value of money). The higher number of times may also be a negative occurrence, signaling that credit extension terms are too tight, and it may exclude qualified consumers from purchasing. Excluding these customers means that they may take their business to a competitor, thus reducing potential sales.

In contrast, a lower number of times indicates that receivables are collected at a slower rate. A slower collection rate could signal that lending terms are too lenient; management might consider tightening lending opportunities and more aggressively pursuing payment from its customers. The lower turnover also shows that the company has cash tied up in receivables longer, thus hindering its ability to reinvest this cash in other current projects. The lower turnover rate may signal a high level of bad debt accounts. The determination of a high or low turnover rate really depends on the standards of the company's industry. It's key to note the tradeoff in adjusting credit terms. Loose credit terms may attract more customers but may also increase bad debt expense. Tighter credit terms may attract fewer customers but may also reduce bad debt expense.

The formula for accounts receivable turnover is

$$\text{Accounts Receivable Turnover} = \frac{\text{Net Credit Sales}}{\text{Average Accounts Receivable}}$$

$$\text{Average Accounts Receivable} = \frac{\text{Beginning Accounts Receivable} + \text{Ending Accounts Receivable}}{2}$$

Net credit sales are sales made on credit only; cash sales are not included because they do not produce receivables. However, many companies do not report credit sales separately from cash sales, so "net sales" may be substituted for "net credit sales" in this case. Beginning and ending accounts receivable refer to the

beginning and ending balances in accounts receivable for the period. The beginning accounts receivable balance is the same figure as the ending accounts receivable balance from the prior period.

When computing the accounts receivable turnover for Clear Lake Sporting Goods, let's assume net credit sales make up \$100,000 of the \$120,000 of the net sales found on the income statement in the current year.

$$\text{Average Accounts Receivable} = \frac{\$20,000 + \$30,000}{2} = \$25,000$$

$$\text{Accounts Receivable Turnover} = \frac{\$100,000}{\$25,000} = 4 \text{ times}$$

To gain a better understanding of its ratio performance, Clear Lake Sporting Goods can compare its turnover to industry averages, key competitors, and its own historical ratios. Given this outcome, the managers may want to consider stricter credit lending practices to make sure credit customers are of a higher quality. They may also need to be more aggressive with collecting any outstanding accounts.

THINK IT THROUGH

Accounts Receivable Turnover

You are a consultant assessing cash management practices for two firms, Company A and Company B (see [Figure 6.3](#)).

	Company A	Company B
Beginning Accounts Receivable	\$ 50,000	\$ 60,000
Ending Accounts Receivable	80,000	90,000
Net Credit Sales	\$550,000	\$460,000

Figure 6.3 Financial Information for Company A and Company B

Based on the information provided, do the following:

- Compute the accounts receivable turnover ratio.
- Interpret the outcomes, indicating how each company is performing

Solution:

Company A: ART = 8.46 times, Company B: ART = 6.13 times. Upon initial review of this limited information, Company A seems to be performing better since its turnover ratio is higher. Accounts receivable turnover has a significant impact on cash flows. One might want more information on trends for each company with these ratios and a comparison to others in the same industry. More information is helpful in assessing performance.

LINK TO LEARNING

American Superconductor Corporation

American Superconductor Corporation specializes in the production and service of energy-efficient wind turbine systems, as well as energy grid construction solutions. On the company's [2019 financial statement \(https://openstax.org/r/financial-statement\)](https://openstax.org/r/financial-statement), the accounts receivable turnover ratio is approximately 6.32 times.

Total Asset Turnover

Total asset turnover measures the ability of a company to use its assets to generate revenues. A company would like to use as few assets as possible to generate the most net sales. Therefore, a higher total asset turnover means the company is using their assets very efficiently to produce net sales. The formula for total asset turnover is

$$\text{Total Asset Turnover} = \frac{\text{Net Sales}}{\text{Average Total Assets}}$$

$$\text{Average Total Assets} = \frac{\text{Beginning Total Assets} + \text{Ending Total Assets}}{2}$$

Average total assets are found by dividing the sum of beginning and ending total assets balances found on the balance sheet. The beginning total assets balance in the current year is taken from the ending total assets balance in the prior year.

Clear Lake Sporting Goods' total asset turnover is

$$\text{Average Total Assets} = \frac{\$200,000 + \$250,000}{2} = \$225,000$$

$$\text{Total Asset Turnover} = \frac{\$120,000}{\$225,000} = 0.53 \text{ times (rounded)}$$

The outcome of 0.53 means that for every \$1 of assets, \$0.53 of net sales are generated. Over time, Clear Lake Sporting Goods would like to see this turnover ratio increase.

Inventory Turnover

Inventory turnover measures how many times during the year a company has sold and replaced inventory. This can tell a company how well inventory is managed. A higher ratio is preferable; however, an extremely high turnover may mean that the company does not have enough inventory available to meet demand. A low turnover may mean the company has too much supply of inventory on hand. The formula for inventory turnover is

$$\text{Inventory Turnover} = \frac{\text{Cost of Goods Sold}}{\text{Average Inventory}}$$

$$\text{Average Inventory} = \frac{\text{Beginning Inventory} + \text{Ending Inventory}}{2}$$

Cost of goods sold for the current year is found on the income statement. Average inventory is found by dividing the sum of beginning and ending inventory balances found on the balance sheet. The beginning inventory balance in the current year is taken from the ending inventory balance in the prior year.

Clear Lake Sporting Goods' inventory turnover is

$$\text{Average Inventory} = \frac{\$35,000 + \$40,000}{2} = \$37,500$$

$$\text{Inventory Turnover} = \frac{\$60,000}{\$37,500} = 1.6 \text{ times}$$

A ratio of 1.6 times seems to be a very low turnover rate for Clear Lake Sporting Goods. This may mean the company is maintaining too high an inventory supply to meet a low demand from customers. Managers may want to decrease their on-hand inventory to free up more liquid assets to use in other ways. Keep in mind, ratios should not be taken out of context. One ratio alone can't tell the whole story. Ratios should be used with caution and in conjunction with other ratios and additional financial and contextual information.

As with accounts receivable, there is a trade-off to consider in managing inventory. Low turnover will usually

mean a low risk of stockouts and the ability to carry more of what customers are looking for. But high inventory levels will mean that more cash is tied up in inventory. High turnover will mean carrying less inventory and the higher risk of stockouts, causing customers to go elsewhere to find what they need.



Figure 6.4 Inventory turnover can help determine how well a company manages its inventory. (credit: "Untitled" by Marcin Wichary/flickr, CC BY 2.0)

LINK TO LEARNING

Target Corporation

As we have learned, the inventory turnover ratio shows how well a company manages its inventory. Look through the financial statements in the [2019 Annual Report for Target \(https://openstax.org/r/annual-report-for-target\)](https://openstax.org/r/annual-report-for-target) and calculate the inventory turnover ratio. What does the outcome mean for Target?

Days' Sales in Inventory

Days' sales in inventory expresses the number of days it takes a company to turn inventory into sales. The fewer the number of days, the more quickly the company can sell its inventory. The greater the number of days, the longer it takes to sell its inventory. The formula for days' sales in inventory is

$$\text{Days' Sales in Inventory} = \frac{\text{Ending Inventory}}{\text{Cost of Goods Sold}} \times 365$$

Clear Lake Sporting Goods' days' sales in inventory is

$$\text{Days' Sales in Inventory} = \frac{\$40,000}{\$60,000} \times 365 = 243 \text{ days (rounded)}$$

Depending on the industry, 243 days may be a long time to sell inventory. While industry dictates what is an acceptable number of days to sell inventory, 243 days is likely to be unsustainable long-term. Remember, it's important to not take one ratio out of context. Review the ratio in conjunction with other ratios and other financial data. For example, we might review the days' sales in inventory along with accounts receivable turnover for Clear Lake Sporting Goods relative to the industry average to get a better picture of Clear Lake's performance in this area.

6.3 Liquidity Ratios

By the end of this section, you will be able to:

- Calculate current, quick, and cash ratios to assess a firm's liquidity and make informed business decisions.
- Assess organizational performance using liquidity ratios.

Liquidity refers to the business's ability to manage current assets or convert assets into cash in order to meet short-term cash needs, another aspect of a firm's financial health. Examples of the most liquid assets include cash, accounts receivable, and inventory for merchandising or manufacturing businesses. The reason these are among the most liquid assets is that these assets will be turned into cash more quickly than land or buildings, for example. Accounts receivable represents goods or services that have already been sold and will typically be paid/collected within 30 to 45 days.

Inventory is less liquid than accounts receivable because the product must first be sold before it generates cash (either through a cash sale or sale on account). Inventory is, however, more liquid than land or buildings because, under most circumstances, it is easier and quicker for a business to find someone to purchase its goods than it is to find a buyer for land or buildings.

Current Ratio

The **current ratio** is closely related to working capital; it represents the current assets divided by current liabilities. The current ratio utilizes the same amounts as working capital (current assets and current liabilities) but presents the amount in ratio, rather than dollar, form. That is, the current ratio is defined as current assets/current liabilities. The interpretation of the current ratio is similar to working capital. A ratio of greater than one indicates that the firm has the ability to meet short-term obligations with a buffer, while a ratio of less than one indicates that the firm should pay close attention to the composition of its current assets as well as the timing of the current liabilities.

$$\text{Current Ratio} = \frac{\text{Current Assets}}{\text{Current Liabilities}}$$

The current ratio in the current year for Clear Lake Sporting Goods is

$$\text{Current Ratio} = \frac{\$200,000}{\$100,000} = 2 \text{ or } 2:1$$

A 2:1 ratio means the company has twice as many current assets as current liabilities; typically, this would be plenty to cover obligations. A 2:1 ratio is actually quite high for most companies and most industries. Again, it's recommended that ratios be used in conjunction with one another. An analyst would likely look at the high current ratio and low accounts receivable turnover to begin asking questions about management performance, as this might indicate a trouble area (high inventory and slow collections).

LINK TO LEARNING

Target Corporation

As we have learned, the current ratio shows how well a company can cover short-term liabilities with short-term assets. Look through the balance sheet in the [2019 Annual Report for Target \(https://openstax.org/r/annual-report-for-target\)](https://openstax.org/r/annual-report-for-target) and calculate the current ratio. What does the outcome mean for Target?

Quick Ratio

The **quick ratio**, also known as the *acid-test ratio*, is similar to the current ratio except current assets are more narrowly defined as the most liquid assets, which exclude inventory and prepaid expenses. The conversion of

inventory and prepaid expenses to cash can sometimes take more time than the liquidation of other current assets. A company will want to know what it has on hand and can use quickly if an immediate obligation is due. The formula for the quick ratio is

$$\text{Quick Ratio} = \frac{\text{Cash} + \text{Short-Term Investments} + \text{Accounts Receivable}}{\text{Current Liabilities}}$$

The quick ratio for Clear Lake Sporting Goods in the current year is

$$\text{Quick Ratio} = \frac{\$110,000 + \$20,000 + \$30,000}{\$100,000} = 1.6 \text{ or } 1.6:1$$

A 1.6:1 ratio means the company has enough quick assets to cover current liabilities. It's again key to note that a single ratio shouldn't be used out of context. A 1.6 ratio is difficult to interpret on its own. Industry averages and trend analysis for Clear Lake Sporting Goods would also be helpful in giving the ratio more meaning.

LINK TO LEARNING

Target Corporation

As we have learned, the quick ratio shows how quickly a company can liquidate current assets to cover current liabilities. Look through the financial statements in the [2019 Annual Report for Target](https://openstax.org/r/annual-report-for-target) (<https://openstax.org/r/annual-report-for-target>) and calculate the quick ratio. What does the outcome mean for Target?

Cash Ratio

Cash is the most liquid asset a company has, and **cash ratio** is often used by investors and lenders to assess an organization's liquidity. It represents the firm's cash and cash equivalents divided by current liabilities and is a more conservative look at a firm's liquidity than the current or quick ratios. The ratio is reflected as a number, not a percentage. A cash ratio of 1.0 means the firm has enough cash to cover all current liabilities if something happened and it was required to pay all current debts immediately. A ratio of less than 1.0 means the firm has more current liabilities than it has cash on hand. A ratio of more than 1.0 means it has enough cash on hand to pay all current liabilities and still have cash left over. While a ratio greater than 1.0 may sound ideal, it's important to consider the specifics of the company. Sitting on idle cash is not ideal, as the cash could be used to earn a return. And having a ratio less than 1.0 isn't always bad, as many firms operate quite successfully with a ratio of less than 1.0. Comparing the company ratio with trend analysis and with industry averages will help provide more insight.

$$\text{Cash Ratio} = \frac{\text{Cash and Cash Equivalents}}{\text{Current Liabilities}}$$

The cash ratio for Clear Lake Sporting Goods in the current year is:

$$\text{Cash Ratio} = \frac{\$110,000}{\$100,000} = 1.1$$

A 1.1 ratio means the company has enough cash to cover current liabilities.



Figure 6.5 Cash is the most liquid asset a company has and is often used by investors and lenders to assess an organization's liquidity. (credit: "20 US Dollar" by Jack Sem/flickr CC BY 2.0)

6.4 Solvency Ratios

By the end of this section, you will be able to:

- Evaluate organizational solvency using the debt-to-assets and debt-to-equity ratios.
- Calculate the times interest earned ratio to assess a firm's ability to cover interest expense on debt as it comes due.

Solvency implies that a company can meet its long-term obligations and will likely stay in business in the future. Meeting long-term obligations includes the ability to pay any interest incurred on long-term debt. Two main solvency ratios are the debt-to-equity ratio and the times interest earned ratio.

Debt-to-Assets Ratio

The **debt-to-assets ratio** shows the relationship between debt and assets. It reflects how much of the assets of the business was financed through debt. It reflects the company's leverage and is helpful to analysts in comparing how leveraged one company is compared to another.

Debts normally carry interest expense and must be repaid. The debt-to-assets ratio includes all debt—both long-term debt and current liabilities. The formula for the debt-to-assets ratio is

$$\text{Debt-to-Assets Ratio} = \frac{\text{Current Liabilities} + \text{Long-Term Liabilities}}{\text{Total Assets}}$$

The information needed to compute the debt-to-assets ratio for Clear Lake Sporting Goods in the current year can be found on the balance sheet. The debt-to-assets ratio for Clear Lake Sporting Goods in the current year is

$$\text{Debt-to-Assets Ratio} = \frac{\$100,000 + \$50,000}{\$250,000} = 0.6 \text{ or } 60\%$$

This means that 60 percent of Clear Lake's assets are financed by debt. We can also then infer that the other 40 percent is financed by equity. A ratio higher than 1.0 means the company has more debts than assets, which means it has negative equity. In Clear Lake's case, a 60 percent debt-to-assets ratio indicates some risk, but perhaps not a high risk. Comparing Clear Lake's ratio to industry averages would provide better insight.

LINK TO LEARNING

Target Corporation

As we have learned, the debt-to-assets ratio shows the relationship between a firm's debt and assets. Look through the financial statements in the [2019 Annual Report for Target \(https://openstax.org/r/annual-report-for-target\)](https://openstax.org/r/annual-report-for-target) and calculate the debt-to-assets ratio. What does the outcome mean for Target?

Debt-to-Equity Ratio

The **debt-to-equity ratio** shows the relationship between debt and equity as it relates to business financing. A company can take out loans, issue stock, and retain earnings to be used in future periods to keep operations running. A key difference in debt and equity is the interest expense repayment that a loan carries as opposed to equity, which does not have this requirement. Therefore, a company wants to know how much debt and equity contribute to its financing. The formula for the debt-to-equity ratio is

$$\text{Debt-to-Equity Ratio} = \frac{\text{Total Liabilities}}{\text{Total Stockholder Equity}}$$

The information needed to compute the debt-to-equity ratio for Clear Lake Sporting Goods in the current year can be found on the balance sheet.

$$\text{Debt-to-Equity Ratio} = \frac{\$150,000 + \$50,000}{\$100,000} = 1.5 \text{ or } 1.5:1$$

This means that for every one dollar of equity contributed toward financing, \$1.50 is contributed from lenders. Recall that total assets equal total liabilities plus total equity. Both the debt-to-assets and debt-to-equity ratio have total liabilities in the numerator. The difference in the two ratios is the denominator. The denominator for the debt-to-equity ratio is total stockholder equity. The denominator for the debt-to-assets ratio is total assets, or total liabilities plus total equity. Thus, the two ratios contain the same information, making calculating both ratios redundant. A financial analyst may prefer to calculate one ratio over the other because of the format of readily available industry data to use for comparison purposes or for consistency with other calculations the analyst is performing.

THINK IT THROUGH

Financing a Business Expansion

You are the CFO of a small corporation. The president, who is one of five shareholders, has created an innovative new product that is testing well with substantial demand. To begin manufacturing, \$400,000 is needed to acquire the equipment. The corporation's balance sheet shows total assets of \$2,400,000 and total liabilities of \$600,000. Most of the liabilities relate to debt that carries a covenant requiring that the company maintain a debt-to-equity ratio not exceeding 0.50. Determine the effect that each of the two options of obtaining additional capital will have on the debt covenant.

Solution:

We know the total liabilities for the firm to be \$600,000. Using the accounting equation, we can find that the firm has \$1,800,000 in equity. $\$600,000/\$1,800,000 =$ current debt-to-equity ratio of 0.33, which is well below the requirement for the debt covenant. If the firm issues debt, the ratio changes to $\$1,000,000/\$1,800,000$, which is 0.55 and would violate the debt covenant. If the firm chooses to issue additional stock, the new debt-to-equity ratio would be $\$600,000/\$2,200,000$, which is 0.27. This is well below the requirements in the debt covenant.

Times Interest Earned (TIE) Ratio

The **times interest earned (TIE) ratio** measures the company's ability to pay interest expense on all debt incurred. This ability to pay is determined by the available earnings before interest and taxes (EBIT) are deducted. These earnings are considered the operating income. Lenders will pay attention to this ratio before extending credit. The more times over a company can cover interest, the more likely a lender will extend long-term credit. The formula for times interest earned is

$$\text{Times Interest Earned} = \frac{\text{Earnings Before Interest and Taxes (EBIT)}}{\text{Interest Expense}}$$

The information needed to compute times interest earned for Clear Lake Sporting Goods in the current year can be found on the income statement.

$$\text{Times Interest Earned} = \frac{\$43,000}{\$2,000} = 21.5 \text{ times}$$

The \$43,000 is the operating income, representing earnings before interest and taxes. The 21.5 times outcome suggests that Clear Lake Sporting Goods can easily repay interest on an outstanding loan and creditors would have little risk that Clear Lake Sporting Goods would be unable to pay.

LINK TO LEARNING

Times Interest Earned

This [video about times interest earned \(https://openstax.org/r/video-about-times\)](https://openstax.org/r/video-about-times) explains how to calculate it and why the ratio is useful, and it provides an example.

6.5 Market Value Ratios

By the end of this section, you will be able to:

- Calculate earnings per share to determine the portion of profit allocated to each outstanding share of common stock.
- Evaluate firm value using the price/earnings ratio and book value per share.

In this section we will turn our attention to **market value ratios**, measures used to assess a firm's overall market price. Common ratios used include earnings per share, the price/earnings ratio, and book value per share.

Earnings per Share (EPS)

Earnings per share (EPS) measures the portion of a corporation's profit allocated to each outstanding share of common stock. An increasing earnings per share can drive up a stock price. Conversely, falling earnings per share can lower a stock's market price. Earnings per share is also a component in calculating the price-to-earnings ratio (the market price of the stock divided by its earnings per share), which many investors find to be a key indicator of the value of a company's stock.

It's key to note, however, that EPS, like any ratio, should be used with caution and in tandem with other ratios and contextual data. Many financial professionals choose not to rely on income statement data and, similarly, EPS because they feel the cash flow statement provides more reliable and insightful information.

CONCEPTS IN PRACTICE

Alibaba Group Earnings Announcements Continue to Exceed Market Expectations

Alibaba, a Chinese-based company traded in the United States, exceeded market expectations in 2020 quarterly earnings releases. In the November 2020 earnings release, Alibaba reported earnings per share of 17.97 yuan versus market estimates of 14.33. Despite many companies struggling due to the pandemic, Alibaba reported strong earnings as a result of the surge in online shopping and remote work.

(sources: "Alibaba Beats Estimates as Pandemic Fuels Online, Cloud Computing Demand." *CNBC*. August 20, 2020. <https://www.cnbc.com/2020/08/20/alibaba-beats-quarterly-revenue-estimates.html>; Emily Bary. "Alibaba Earnings Top Expectations as Pandemic Drives Increased Digital Purchases." *Market Watch*. August 20, 2020. <https://www.marketwatch.com/story/alibaba-earnings-top-expectations-as-pandemic-drives-increased-digital-purchases-2020-08-20>; Matthew Johnston. "Alibaba Earnings: What Happened." *Investopedia*. November 5, 2020. <https://www.investopedia.com/alibaba-q2-2021-earnings-5085444>; Chris Versace. "Why S&P 500 EPS Expectations Showcase the Need for Thematic Investing." *Tematica Research*. June 3, 2020. <https://www.tematicaresearch.com/why-sp-500-eps-expectations-showcase-the-need-for-thematic-investing>)

Calculating Earnings per Share

Earnings per share is the profit a company earns for each of its outstanding common shares. Both the balance sheet and income statement are needed to calculate earnings per share. The balance sheet provides details on the preferred dividend rate, the total par value of the preferred stock, and the number of common shares outstanding. The income statement indicates the net income for the period. The formula to calculate basic earnings per share is

$$\text{Earnings per Share} = \frac{\text{Net Income} - \text{Preferred Dividends}}{\text{Weighted Average Common Shares Outstanding}}$$

By removing the preferred dividends from net income, the numerator represents the profit available to common shareholders. Because preferred dividends represent the amount of net income to be distributed to preferred shareholders, this portion of the income is obviously not available for common shareholders. While a number of variations of measuring a company's profit, such as NOPAT (net operating profit after taxes) and EBITDA (earnings before interest, taxes, depreciation, and amortization), are used in the financial world, GAAP requires companies to calculate earnings per share based on a corporation's net income, as this amount appears directly on a company's income statement, which for public companies must be audited.

In the denominator, only common shares are used to determine earnings per share because earnings per share is a measure of earnings for each common share of stock. The denominator can fluctuate throughout the year as a company issues and buys back shares of its own stock. The weighted average number of shares is used on the denominator because of this fluctuation. To illustrate, assume that a corporation began the year with 600 shares of common stock outstanding and then on April 1 issued 1,000 more shares. During the period January 1 to March 31, the company had the original 600 shares outstanding. Once the new shares were issued, the company had the original 600 plus the new 1,000 shares, for a total of 1,600 shares for each of the next nine months—from April 1 to December 31. To determine the weighted average shares, apply these fractional weights to both of the stock amounts (see [Figure 6.6](#)).

Number of Shares	×	Portion of Year	=	Weighted Shares
600	×	3/12	=	150
1,600	×	9/12	=	1,200
Weighted Average Shares				<u>1,350</u>

Figure 6.6 Weighted Shares

If the shares were not weighted, the calculation would not consider the time period during which the shares were outstanding.

To illustrate how earnings per share is calculated, assume Clear Lake Sporting Goods earns \$35,000 in net income during the current year. During the year, the company also declared a \$5,000 dividend on preferred stock and a \$6,000 dividend on common stock. The company had 8,000 common shares outstanding the entire year. Clear Lake Sporting Goods has generated \$3.75 of earnings (\$35,000 less the \$5,000 of preferred dividends) for each of the 8,000 common shares of stock it has outstanding.

$$\text{Earnings per Share} = \frac{\$35,000 - \$5,000}{8,000} = \$3.75$$

Measuring Performance with Earnings per Share

Earnings per share is a key profitability measure that both current and potential common stockholders monitor. Its importance is accentuated by the fact that GAAP requires public companies to report earnings per share on the face of a company's income statement. This is the only ratio that requires such prominent reporting. In fact, public companies are required to report two different earnings per share amounts on their income statements—basic and diluted. We've illustrated the calculation of basic earnings per share. Diluted earnings per share, which is not demonstrated here, involves the consideration of all securities, such as stocks and bonds, that could potentially dilute, or reduce, the basic earnings per share.

LINK TO LEARNING

Finding Earnings per Share for Public Companies

Where can you find earnings per share information on public companies? Use the [Yahoo! Finance](https://openstax.org/r/yahoo-finance) (<https://openstax.org/r/yahoo-finance>) website to look up stock and earnings per share data for Tellurian Inc. (TELL), Amazon (AMZN), or CVS Pharmacy (CVS). Or use the search function to search for earnings per share data for your favorite corporation. Enter the ticker for the company you are looking up, and a basic chart and graph will display with stock price data along with several commonly used ratios (including earnings per share).

As you review data, keep in mind that a company can manipulate or impact its earnings per share by issuing new shares or buying back issued shares. What are the ethical implications of earnings per share calculations?

Common stock shares are normally purchased by investors to generate income through dividends or to sell at a profit in the future. Investors realize that inadequate earnings per share can result in poor or inconsistent dividend payments and fluctuating stock prices. As such, companies seek to produce earnings per share amounts that rise each period. However, an increase in earnings per share may not always reflect favorable performance, as there are multiple reasons that earnings per share may increase. One way earnings per share can increase is through increased net income. On the other hand, it can also increase when a company buys back its own shares of stock.

For example, assume that Clear Lake Sporting Goods generated net income of \$30,000 and paid out \$3,000 in

preferred shareholder dividends last year. In addition, 10,550 shares of common stock were outstanding throughout the entire year. In January of the current year, the company buys back shares of its common stock and holds them as treasury shares, making its current weighted average shares outstanding for this year 8,000. Net income for the current year is \$35,000, \$5,000 of which was paid to preferred shareholders in dividends. In the prior year, the company's earnings per share was

$$\text{Earnings per Share} = \frac{\$30,000 - \$3,000}{10,550} = \$2.56$$

Clear Lake Sporting Goods' current year earnings per share is

$$\text{Earnings per Share} = \frac{\$35,000 - \$5,000}{8,000} = \$3.75$$

The purchase of treasury stock in the current year reduces the common shares outstanding to 8,000 because treasury shares are considered issued but not outstanding. Earnings per share for the current year is now \$3.75 per share even though earnings only increased by \$5,000. It's key to note the impact of purchasing treasury stock and the intentions in doing so. Treasury stock is commonly purchased for a variety of reasons, but doing so to intentionally manipulate earnings per share should not be a primary reason.

This increase in earnings per share occurred because the net income is now spread over fewer shares of stock. Similarly, earnings per share can decline even when a company's net income increases if the number of shares increases at a higher degree than net income.

CONCEPTS IN PRACTICE

Stock Buybacks Can Drive Up Earnings per Share: Ethical?

As many companies struggled to make ends meet or meet their cash flow needs amid the COVID-19 pandemic, some companies continued to thrive. Apple continued to have a healthy financial position with ample cash supply. It repurchased \$18.5 billion of its own stock in the second quarter of 2020.¹ The total stock buyback over the preceding five years was \$282.87 billion, which is 3.5 times higher than any other company. Since the earnings per share calculation is earnings divided by average outstanding shares, the fewer shares there are outstanding, the higher earnings per share goes without the firm having to actually raise earnings.

What do you think? Did Apple act ethically in repurchasing large quantities of its own shares? Is it ethical for any company to do so? If you were an investor or analyst, what questions would you ask or what cautions would you take in assessing and comparing earnings per share data?

(sources: Wayne Duggan. "7 S&P 500 Companies with Stock Buybacks." *US News & World Report*. December 14, 2020. <https://money.usnews.com/investing/stock-market-news/slideshows/s-p-500-companies-with-stock-buybacks?slide=2>; "Apple's \$460 Billion Stock Buyback." *Above Avalon*. April 23, 2020. <https://www.aboveavalon.com/notes/2020/4/23/apples-460-billion-stock-buyback>; "Apple Stock Buybacks (Quarterly)." *Ycharts*. n.d. https://ycharts.com/companies/AAPL/stock_buyback)

LINK TO LEARNING

Stock Buybacks

This *Wall Street Journal* [video about stock buybacks \(https://openstax.org/r/video-about-stock-buybacks\)](https://openstax.org/r/video-about-stock-buybacks)

¹ Bill Maurer. "Apple: New Highs Seem Likely." *Seeking Alpha*. May 11, 2020. <https://seekingalpha.com/article/4346246-apple-new-highs-seem-likely>

explains the various perspectives on the subject. It walks through the basic concepts of how buybacks work and explores some viewpoints on whether buybacks are good, bad, or otherwise.

To put a firm's earnings per share into perspective and allow for a more meaningful analysis, earnings per share is often tracked over a number of years, such as when presented in the comparative income statements for Clear Lake Sporting Goods (see [Figure 6.7](#)).

Clear Lake Sporting Goods Comparative Year-End Income Statements			
	Current Year	Prior Year	2 Years Prior
Net Sales	\$120,000	\$100,000	\$90,000
Cost of Goods Sold	60,000	50,000	45,000
Gross Profit	60,000	50,000	45,000
Rent Expense	5,500	5,000	5,000
Depreciation Expense	3,600	2,500	2,000
Salaries Expense	5,400	3,000	2,750
Utility Expense	2,500	1,500	1,250
Operating Income	43,000	38,000	34,000
Interest Expense	2,000	3,000	2,000
Income Tax Expense	6,000	5,000	5,000
Net Income	\$ 35,000	\$ 30,000	\$27,000
Basic Weighted Shares Outstanding	8,000	10,550	11,100
Basic Net Income per Share (EPS)	\$ 3.75	\$ 2.56	\$ 2.21
Common Dividends	\$ 6,000	\$ 4,000	\$ 3,500
Preferred Dividends	\$ 5,000	\$ 3,000	\$ 2,500

Figure 6.7 Comparative Year-End Income Statements Earnings per share year after year can be a good indication of a company's financial health.

Most analysts believe that a consistent improvement in earnings per share year after year is an indication of continuous improvement in the earning power of a company. This is what is seen in Clear Lake Sporting Goods' earnings per share amounts over each of the three years reported, moving from \$2.21 to \$2.56 to \$3.75. However, it is important to remember that earnings per share is calculated on historical data, which is not always predictive of the future.

THINK IT THROUGH

Would You Have Invested?

What if, in 1997, you invested \$5,000 in Amazon? Today, your investment would be worth nearly \$6 million. Potential investors viewing Amazon's income statement in 1997 would have seen earnings per share of negative \$1.27. In other words, Amazon lost \$1.27 for each share of common stock outstanding. Would you have invested?

Solution:

Answers will vary. A strong response would include the idea that a negative or small earnings per share reflects upon the historical operations of a company. Earnings per share does not predict the future. Investors in 1997 looked beyond Amazon's profitability and saw its business model having strong future potential.

Price/Earnings (P/E) Ratio

The **price/earnings (P/E) ratio** measures the current market share price of a company's stock relative to its earnings per share (EPS). The ratio is helpful in comparing performance and stock price of a company to other companies. It's also helpful in evaluating how much investors are willing to pay for earnings performance. Investors, in particular, use this ratio and rely on two key characteristics: past performance (trailing) and future estimates (forward). Trailing data can be calculated but is also easily found online, as it's a common measure reported on financial sites. Investors will often look for P/E TTM, which is the price/earnings ratio for the trailing 12 months (last year worth of earnings data). This helps investors assess one day's stock price relative to the earnings per share over the past 12 months. P/E ratio is widely used by investors to determine if a stock is over- or undervalued. It also helps them compare one firm to that of the industry average or index, such as the S&P 500.

$$\text{Price/Earnings Ratio} = \frac{\text{Market Value per Share}}{\text{Earnings per Share}}$$

In the prior section we saw earnings per share data for Clear Lake Sporting Goods. Using its current year earnings per share of \$3.75 and the current stock price of \$69.41, we can calculate price/earnings ratio for Clear Lake Sporting Goods:

$$\text{Price Earnings Ratio} = \frac{\$69.41}{\$3.75} = 18.51$$

An 18.51 ratio means an investor would expect to invest \$18.51 to gain \$1 of earnings.

Book Value per Share

Book value per share is often used hand in hand with market value per share. Investors compare the two in order to see if the stock is possibly over- or undervalued. Book value is derived from accounting practices and shows the value of the firm on paper. Market value, on the other hand, is determined by supply and demand, based on what investors are willing to pay for the stock. If the market value per share is higher than the book value, the stock is considered overvalued. If the market value is lower than the book value, it's considered undervalued.

In theory, book value per share represents the total value common shareholders would receive if the firm were liquidated. It is total equity less preferred equity, spread across the total shares outstanding. The formula to calculate book value per share is

$$\text{Book Value per Share} = \frac{\text{Total Equity} - \text{Preferred Equity}}{\text{Total Shares Outstanding}}$$

The book value per share for Clear Lake Sporting Goods is

$$\text{Book Value per Share} = \frac{\$100,000 - \$20,000}{8,000} = \$10$$

If investors compared the book value per share of \$10.00 for Clear Lake Sporting Goods to the P/E ratio of \$18.51, they would likely conclude that the stock was undervalued in the year of analysis.

LINK TO LEARNING

Book Value versus Market Value of Shares

This [video about book value and market value \(https://openstax.org/r/video-about-book-value\)](https://openstax.org/r/video-about-book-value) explains the basic concepts and discusses how the two differ. Samples of the concept are then explored using Apple Inc. as an example.

6.6 Profitability Ratios and the DuPont Method

By the end of this section, you will be able to:

- Calculate profit margin to determine how much sales revenues the firm has translated into income.
- Evaluate firm performance by calculating return on total assets and return on equity.
- Analyze organizational performance using DuPont method calculations.

Profitability considers how well a company produces returns given its operational performance. The company needs to use its assets and operations efficiently to increase profit. To assist with profit goal attainment, company revenues need to outweigh expenses. Let's consider three profitability measurements and ratios: profit margin, return on total assets, and return on equity.

Profit Margin

Profit margin represents how much of sales revenue has translated into income. This ratio shows how much of each \$1 of sales is returned as profit. The larger the ratio figure (the closer it gets to 1), the more of each sales dollar is returned as profit. The portion of the sales dollar not returned as profit goes toward expenses. The formula for profit margin is

$$\text{Profit Margin} = \frac{\text{Net Income}}{\text{Net Sales}}$$

For Clear Lake Sporting Goods, the profit margin in the current year is

$$\text{Profit Margin} = \frac{\$35,000}{\$120,000} = 0.29 \text{ (rounded) or } 29\%$$

This means that for every dollar of sales, \$0.29 returns as profit. If Clear Lake Sporting Goods thinks this is too low, the company would try to find ways to reduce expenses and increase sales.

Return on Total Assets

The **return on total assets** measures the company's ability to use its assets successfully to generate a profit. The higher the return (ratio outcome), the more profit is created from asset use. Average total assets are found by dividing the sum of beginning and ending total assets balances found on the balance sheet. The beginning total assets balance in the current year is taken from the ending total assets balance in the prior year. The formula for return on total assets is

$$\begin{aligned} \text{Return on Total Assets} &= \frac{\text{Net Income}}{\text{Average Total Assets}} \\ \text{Average Total Assets} &= \frac{(\text{Beginning Total Assets} + \text{Ending Total Assets})}{2} \end{aligned}$$

For Clear Lake Sporting Goods, the return on total assets for the current year is

$$\begin{aligned} \text{Average Total Assets} &= \frac{(\$200,000 + \$250,000)}{2} = \$225,000 \\ \text{Return on Total Assets} &= \frac{\$35,000}{\$225,000} = 0.16 \text{ (rounded) or } 16\% \end{aligned}$$

The higher the figure, the better the company is using its assets to create a profit. Industry standards can dictate what an acceptable return is.

LINK TO LEARNING

Return on Assets

This [video explains how to calculate return on assets \(https://openstax.org/r/video-explains-how-to-calculate\)](https://openstax.org/r/video-explains-how-to-calculate) and how to interpret the results. The video provides the formula, a discussion of the concept, and the importance of the ratio.

Return on Equity

Return on equity measures the company's ability to use its invested capital to generate income. The invested capital comes from stockholders' investments in the company's stock and its retained earnings and is leveraged to create profit. The higher the return, the better the company is doing at using its investments to yield a profit. The formula for return on equity is

$$\text{Return on Equity} = \frac{\text{Net Income}}{\text{Average Stockholder Equity}}$$

$$\text{Average Stockholder Equity} = \frac{\text{Beginning Stockholder Equity} + \text{Ending Stockholder Equity}}{2}$$

Average stockholders' equity is found by dividing the sum of beginning and ending stockholders' equity balances found on the balance sheet. The beginning stockholders' equity balance in the current year is taken from the ending stockholders' equity balance in the prior year. Keep in mind that the net income is calculated after preferred dividends have been paid.

For Clear Lake Sporting Goods, we will use the net income figure and deduct the preferred dividends that have been paid. The return on equity for the current year is

$$\text{Average Stockholder Equity} = \frac{90,000 + 100,000}{2} = 95,000$$

$$\text{Return on Equity} = \frac{\$35,000 - \$5,000}{\$95,000} = 0.32 \text{ (rounded) or } 32\%$$

The higher the figure, the better the company is using its investments to create a profit. Industry standards can dictate what an acceptable return is.

The DuPont Method

ROE in its basic form is useful; however, there are really three components of ROE: operating efficiency (profit margin), asset usage (total asset turnover), and leverage (equity ratio). This is known as the **DuPont method**. It originated in 1919 when the DuPont company implemented it for internal measurement purposes.² The DuPont method can be expressed using this formula:

$$\text{ROE} = \text{Profit Margin} \times \text{Total Asset Turnover} \times \text{Equity Multiplier}$$

Profit margin indicates how much profit is generated by each dollar of sales and is computed as shown:

$$\text{Profit Margin} = \frac{\text{Net Income}}{\text{Net Sales}}$$

Total asset turnover indicates the number of sales dollars produced by every dollar invested in capital assets—in other words, how efficiently the company is using its capital assets to generate sales. It is computed as shown:

² Joshua Kennon. "What Is the DuPont Method Return on Equity, or ROE, Formula?" The Balance. December 16, 2020. <https://thebalance.com/the-dupont-model-return-on-equity-formula-for-beginners-357494>

$$\text{Equity Multiplier} = \frac{\text{Average Total Assets}}{\text{Average Stockholders' Equity}}$$

The equity multiplier measures leverage. It is computed as shown:

$$\text{Equity Multiplier} = \frac{\text{Average Total Assets}}{\text{Average Stockholders' Equity}}$$

Using DuPont analysis, investors can see overall performance broken down into smaller pieces, which helps them better understand what is driving ROE. We already have the computations for Clear Lake Sporting Goods' profit margin and total asset turnover:

$$\text{Profit Margin} = \frac{\$35,000}{\$120,000} = 0.29 \text{ (rounded) or } 29\%$$

$$\text{Total Asset Turnover} = \frac{\$120,000}{\$225,000} = 0.53 \text{ times (rounded)}$$

We can calculate the equity multiplier using the equity multiplier equation and prior calculations for Clear Lake's average total assets and average stockholder equity:

$$\text{Equity Multiplier} = \frac{\text{Average Total Assets}}{\text{Average Stockholders' Equity}}$$

$$\text{Average Total Assets} = \frac{\$200,000 + \$250,000}{2} = \$225,000$$

$$\text{Average Stockholder Equity} = \frac{\$90,000 + \$100,000}{2} = \$95,000$$

$$\text{Equity Multiplier} = \frac{\$225,000}{\$95,000} = 2.37$$

Now that we have all three elements, we can complete the DuPont analysis for Clear Lake Sporting Goods:

$$\text{ROE} = \text{Profit Margin} \times \text{Total Asset Turnover} \times \text{Equity Multiplier}$$

$$\text{ROE} = 29\% \times 0.53 \times 2.37 = 0.36 \text{ or } 36.4\%$$

LINK TO LEARNING

The DuPont Method

This [video about the DuPont method \(https://openstax.org/r/video-about-DuPont-method\)](https://openstax.org/r/video-about-DuPont-method) walks through its history, discusses its basic components, and shows how to calculate and interpret each measurement.

Performance Analysis

ROE captures the nuances of all three elements. A good sales margin and a proper asset turnover are both needed for a successful operation. Like all ratios, assessing performance is relative. It's important to look at the ratio in context of the organization, its history, and the industry. If we compare Clear Lake's ROE of 26.4% to the recreational products industry average of 12.56% for the same year, it would appear as though Clear Lake Sporting Goods is outperforming the general industry. However, recreational products can include a wide variety of businesses beyond just the outdoor gear in which Clear Lake Sporting Goods specializes. An analyst could look at other key competitors such as Cabela's or Bass Pro Shops to get even more relevant comparisons.

Clear Lake Sporting Goods is also technically a retail store, albeit a specialized one. An analyst might also consider the industry averages for general or online retail of 20.64% and 27.05%, respectively. Compared to the broader retail industry, Clear Lake Sporting Goods is still performing well, but its performance is not as

disparate to industry average as when compared to recreational products (see [Table 6.1](#)).

Industry	ROE (%)
Advertising	2.93
Air Transportation	-47.03
Computer Services	13.50
Banking	8.22
Financial Services (nonbanking)	64.28
Food Processing	10.12
Renewable Energy	-20.59
Hospitals/Health Care Facilities	70.64
Hotels/Gaming	-30.40
Publishers	-14.18
Recreational Products	12.56
Real Estate (general)	2.00
Retail:	0.00
Automotive	36.28
Building Supply	0.27
General	20.64
Grocery	30.63
Online	27.05
Rubber and Tires	-26.69
Shoes	23.70
Software (systems and applications)	28.09
Transportation	21.47
Total Market Average	8.25

Table 6.1 Return on Equity by Industry in 2020 It's

important to look at any ratio in context of the organization, its history, and the industry. (data source: Aswath Damodaran Online)



Summary

6.1 Ratios: Condensing Information into Smaller Pieces

Ratios used in financial analysis can help investors and other analysts identify trends over time, compare companies to one another, and make informed decisions about a company. There are, however, limitations to analysis, so it should be used wisely and in conjunction with other contextual information available.

6.2 Operating Efficiency Ratios

Efficiency ratios measure how well management uses the assets of the organization to earn a profit. Common efficiency ratios include accounts receivable turnover, total asset turnover, inventory turnover, and days' sales in inventory.

6.3 Liquidity Ratios

Liquidity ratios help analysts measure how well an organization can meet its short-term obligations (liabilities) as they come due. Common ratios to measure liquidity include the current ratio, the quick ratio, and the cash ratio. Each of these three ratios includes more (cash ratio) or less liquid (current ratio) current assets in its measure of liquidity.

6.4 Solvency Ratios

Solvency ratios measure how well an organization can meet its long-term obligations (liabilities) as they come due, or in more general terms, its ability to stay in business. Common solvency ratios include the debt-to-assets ratio and the debt-to-equity ratio.

6.5 Market Value Ratios

Market value ratios help analysts assess the value of publicly traded firms in the market. The most commonly used market value ratios include earnings per share (EPS), the price/earnings ratio, and book value per share.

6.6 Profitability Ratios and the DuPont Method

Profitability ratios help measure how effectively the organization earns a profit. Common profitability ratios include profit margin, return on total assets, and return on equity. The DuPont method breaks down return on equity into three smaller components for a more thorough assessment of performance: profit margin, total asset turnover, and an equity multiplier.



Key Terms

accounts receivable turnover ratio measures how many times in a period (usually a year) a company will collect cash from accounts receivable

book value per share total book value (assets – liabilities) of a firm expressed on a per-share basis

cash ratio represents the firm's cash and cash equivalents divided by current liabilities; often used by investors and lender to assess an organization's liquidity

current ratio current assets divided by current liabilities; used to determine a company's liquidity (ability to meet short-term obligations)

days' sales in inventory the number of days it takes a company to turn inventory into sales

debt-to-assets ratio measures the portion of debt used by a company relative to the amount of assets

debt-to-equity ratio measures the portion of debt used by a company relative to the amount of stockholders' equity

DuPont method framework for financial analysis that breaks return on equity down into smaller elements

earnings per share (EPS) measures the portion of a corporation's profit allocated to each outstanding share of common stock

efficiency ratios ratios that show how well a company uses and manages its assets